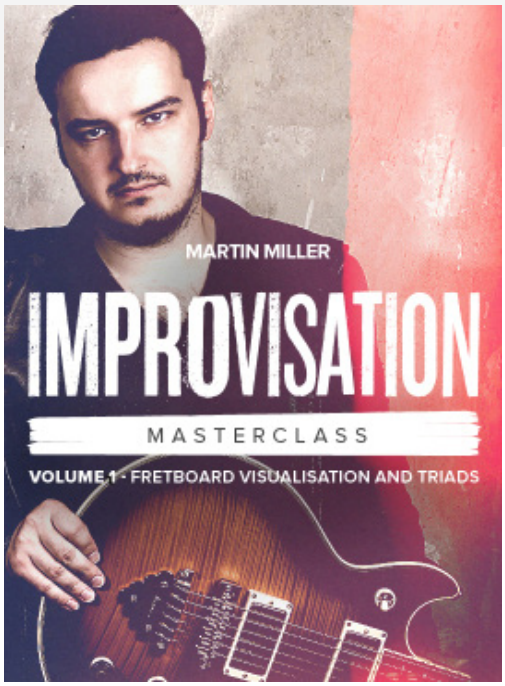


Martin Miller

IMPROVISATION MASTERCLASS VOL.1

FRETBOARD VISUALISATION & TRIADS



1. INTRODUCTION

Hello fellow guitar player, thanks for checking out my Improvisation Masterclass Vol. 1: Fretboard Visualisation & Triads. This package is the first volume of a new series of lessons that will cover all kinds of topics regarding improvisation on guitar.

It is my idea to turn this Improvisation Masterclass into a huge encyclopaedia, adding one volume at a time until it becomes a huge resource for the improvising guitar player.

There is a logical order to this, and future masterclasses. To get the maximum understanding from the ideas and concepts presented here, I recommend that you read (and practise!) through the pages in order as there is a common thread going from one topic to the next. However, each chapter is also designed to be self-contained. This way, the masterclass can be used as a long-term reference (for when you might decide to learn more about triads, for instance).

I highly recommend that you take a lot of time reading, digesting and applying this material. Unlike some of my other releases, which were based more on licks and vocabulary, this product is much more in-depth and conceptual. It is less about giving your playing a quick boost or making you sound better in an instant (not there is anything wrong with that!)

The ideas presented here need a lot more time and effort on your part to be understood, absorbed and applied. It is more about academia than artistry and performance, but if -

- studied thoroughly it will pay off hugely in the long run. It'll help you gain a more thorough understanding of musical concepts and ultimately turn you into a better musician on a more fundamental level. More than learning and mechanically repeating a few licks could ever do. You'll find that the most incredible musicians have the seeming ability to do whatever they want, pretty much on the fly. My masterclass is intended to help you get to that point!

This material is a collection of concepts and methods that I've collected and refined through many years of academic studies; self tuition; recording and gigging; conversing with musicians of all instruments and styles; and teaching guitarists ranging from absolute beginners to college students and professionals. It is the closest reflection of my thought/learning process that I can possibly give you. So if you want to play like me, you should study like me!

Remember to take your time, study carefully and thoroughly and most importantly have fun stepping into the exciting depths of guitar improvisation!

2. Fretboard Visualisation

2.1. What is it and why is it important?

Consider fretboard visualisation as the roadmap of your guitar. It's a key element when navigating the fretboard during your improvisations (and it also helps you memorise composed parts better). It helps you find the right notes you're looking for in an impromptu situation and organises any devices that you know (triads, pentatonics, diatonic scales etc) in a logical fashion.

My preferred method for this is the CAGED system. Of course there are other systems that serve the same purpose (3-note-per-string is probably the most popular) and it can never hurt to study alternative approaches and eventually morph them into a method that works for you. Like many new students I see, you may have never heard of CAGED before, but you may have subconsciously adopted elements of it into your fretboard visualisation. I now want to provide you with a systematic, thorough approach to it, leaving the fewest possible gaps in your fretboard knowledge.

Keep in mind that CAGED is a relative system and is fully transposable, meaning that anything you learn in one key can be transposed to another key, by moving the same shapes to a different fretboard position. It is NOT for people who want to learn the note names on the fretboard. This is something you need to do before getting serious about studying CAGED. Also, CAGED is not a system set in stone. There are all kinds of variations in fingerings, names and basic rules. What I'm presenting here is my interpretation of CAGED; the version that I found most useful for myself and my students.

One of the advantages of the CAGED system is how it divides the fretboard into five bite-sized pieces, covering every note on every string. Each position has a name, and these names are derived from the classic open (“campfire”) guitar chords: C major, A major, G major, E major and D major. This allows for easier communication among guitar players, particularly between student and teacher. Also, we humans tend to memorize things more easily when we attach a name and category to them.

My thought process when improvising single note lines goes through 3 steps:

1. “Pre-hear” a melodic line in your head. This is a culmination of your personal taste, artistic vision, listening and playing experience, musical upbringing and knowledge, etc.

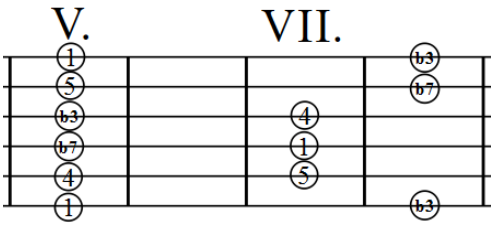
2. Once you have an artistic idea of what you want to play, formulate the idea. This means that you turn the melody into an intervallic formula against the underlying harmony. For example, if I’m playing over Dmaj7 and hear a melody that goes A-D-E-F#-D-E-C#-D, in my mind I'll turn that into 5-1-2-3-1-2-7-1. These are the relationships between the notes and the chords (1=root, etc).

3. The final step is to translate that formula onto the fretboard to find the notes you were looking for. This is where your fretboard knowledge kicks in. You can have the greatest possible musical ideas in your head, but if you don't know where to find them on a guitar neck, you will not be able to express them.

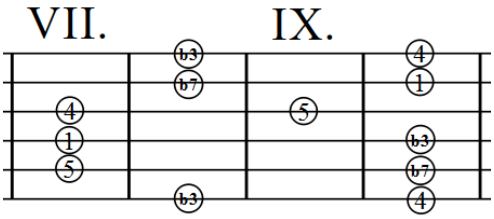
2.2. How CAGED works...

If you have ever learned any of these five typical pentatonic shapes, you have (consciously or not) used the CAGED system before: *(Note: the numbers in all diagrams reflect the interval, not the finger used)*

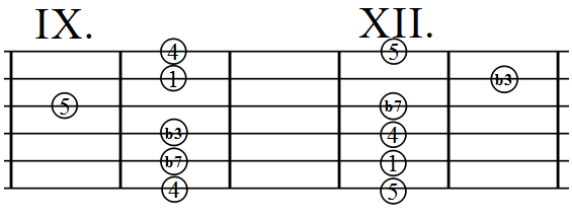
A Minor Pentatonic Position 1 (E Shape)



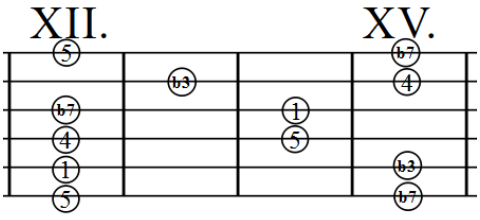
A Minor Pentatonic Position 2 (D Shape)



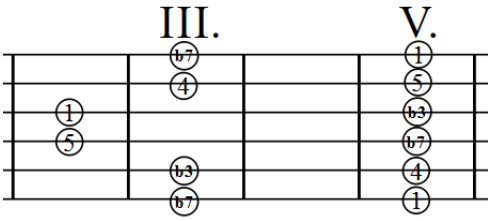
A Minor Pentatonic Position 3 (C Shape)



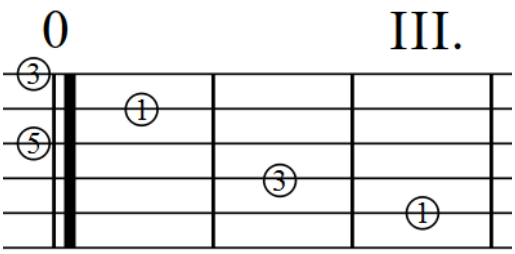
A Minor Pentatonic Position 4 (A Shape)



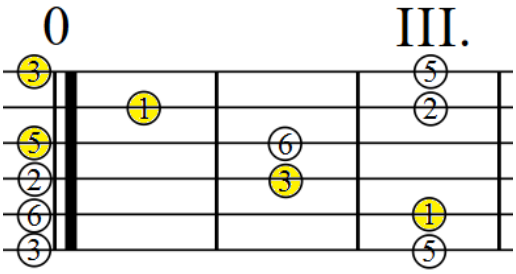
A Minor Pentatonic Position 5 (G Shape)



By using these shapes, you have divided the fretboard into 5 distinct positions just like in CAGED. Another integral aspect of CAGED is how each position is not only named after the open chords, but also based around the geometrical shape of those chords. Look at the how the open C major chord (image #1) sits within the C major pentatonic shape (image #2). They share three notes (root, 3rd, 5th) and in order to turn the C chord into the pentatonic you add two more notes (2nd & 6th). The coloured dots represent the original open chord shape.

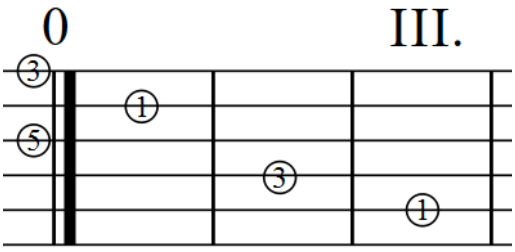


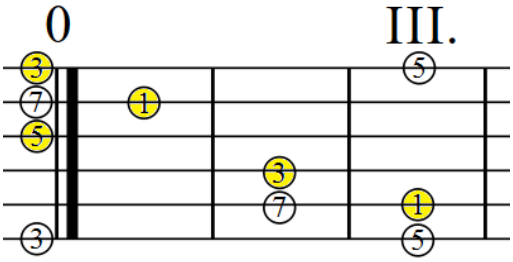
Open C major chord



Open C major pentatonic, C-position

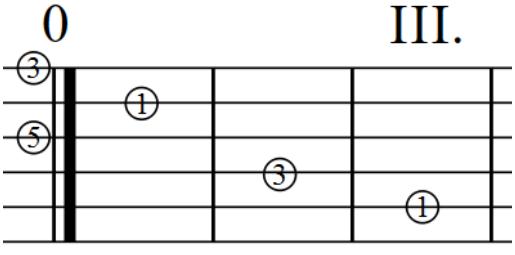
You can think of the open C major chord as a framework or skeleton, around which you build the major pentatonic. The same goes for the C major 7 arpeggio...

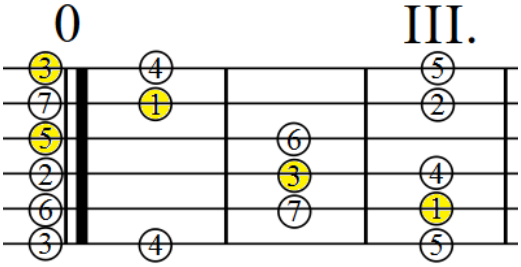




Open C major 7 arpeggio, C-position

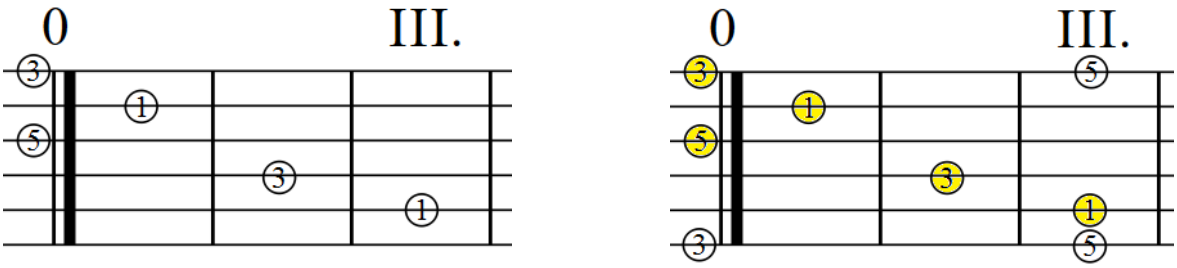
... and the C major scale. In all these cases the C major chord is your point of orientation.





Open C major scale, C-position

And of course you can expand the C major triad across all strings by using the C major open chord shape as a reference:



Open C major triad, C-position

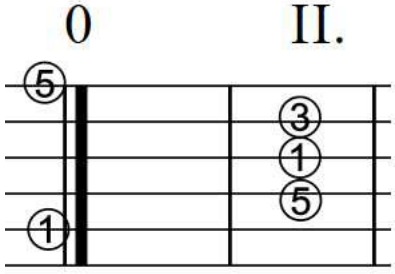
All of the above examples are played in the C-position. This means they are based on the shape of the C major chord, NOT because they are in the key of C major. The C-position can be used in any of the 12 keys by moving the root note to the desired position. If you wanted to play that major pentatonic shape (see above) in E major, you'd start off by placing the root note (the “1” in our diagrams) on an E note. The “1” on the A string would move from the 3rd fret to the 7th fret, and the whole shape would follow (of course then you won't be using open strings, but fretting each note. This gives you the E major pentatonic in the C-position.

2.3. Finding the positions

Next we want to learn how to find and identify each of the five positions on the neck. We're going to stick to the key of C major for now. We’ve already discovered the C-position, which is down in the open position, but can also be played 12 frets higher if you want to avoid using open strings. Let's try for the A-position next!

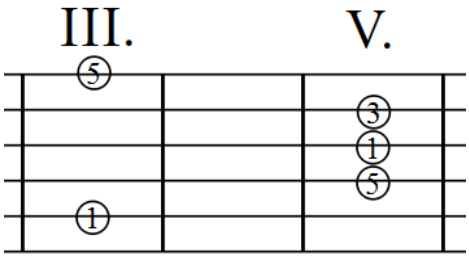
When finding a CAGED position, before it becomes second nature, start out by playing the equivalent open chord for reference. In this case we need an open A major chord, our “parent chord”. Now we have to move that open A major chord up to the key of C major. Do that by moving it up chromatically step by step. Use your index finger as a bar as you go up instead of the open strings. Move up till you hit the C major chord (the one with the index barred on the 3rd fret).

If you've done it correctly you'll have gone from this chord...



Open A major chord (parent chord)

To this chord:



C major chord, A-shape

Congratulations, you've found the A-position in the key of C major! All your practice work in this position (triads, arpeggios, phrases etc) is going to visually revolve around this chord shape... this “framework”.

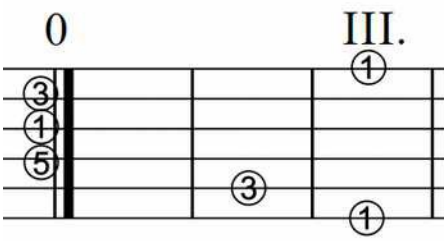
To be totally clear about what's happening: you're now playing a C major chord using the A-shape. If you were to play e.g. the C major scale around that transposed A-shape, you'd be playing the C major scale in the A-position.

[The term “shape“ refers to the geometrical layout of the actual notes that you're playing. “Position“ refers to the part of the CAGED system, that you're in.]

Feel free to experiment a little by simply moving around the A-shape bar chord up and down the neck. Whatever fret you land on, the note on the A-string (with the “1” in the diagram above) will be the root note of the A-position. As an example, if you move this chord shape to the 10th fret, you will be in the key of G major in the A-position. All the other chord shapes of the CAGED system transpose up and down the fretboard in the same way.

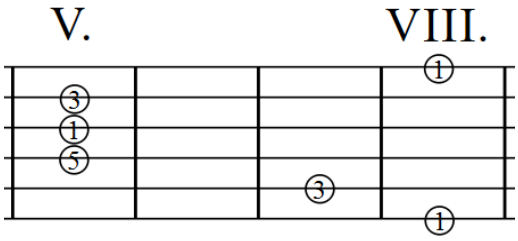
We will start playing and practising within these positions later. For now let's keep trying to find the other positions.

The next task will be to find the G-position of C major. That means we’ll be using an open G chord shape as our point of reference will be the G root on the low E-string. We’re going to follow the same process as before. Start out by playing the open parent chord that gives its name to the position (G major) then move it up one fret at a time, until the reference point (the root note on the low E string) ends up at a C note. On the low E string, the C is at the 8th fret:



Open G major chord (parent chord)

Move up to...

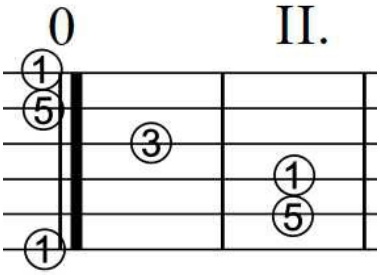


C major chord, G-shape

As you may have guessed, in order to figure out the five CAGED positions in any major key, we're basically just taking the five open chord types and moving them to the desired key.

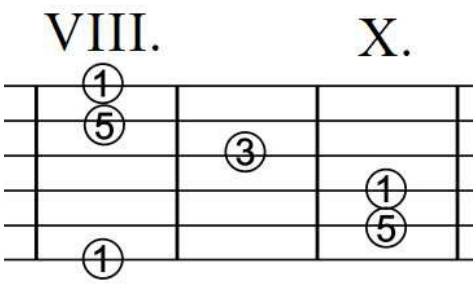
By now you should have understood this concept and I'm going to fast forward a little by showing you where to move the remaining open parent chords, so they end up in C major. These chords will reveal the location of each CAGED position in the key of C major. We’ll be using these five positions to visualise scales, arpeggios etc.

Again, when transposing these chords, I use the lowest root note of the parent open chords as a reference point, moving that to the desired key. For each open shape, move their lowest root note to a C (that root note will be either on the E-, A- or D-string, depending on the parent chord), maintaining the chord shape at the new position.



Open E major chord (parent chord)

Move to...

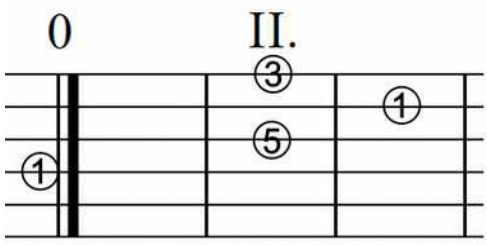


C major chord, E-shape

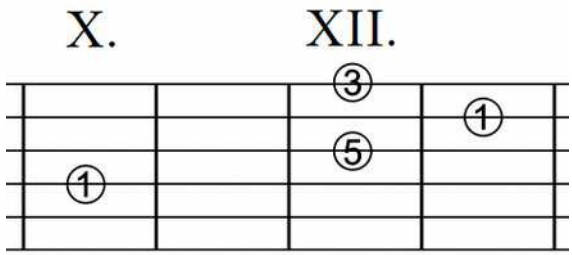
Note how the E-position and G-position share the same lowest root note on the same fret, but the geometric direction in which the notes are laid out across the fretboard is different. In very basic terms: the E-position moves "straight across" the strings, whereas the G-position moves "diagonally towards the lower frets". This visual reference is very important for later, when we'll investigate scales and arpeggios in those positions.

The visual aspect is one of the key elements of the CAGED system. Just as the root notes of the E-position and G-position "overlap", the same thing happens between the A- and C-positions, where the rest of the notes go into different directions over the neck, starting from the same root.

Finding the D-position of C Major:



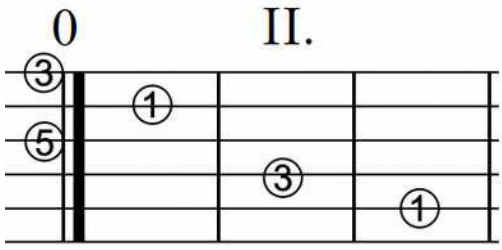
Move to...



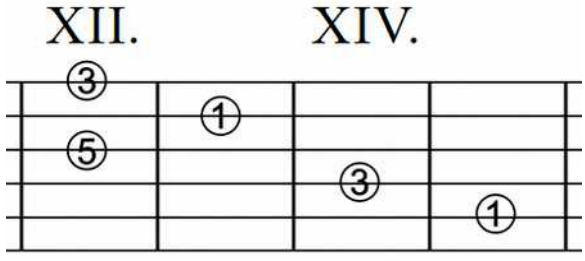
Open D major chord (parent chord)

C major chord, D-shape

Lastly we're going to look for the C-position again, only this time 12 frets higher, so we won't be using any open strings.



Move to...



Open C major chord (parent chord

C major chord, C-shape

Video/tab file: "moving_chords_C"

2.4. The CAGED-workout

2.4.1. Introduction

Now it's time to show you how to study the individual positions. I'm going to show you some basic exercises first and will then explain to you how to expand them into a workout that is going to keep you busy for months and years. The beauty of fretboard visualisation is that it doesn't matter how good you are. All of these exercises are designed to benefit you, whether you're a beginner or a seasoned pro. I recommend working on your fretboard knowledge on a regular basis. These skills tend to fade rather quickly if you don't renew them every once in a while.

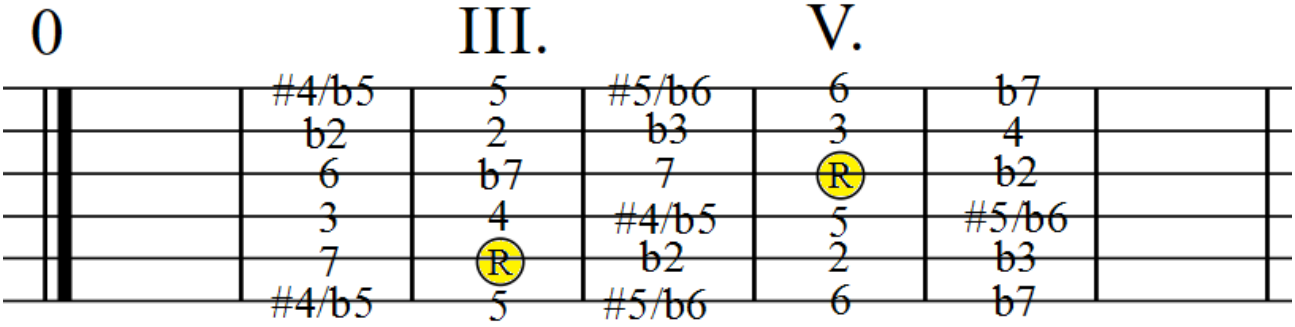
With the first set of exercises, we're going to cover the entire key of C major all over the neck, before moving on to other keys and tonalities to further expand our understanding of the fretboard. We're going to skip fingerings involving open string, since it's not possible to transpose them. This is why we're starting with the A-position as opposed to the open C-position. When trying to study a position in my "workout", I usually play through the following (musically crucial) devices:

- **triad arpeggio built from the root note of the key**
- **7th arpeggio built from the root note of the key**
- **pentatonic**
- **diatonic scale/mode**

In all four cases, I start on the lowest possible root note in the position. I play up to the highest note within that position, down to the lowest note of the position and back to the starting note. When applied to all five positions, this method covers every note on the fretboard. My CAGED-workout is designed to get rid of all the blank spots in your fretboard knowledge.

Another guideline for me is not to use any stretches. A stretch by definition is anything that takes your fingers out of a four-fret space. Use position shifts only if they can't be avoided and choose the way that deviates the least from your position. The less you move away from your initial point of orientation, the easier it will be to memorise the fretboard.

For every note in a CAGED position, think about its function in relation to the lowest root note in that position. In other words, what is its interval? I like to do this for all notes along a 4-5 fret range. So in my mind, the A-position in the key of C would look something like this:



A-position interval mind map (key of C)

The further you stray away from your initial reference note by position shifting or stretching, the harder it becomes to visualise the function of each note. If I move any further from my initial position, I would rather use the root note of the next CAGED position as my new reference point. Also it's entirely up to you, whether you prefer to reference against the lowest root note of your position (like I do) or against the root note, that is the closest to the note you're on.

Use my neck diagrams as a starting point to check that what you're doing is correct. Ultimately you want learn how to play any of these things (arpeggios, pentatonic, full scale, etc) without having to look at these pages. I've written out a lot of shapes for you to get started. If you have trouble finding the right notes for whatever you're trying to play, I've included interval maps like above as a reference. Look them up and follow the advice given later in the masterclass to get a better grasp on recognising individual intervals on the neck.

The beauty of this system is, if you have a good ear, you can pre-hear what an interval sounds like in your head and then translate it to the guitar neck immediately. The combination of great relative pitch and a thorough understanding of the fretboard is the secret weapon for great improvisational freedom.

Some last pieces of advice before you start playing...

Repeat each exercise carefully several times. While doing so, shift your attention and imagination towards different elements:

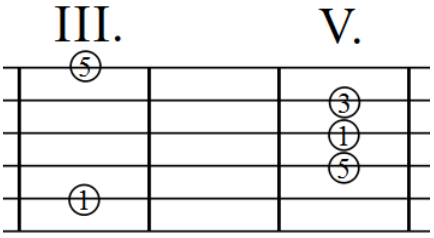
- The geometrical shape of whatever you're practising; both in itself and in comparison to the shape of the parent chord
- The sound of each interval in relation to the key you're in. If possible, hold the chord you're playing over (on a keyboard, loop pedal, invite a friend over, etc)
- The name or function of the interval you're playing in relation to the key (3rd, 7th, whatever).
- The actual name of the note you're playing (E, B, whatever).

But enough talk, on to the playing!

2.4.2. The A-position (C major)

Reference Chord

Our reference chord is this, the C major chord in the A-position:

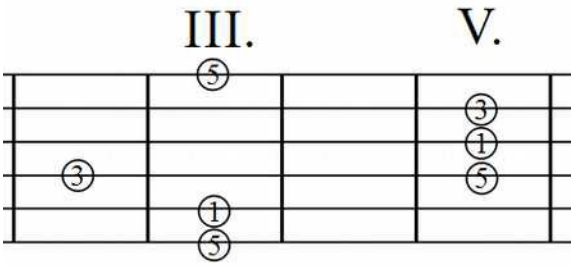


C major chord, A-shape (A-position)

Triad

After finding the position, start out by playing the triad arpeggio from the root note (C major triad: C E G). Remember: start on the low root note, play up to the highest note in the position, all the way back to the lowest and back to the root note you started on. Take all the time in the world you need to find the correct notes.

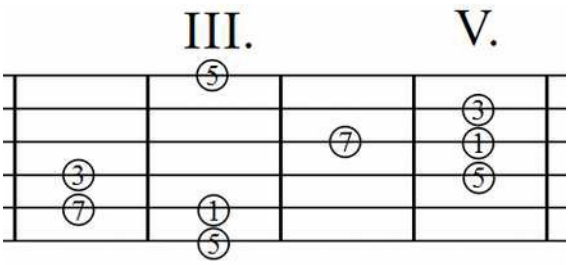
These exercises are about discovering and learning, not playing fast or sounding great. Carefully memorise their shape and the sound of each note.



C major triad, A-position

7th Arpeggio

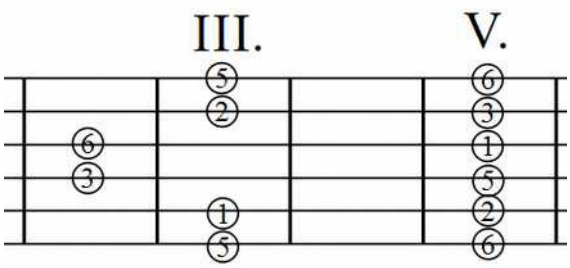
Moving on to the 7th arpeggio built from the tonic of the key (Cmaj7: C E G B). The same process as above – low root to top, then to bottom, back to root.



C *major 7 arpeggio, A-position*

Pentatonic

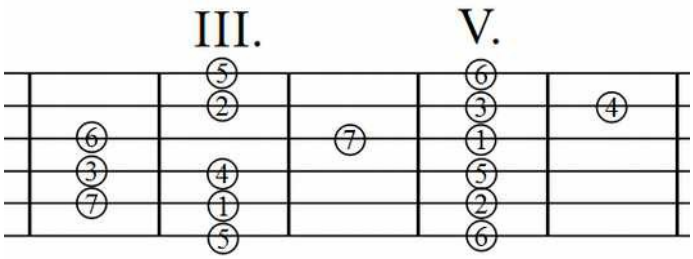
Next is the pentatonic built from the tonic of the key (C major pentatonic: C D E G A). Same rules apply.



C major pentatonic, A-position

Diatonic Scale

Before we move on to the next position we're going to play the entire C major scale now (C D E F G A B). As you probably guessed, the same rules apply.



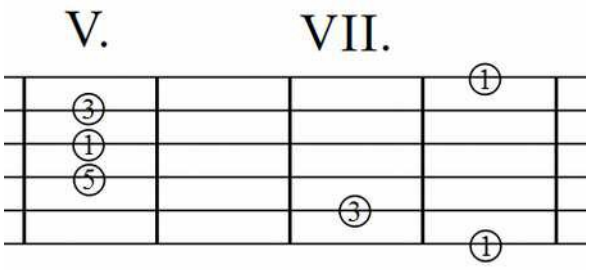
C major scale, A-position

... and that rounds up our exercise of the C major scale in the A-position. But that's just one CAGED position; there's a lot more to do yet!

2.4.3. The G-position (C major)

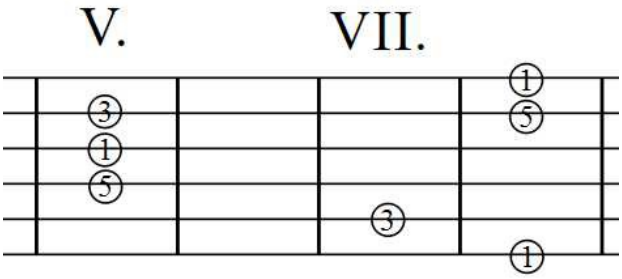
Now we're going to practise the same exercises in the same key (C major) but in the G-position. Please remember these principles and apply them to all of the upcoming exercises, executing them carefully. Practice doesn't make perfect. Perfect practice makes perfect!

Reference Chord



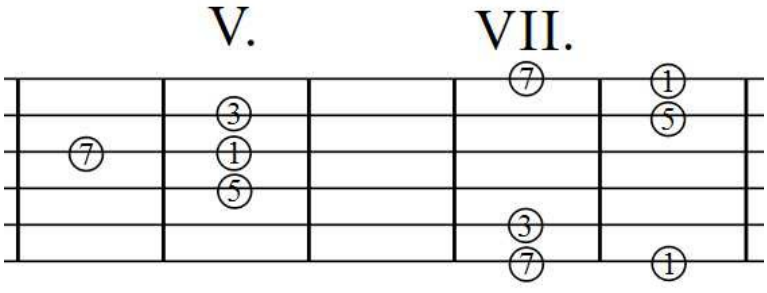
C major chord, G-shape (G-position)

Triad



C major triad, G-position

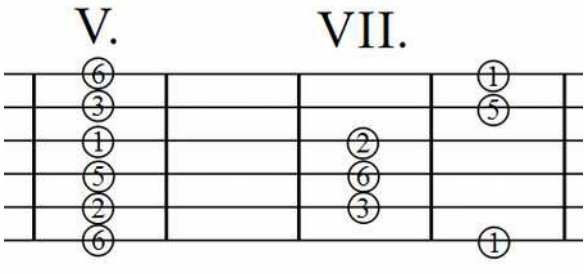
7th Arpeggio



C major 7 arpeggio, G-position

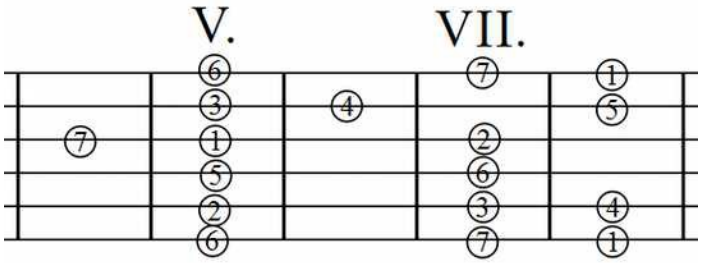
You may find this fingering slightly awkward to play. I normally use my index finger to play both the B and the C on the G-string. Alternatively you might try to play the B on the 9th fret of the D-string. I prefer my method because the G-position has a tendency to move towards the direction of the nut of the guitar as you ascend the scale; I like to maintain this orientation (plus I get to avoid the stretch). The same conflict happens when playing the entire diatonic scale.

Pentatonic



C major pentatonic, G-position

Diatonic Scale

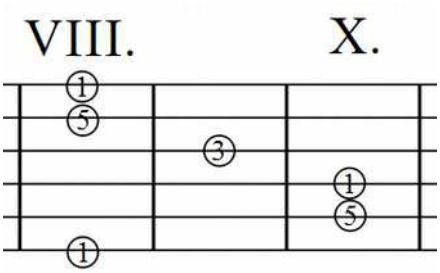


C major scale, G-position

2.4.4. The E-position (C Major)

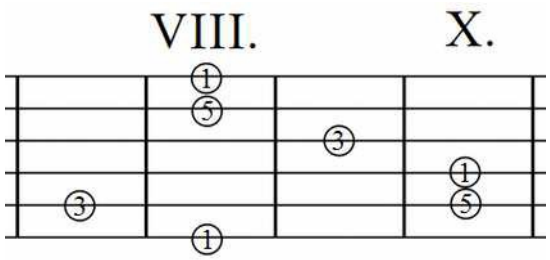
Our workout continues in the E-position. The procedure remains the same.

Reference Chord



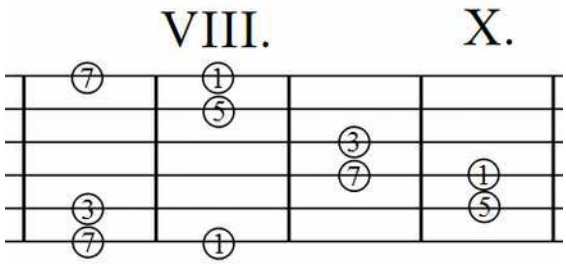
C major chord, E-shape (E-position)

Triad



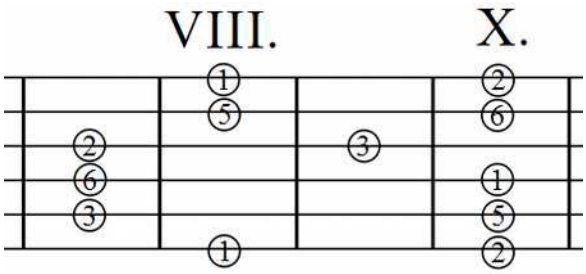
C major triad, E-position

7th Arpeggio



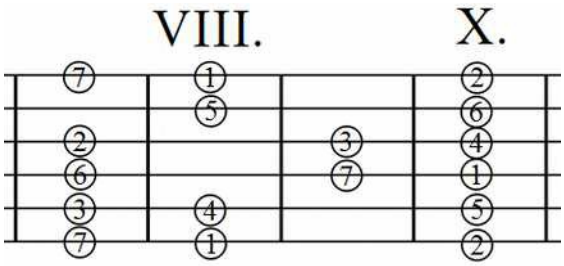
C major 7 arpeggio, E-position

Pentatonic



C major pentatonic, E-position

Diatonic Scale

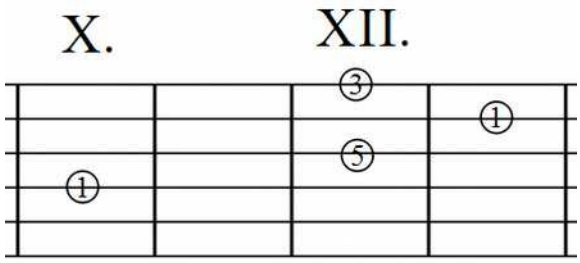


C major scale, E-position

2.4.5. The D-position (C major)

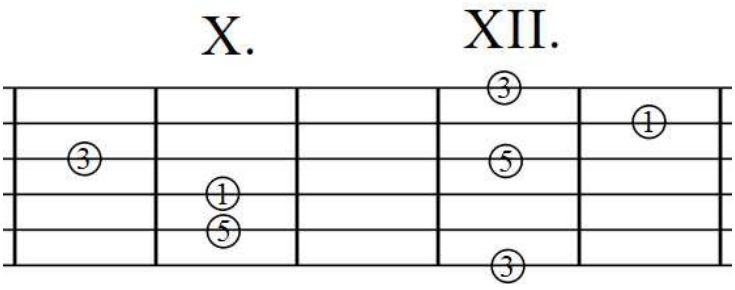
This is the position, that for me personally, requires the most maintenance and is the hardest to visualise. I think that's because it's the only CAGED position where the lowest root note of the key is on the D-string.

Reference Chord



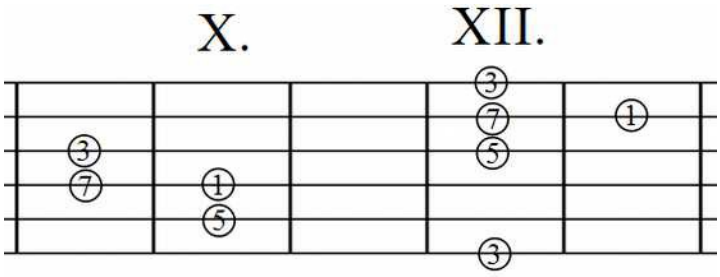
C major chord, D-shape (D-position)

Triad



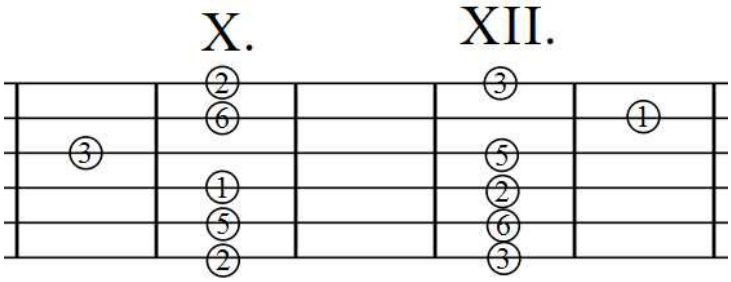
C major triad, D-position

7th Arpeggio



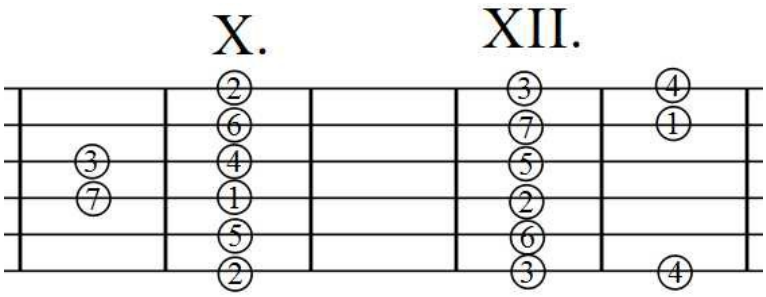
C major 7 arpeggio, D-position

Pentatonic



C major pentatonic, D-position

Diatonic Scale

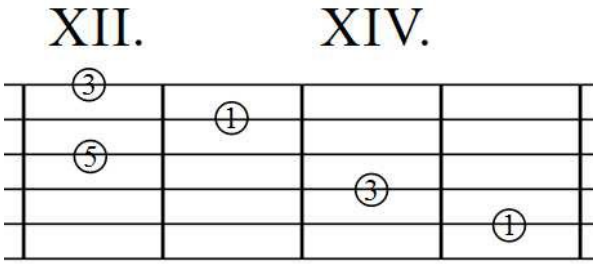


C major scale, D-position

2.4.6. The C-position (C major)

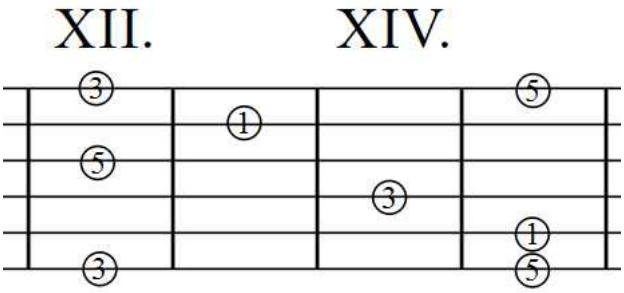
I've sort of covered this position before, but that was using open strings in the open position. In order to be able to learn these shapes in a transposable fashion, we're going to play the same things 12 frets up.

Reference Chord



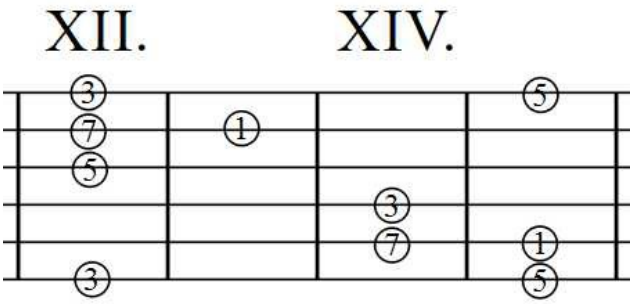
C major chord, C-shape (C-position)

Triad



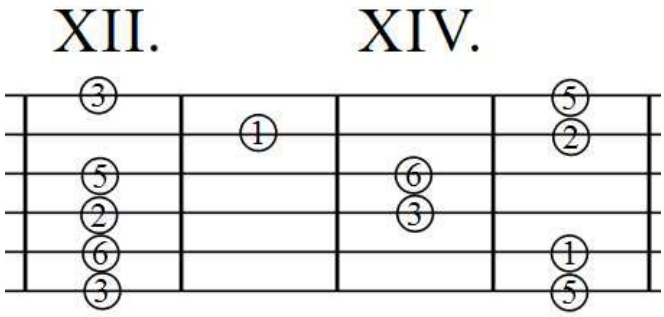
C major triad, C-position

7th Arpeggio



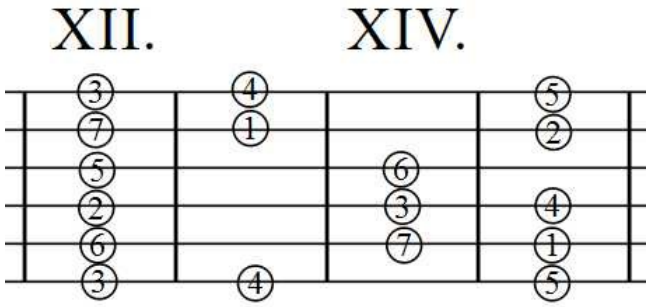
C major 7 arpeggio, C-position

Pentatonic



C major pentatonic, C-position

Diatonic Scale



C major scale, C-position

2.4.7. What's next?

Congratulations, you just covered every note of the C major triad, 7th arpeggio, pentatonic and scale over the entire neck in systematic fashion, using the CAGED system. We only reached the 15th fret, but we don't really need to go higher as the patterns repeat, but it won't harm you to practice that anyway!

We are at a kind of crossroads with our CAGED workout now, that allows us to go into different directions. When practising CAGED, there are multiple parameters that you can change. I would usually modify only one parameter at a time to avoid confusion and to gain a better insight into how that specific parameter affects what I'm doing.

Here's a few things you can do:

1. Change the position. For example, stay in the same key, but move from the A-position to the G-position. This is what we did in the above C major examples.
2. Change the key. For example, take one CAGED position and move from C major to G major.
3. Change the tonality (like #2, but move from C major to C melodic minor, for example).
4. Take one of the exercises (triad, 7th arp, pentatonic, full scale) and modify the key, tonality or position. You could use this approach to focus on your dominant 7th arpeggio knowledge in different keys, for example.

We already did the first exercise by covering all positions of C major. In order to get a hands-on understanding of how the patterns of CAGED can be transposed to any key, we're going to pick a different key now (#2 above) and go through the whole procedure over again (#1 above).

A good idea when changing keys is to go through the cycle of fifths. For the purpose of efficient practice, it can be a bit too simple to just go up or down in whole- or half-steps. It's too easy to move a pattern up or down one fret. Using the cycle of fifths means that you can move systematically through all twelve keys, using quite large jumps between each one.

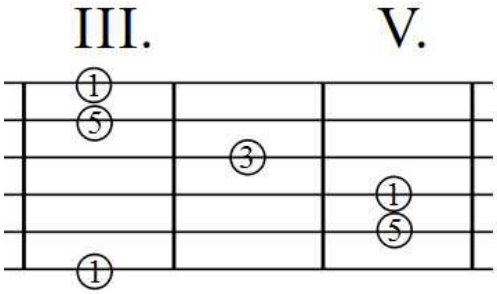
The next key in the cycle of fifths is G major and again I want you to start in the lowest available position, without open strings.

Keep in mind that these exercises are like a sandbox. You can combine them freely and mix things up at will to iron out the weak spots in your fretboard knowledge, because nobody knows better where they are than you. Practice smart!

2.4.8. The E-position (G major)

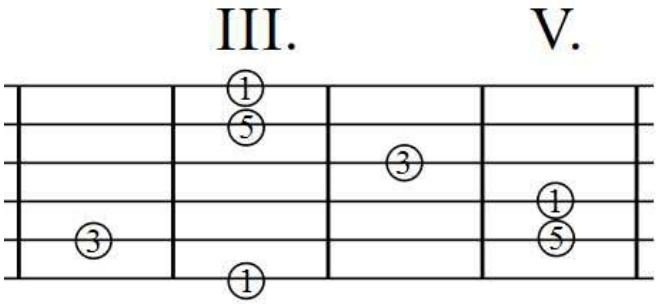
Regardless of key, you will find that all the shapes and relationships of one position to another will be exactly the same. The only thing that changes is the location on the neck. The fret numbers change, but the shapes don't! If you went through all the previous exercises you will recognise these recurring fingerings rather quickly and soon transposition won't be nearly as difficult anymore.

Reference Chord



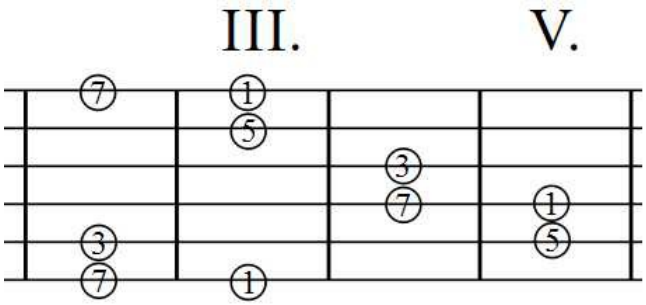
G major chord, E-shape (E-position)

Triad



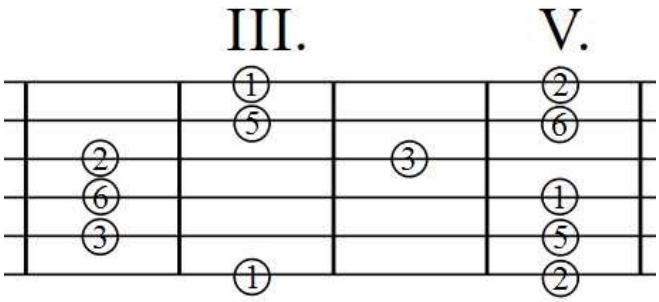
G major triad, E-position

7th Arpeggio



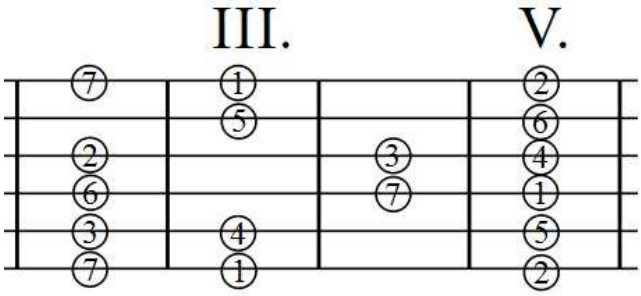
G major 7 arpeggio, E-position

Pentatonic



G major pentatonic, E-position

Diatonic Scale



G major scale, E-position

Now I want you do complete this part of the CAGED-workout on your own! Go through every position, just like we did before, but now in G major. Remember to do this at your own pace. Faster doesn't necessarily mean better. If you need to spend weeks in C major before feeling comfortable with it, so be it!

After you're done in G major, use the cycle of fifths to move on (D major, A major, E major, etc) until you've covered every position in every major key!

2.4.9. Changing the tonality

As soon as you're relatively comfortable with the major scale in several keys and positions, it's time to introduce another tonality to your practice regimen: the minor scale.

The idea is to go through the same practice process that we used for the major scale. That means we'll be playing triads, 7th arpeggios, pentatonics and diatonic scales in all positions and keys but adapted to fit the minor scale.

That means you have to adjust notes for each device to fit the new scale. Let's have a look at the functional/intervallic design of each device (only applies to standard seven-note scales; symmetrical scales or non-functional scales with more notes have their own rules).

1. Triad (built from the root of the key) = 1st, 3rd & 5th note of the scale.
2. 7th arpeggio (built from the root of the key) = 1st, 3rd, 5th & 7th note of the scale.
3. Pentatonic = 1st, 2nd, 3rd, 5th & 6th note of the scale in major keys; 1st, 3rd, 4th, 5th and 7th note in minor keys.
4. Full scale (all seven notes).

I don’t want to get too deep into music theory, as there is other literature specifically addressing that, but here's how that applies to the major and minor scale. The notes in brackets refer to the absolute notes in the key of C.

	Major	Minor
Triad	1-3-5 (C E G)	1-b3-5 (C Eb G)
7th Arpeggio	1-3-5-7 (C E G B)	1-b3-5-b7 (C Eb G Bb)
Pentatonic	1-2-3-5-6 (C D E G A)	1-b3-4-5-b7 (C Eb F G Bb)
Diatonic Scale	1-2-3-4-5-6-7 (C D E F G A B)	1-2-b3-4-5-b6-b7 (C D Eb F G Ab Bb)

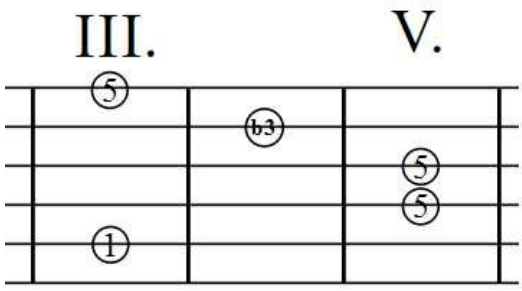
Now it's time to put this knowledge into practice and figure out the CAGED positions for the C minor scale.

2.4.10. C minor (A-position)

Once again, we're starting out in the lowest possible position of the key that doesn't have open strings. Please take note how I applied the same rules as described in 2.4.1. about constructing the fingerings (no stretches, little position shifts, etc). The thought process in general is exactly the same. If you feel comfortable enough, try to proceed without peeking at my fingerings. In case you're having trouble finding the right notes, please take a look at the interval maps in chapter 2.7!

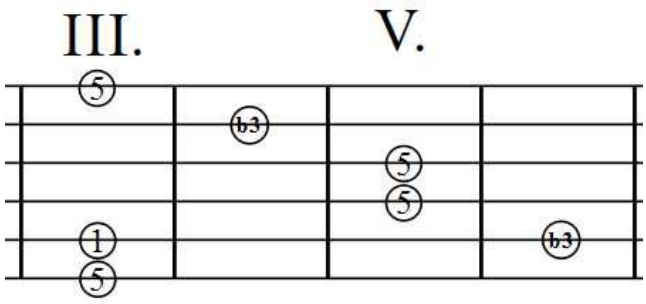
Reference Chord

Now that we're in a minor key, let's use the open minor chord shapes transposed to C minor as a point of orientation.



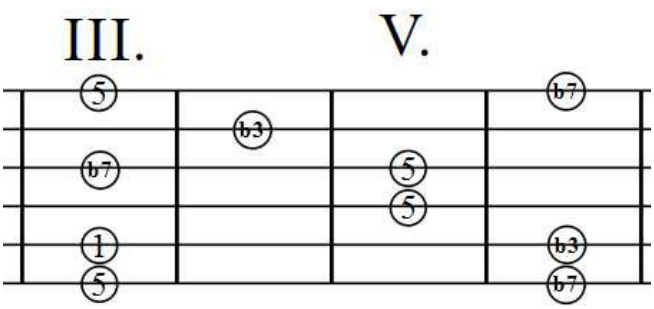
C minor chord, A-shape (A-position)

Triad



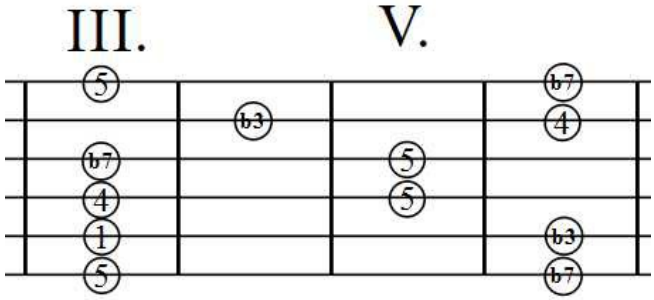
C minor triad, A-position

7th Arpeggio



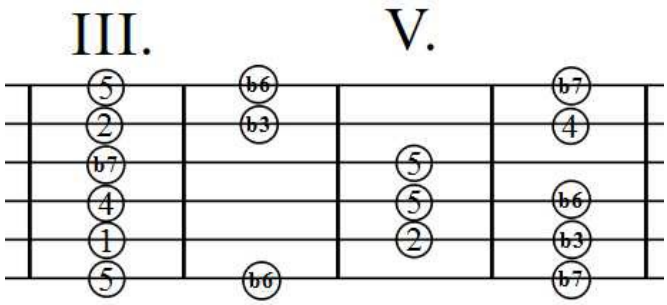
C minor 7 arpeggio, A-position

Pentatonic



C minor pentatonic, A-position

Diatonic Scale

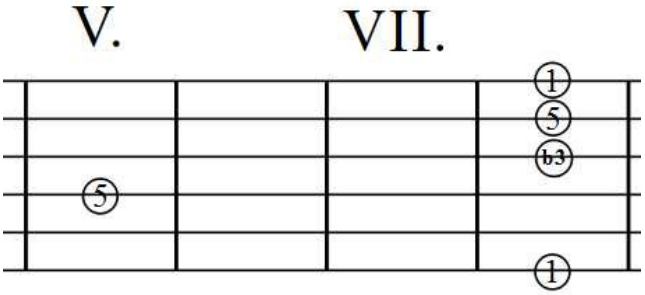


C minor scale, A-position

2.4.11. C minor (G-position)

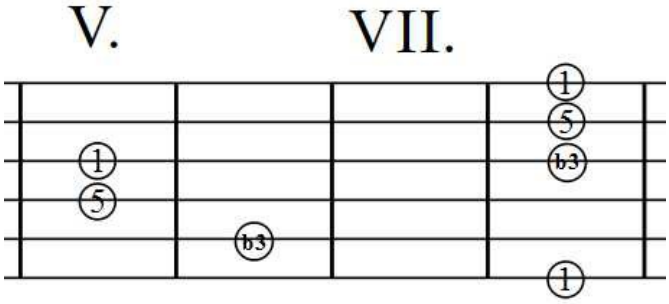
Reference Chord

There isn't really ONE popular open G minor voicing. This reference voicings works best for me, but it is not really set in stone if you find another one that works better for you.



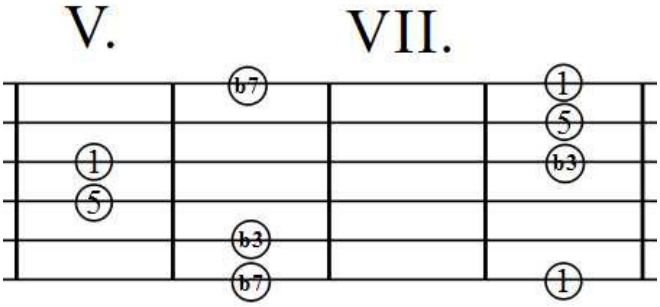
C minor chord, G-shape (G-position)

Triad



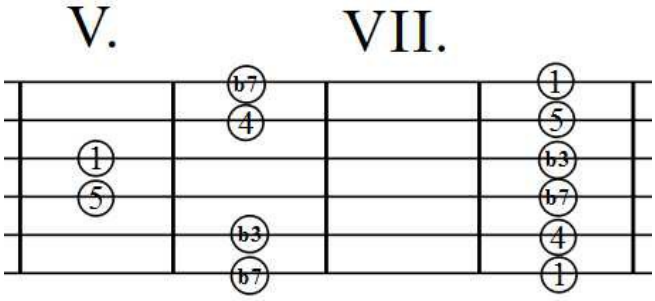
C minor triad, G-position

7th Arpeggio



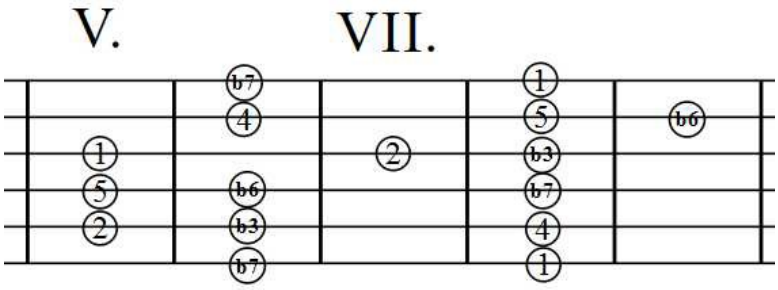
C minor 7 arpeggio, G-position

Pentatonic



C minor pentatonic, G-position

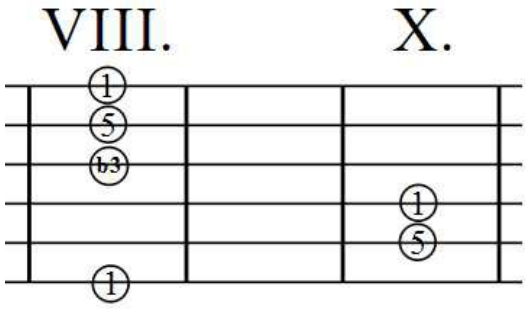
Diatonic Scale



C minor scale, G-position

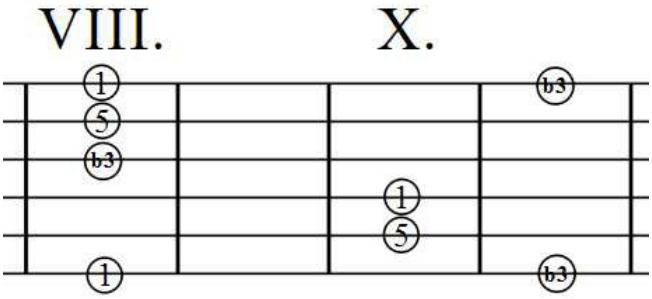
2.4.12. C minor (E-position)

Reference Chord



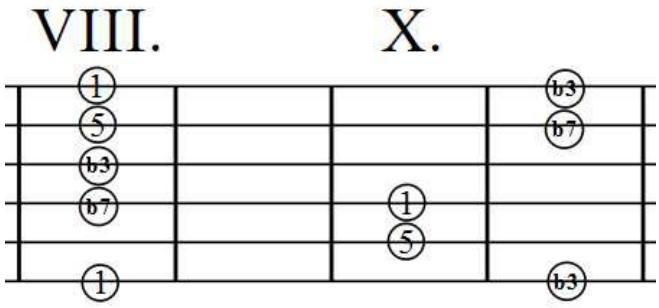
C minor chord, E-shape (E-position)

Triad



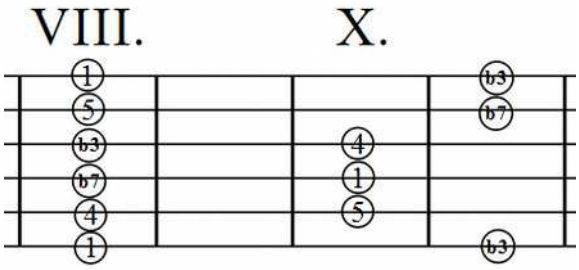
C minor triad, E-position

7th Arpeggio



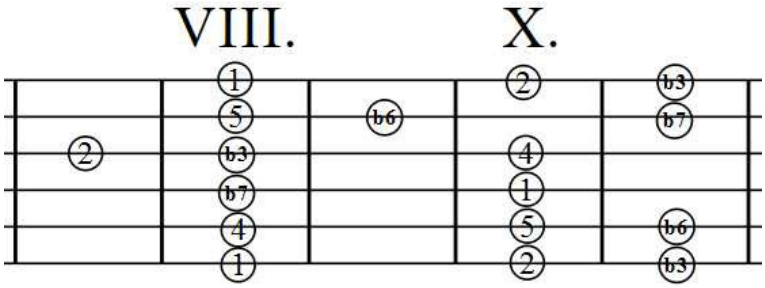
C minor 7 arpeggio, E-position

Pentatonic



C minor pentatonic, E-position

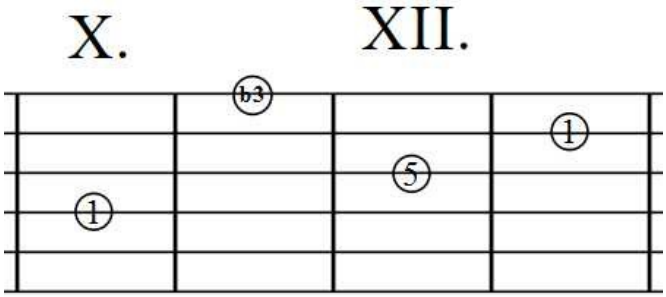
Diatonic Scale



C minor scale, E-position

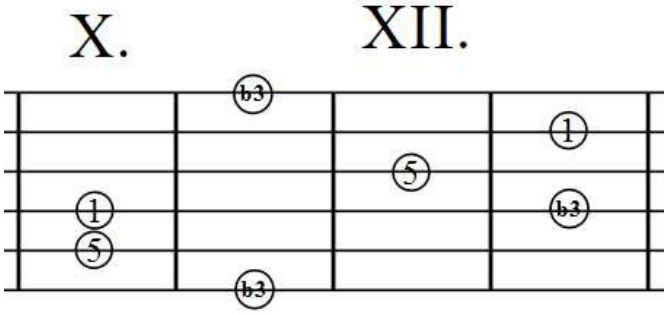
2.4.13. C minor (D-position)

Reference Chord



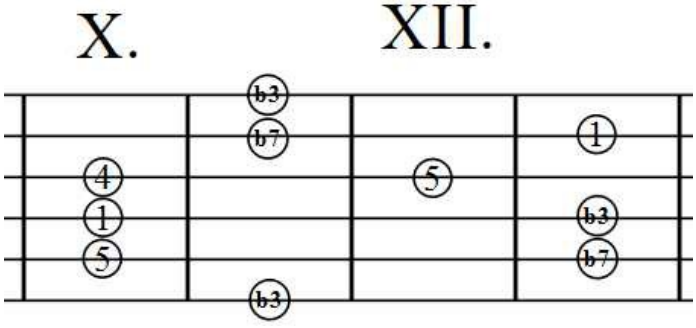
C minor chord, D-shape (D-position)

Triad



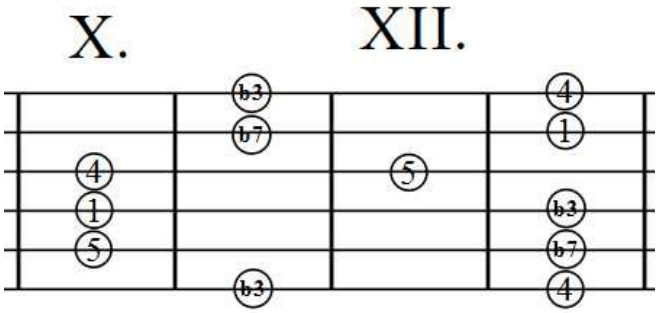
C minor triad, D-position

7th Arpeggio



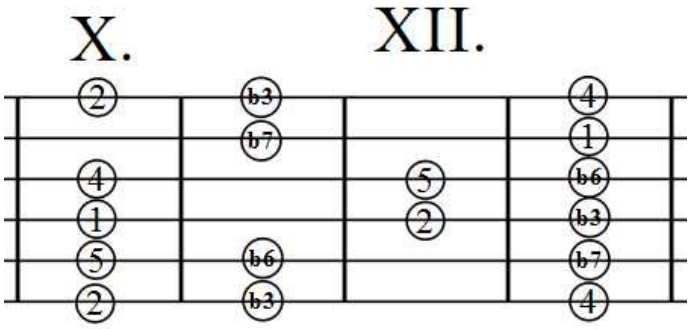
C minor 7 arpeggio, D-position

Pentatonic



C minor pentatonic, D-position

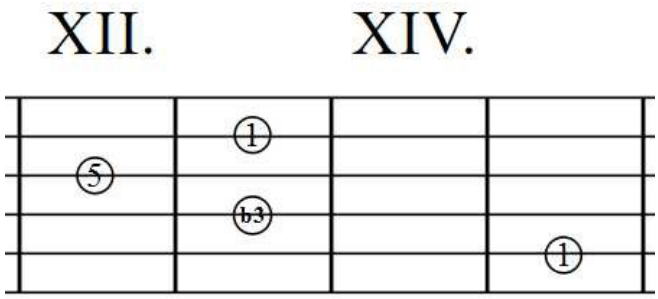
Diatonic Scale



C minor scale, D-position

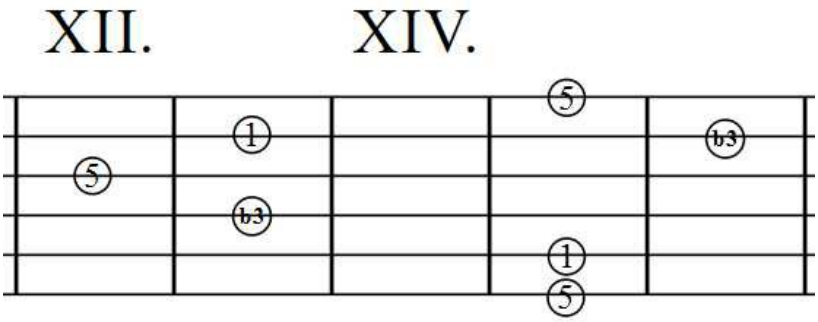
2.4.14 C minor (C-position)

Reference Chord



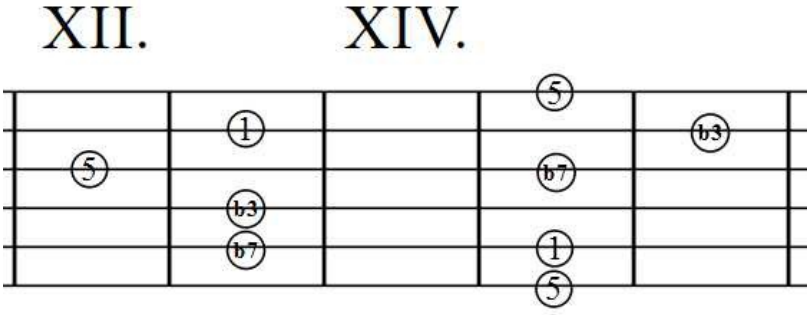
C minor chord, C-shape (C-position)

Triad



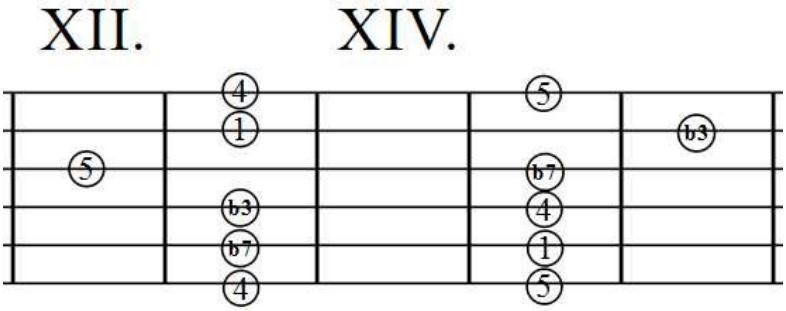
C minor triad, C-position

7th Arpeggio



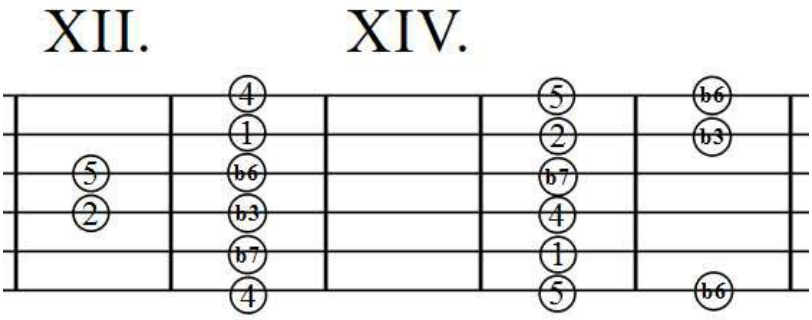
C minor 7 arpeggio, C-position

Pentatonic



C minor pentatonic, C-position

Diatonic Scale



C minor scale, C-position

2.4.15. What's next?

Next up I want you to practise the minor positions in lots of different keys, just like we did with the major scale. If you went through the material carefully and followed the rules I laid out, you should now be able to come up with CAGED fingerings for all the modes of the major scale, the melodic minor scale (plus modes), the harmonic minor scale and so on. The procedure for all these scales and tonalities will be just the same. My goal is to show you how the fundamentals of how CAGED works, so you can figure these things about by yourself, instead of bombarding you with even more neck diagrams.

Also be creative in your practice regimen! Switch around keys, tonalities, positions and exercises at will. For example, why not practise D-position minor 7 triads in all 12 keys, or stay in the G position in the key of A and run through all 7 major scale modes? The combinations are endless and you'll never run out of practice material.

Another thing to remember: the shapes you've learned are not necessarily going to be the only fingerings you'll use in a real music situation, although they will serve most purposes. A remark I've heard from students is that they see me play things that look very different from the shapes I'm teaching. The truth is, CAGED is a tool designed to learn and unlock the fretboard. I might not always use these exact shapes (I might do 3-notes-per-string) but my knowledge of CAGED has still informed the decision of where I place my fingers. In a situation where I might stretch my fingers out a lot or play rather unconventional position shifts, I may be combining two or more different CAGED fingerings in my mind.

2.5. CAGED Video Practice Session

I've recorded a number of videos and tabs for you, to use as play-alongs for your practice sessions. I'm going through all major scale modes in every CAGED position in 2 keys (C and G major) playing triads, 7th arpeggios, pentatonics and diatonic scales. You can also use these videos to confirm you're using the same fingerings as I do.

I didn't do videos and notation for all keys because (a) this would be a HUGE download, and (b) you don't need to run everything through 12 keys in the guitar. If you can do something in maybe 5 keys, you have covered pretty much every area of the fretboard and you can reach the other keys by moving fingerings up or down a fret. The two keys I chose (C and G) are far apart from each other, so they cover a wide range of the neck, which is good for getting you started. As you progress you should however practise these things in all kinds of keys.

Do not play along to these videos in time until you're familiar with the fingerings. The purpose here is to go over and maintain material that you've already learned. I recommend that you play in free time while you're still learning the notes. When you feel that you're ready to go, put the videos on and play along. Software like VLC player or Transcribe! can be used to slow down or speed up the videos without changing pitch. This way you can work at a tempo that suits your current level.

2.6. CAGED over changes

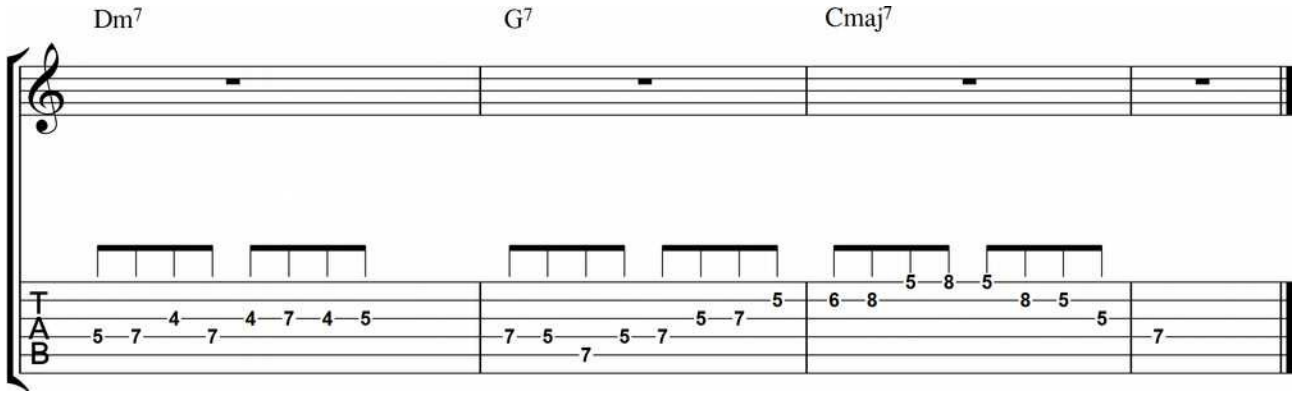
When playing over diatonic chord progressions, less experienced players have a tendency to think in key centres rather than the individual chords. For example, when playing over a major II-V-I in the key of C major (Dm7, G7, Cmaj7) they might think within the C major scale, C major pentatonic or even the A minor scale/pentatonic.

In the above example, all three chords belong to the same key centre (C major) and are derived from the same scale, so this would all be “theoretically correct”. However, when playing over changes, (especially functional harmony like a II-V-I) the key centre approach means that you often fail to outline the harmony; you may land on so-called “avoid notes” (like F over C major) instead of the strong “target notes”.

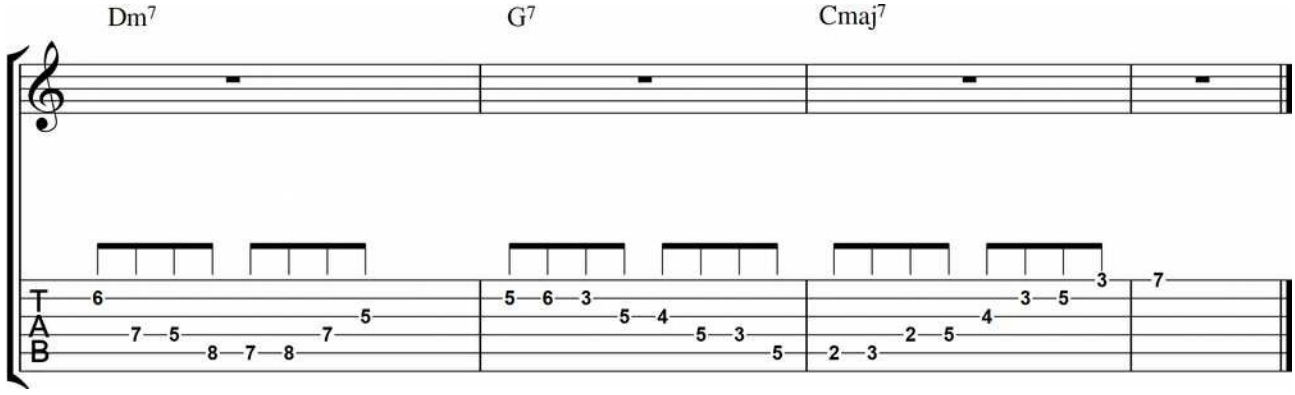
Outlining the harmony means that even if there was no chordal accompaniment, you would still be able to hear the chords just from your solo line. An experienced improviser can imply harmony just from a single note line. The key to that is to land on chord tones (meaning the root, 3rd, 5th and 7th of each chord) on strong rhythmic beats of the bar; for instance, you could resolve to the 3rd of a chord on beat one of the bar.

I've come up with two II-V-I lines to exemplify this. Both strictly use notes from the C major scale. The first one (“bad example”) shows the key centre approach; the second (“good example”) works around chord tones.

Video/Tab Files: 251_Bad & 251_Good

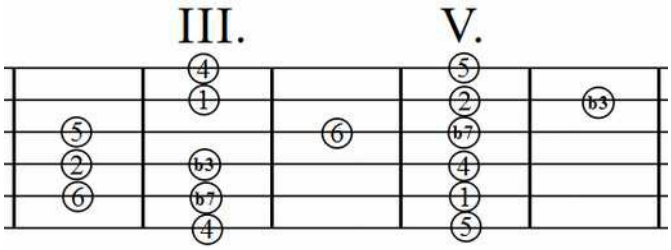


C major II-V-I line, bad example

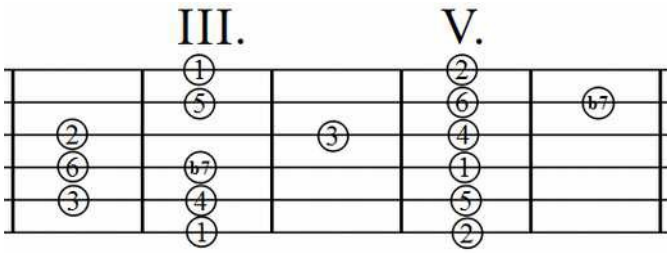


C major II-V-I line, good example

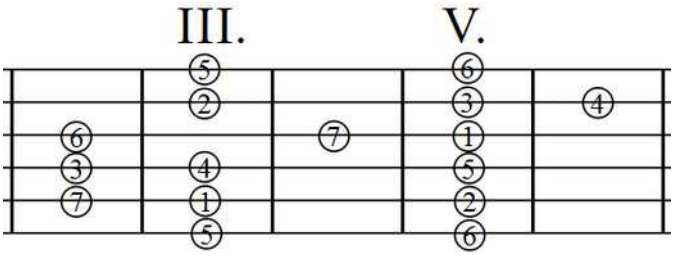
So if you play through chord changes using the CAGED system, you should be changing the positions you're thinking in along with the chord changes. When playing over a II-V-I in C major in the 3rd position (index finger at the 3rd fret) and stay in that position, you should be thinking:



D minor 7 (dorian), C-position

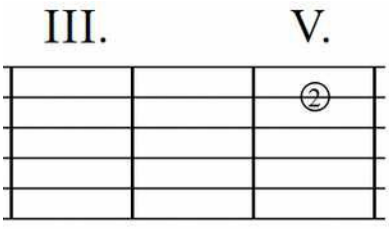


G dominant 7 (mixolydian), E-position

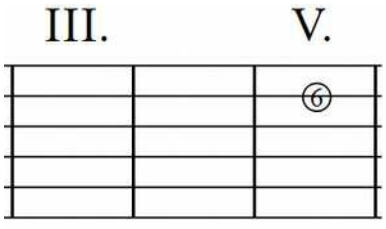


C major 7 (ionian) , A-position

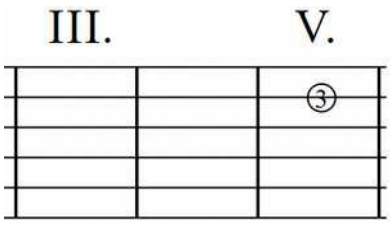
As you can see, the set of absolute notes is the same with each chord, but the relative function of each note changes. Improvising melodic lines is all about the intervals of melody against harmony. To make this even clearer, here's an example of how the same absolute note (E) becomes a different function as chords change. That is also a majority of what's going through my head as I improvise.



E against Dm7 (C-position)

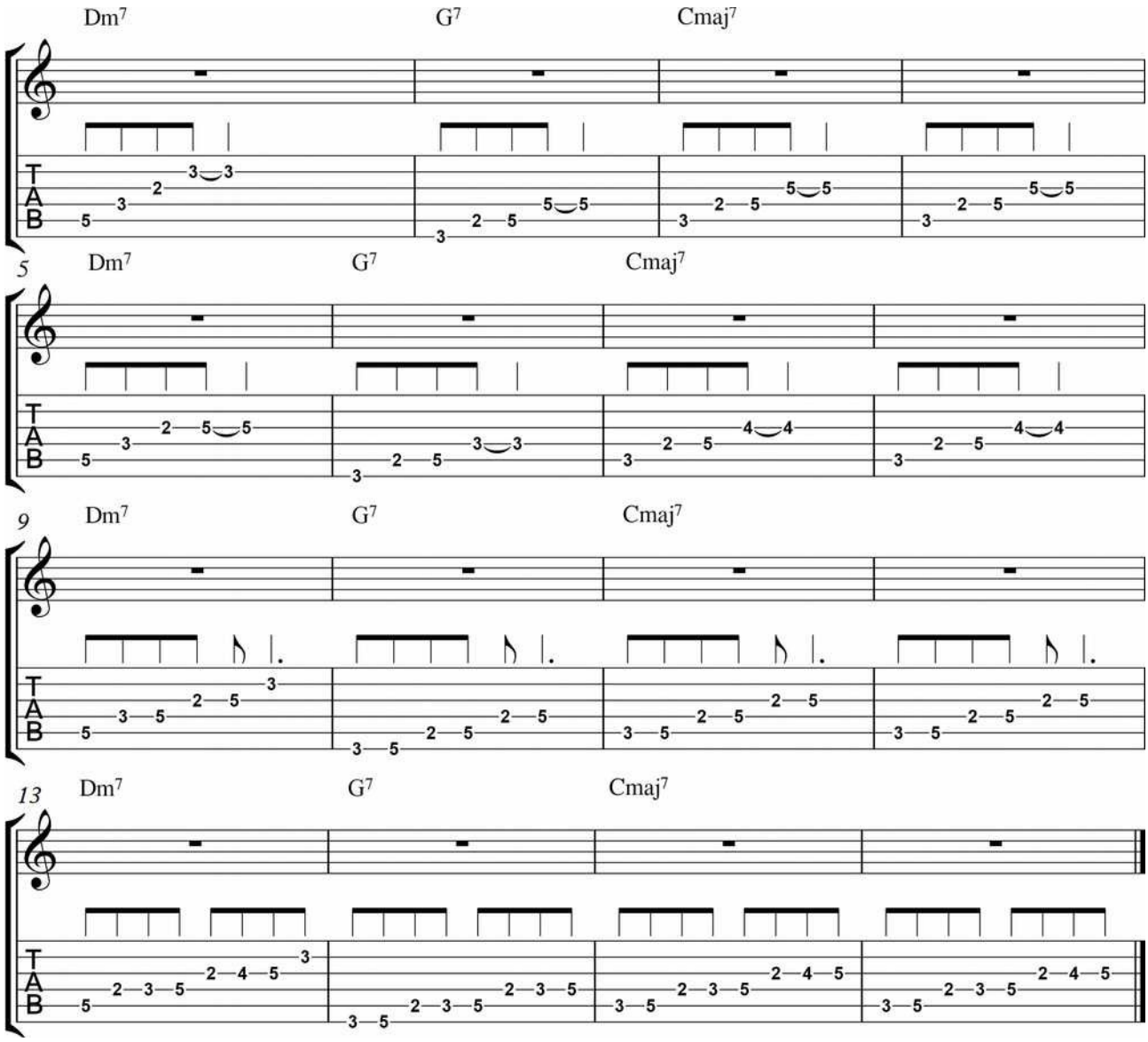


E against G7 (E-position)



E against Cmaj7 (A-position)

You're basically re-thinking the function of each note as the chords go by. Now let's see how I would apply the CAGED-workout procedure (triad, 7th arpeggio, pentatonic, diatonic scale) over a II-V-I.



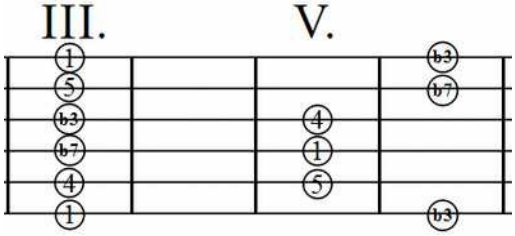
Feel free to practise going through the CAGED shapes in a II-V-I in C major (or any other key you like) through all positions like I did just now. When doing these exercises, remember that when a chord change happens, you stay in the same position as far as fret numbers go, but the relative CAGED position switches accordingly.

Video/tab File: CAGED_over_changes

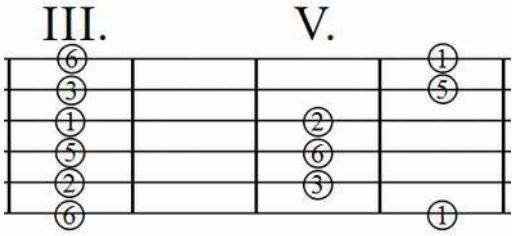
As you can see I'm a proponent of thinking about each chord individually. When playing modally, people tend to relate their thinking back to the parent major or minor scale. Let's say you were to play over an E9 vamp. Some people may understand that the appropriate scale for this chord would be E Mixolydian, which of course is correct. But they will then think of E Mixolydian as the 5th mode of the A major scale and therefore their entire thought process will be in A major, which will not make them sound as if they were playing over an E9 chords. I advocate trying to think of each chord/harmony/mode as a separate entity and treating it for what it actually is. Relating back to the parent scale for me is only an emergency tool, if I'm too unfamiliar with a certain mode or harmony in an impromptu situation, such as when improvising over a tune while reading it for the first time. When studying I'm try to make sure I don't have to rely on it.

A similar idea to the above is so called “minorising”. For the same reasons as above, I don't recommend getting into this. Here, you relate anything you play back to a minor chord or minor scale. Say you were to play over a Bbmaj7 chord and you wanted to use a pentatonic scale to do so, in minorizing you would think of the relative minor key pentatonic (G minor) instead of the Bb major pentatonic. Even though the notes are the same, you’re always thinking in the wrong key, so all the note functions are different. An experienced listener will immediately be able to tell whether you're thinking G minor or Bb major. This can be done with all kinds of tonalities, but yet again I don't recommend to use this as your go-to system.

If this is a habit of yours, here's an idea how you can “unlearn” it. For this purpose let's stick to the example of the Bb major/G minor pentatonic. As you know by now they're the same physical shape on the guitar. Play through the entire shape twice. First time around think “G minor pentatonic” and spell out the function of each note either in your head or verbally. Then switch over to Bb major pentatonic, play the same set of notes and spell out their relation against the new harmony (“root, 3rd, 4th” etc). Do this slowly and carefully and you'll soon develop a sense for where the notes are, that you're looking for.

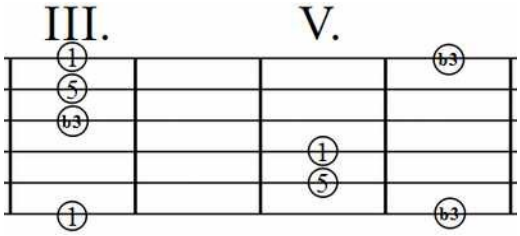


G minor pentatonic, E-shape

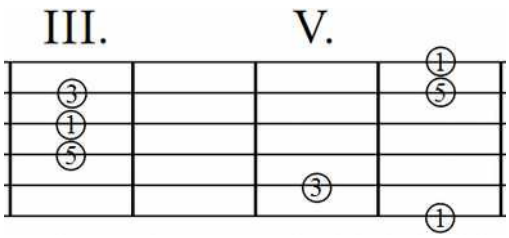


Bb major pentatonic, G-shape

Then in order to familiarise yourself with the location of the strong notes within the pentatonic scale (i.e. the chord tones) practise the pentatonics followed by the corresponding triads and vice versa.
Understand how the minor triad is contained within the minor pentatonic, then do the same in major.



G minor triad, E-shape



Bb major triad, G-shape

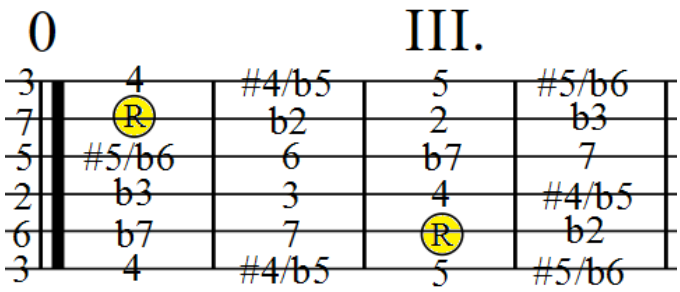
For now we're not going to get any deeper into this topic. This was supposed to give you an overview of how CAGED works marvellously for playing over chord changes.

2.7. Interval mind maps

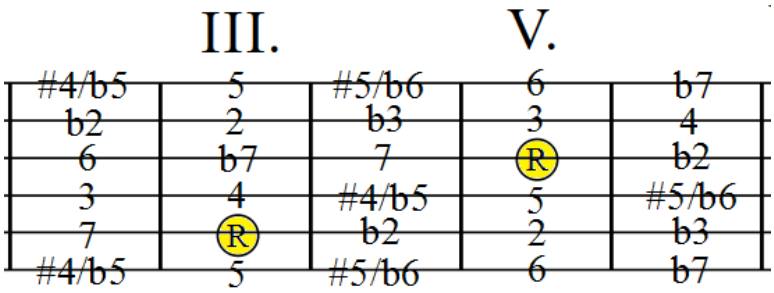
The following neck diagrams can be used as a reference to find out the location of every interval on the fretboard in relation to the root note C. I described the thought process behind this in 2.3.1. In order to stay true to the CAGED system, I split the diagrams up into five positions. It is recommended to thoroughly study the geometrical shape of every interval, so you'll be able to recognise them immediately as you play.

You can turn this into an exercise by randomly picking an interval; let's say the minor third. Now go through each position in the key of your choice and find all the minor thirds available in that area of the neck. Due to the geometry of the guitar, these shapes will be the same regardless of key. Keep in mind however, that due to the standard tuning of the guitar, a minor third played on the A- to D- string is going to be a different shape (or “look” different) than if you played the same interval on the G- and B-strings.

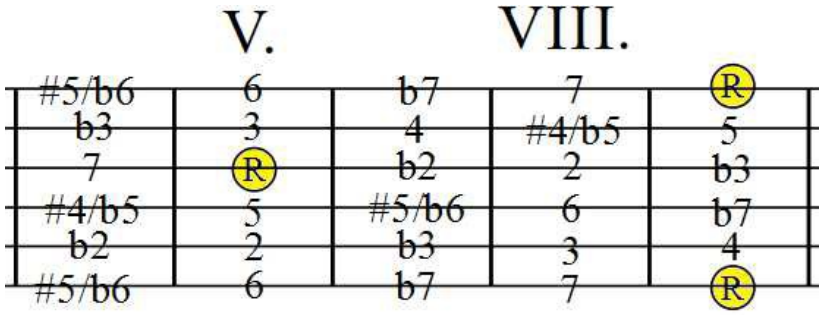
Another perk of having a good understanding of these mind maps is that you can find a fingering for any scale or arpeggio pretty much in an instant, even if you've never played it before. The only thing you'd have to know about is it's intervallic structure.



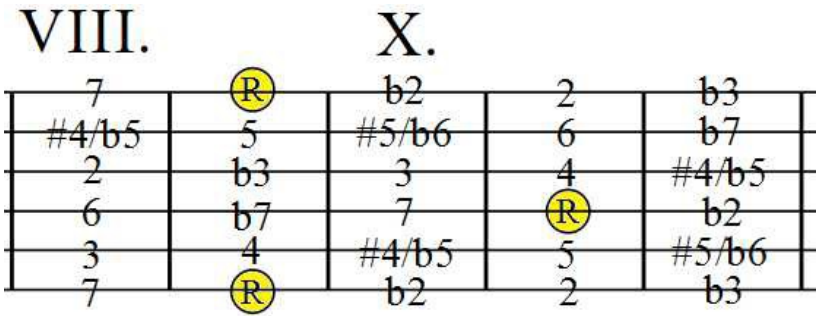
Open C-position interval mind map (key of C)



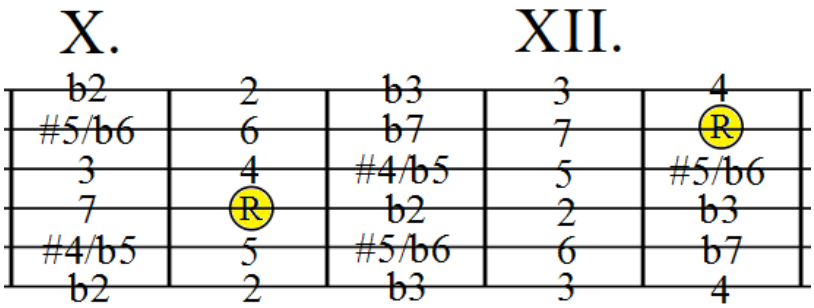
A-position interval mind map (key of C)



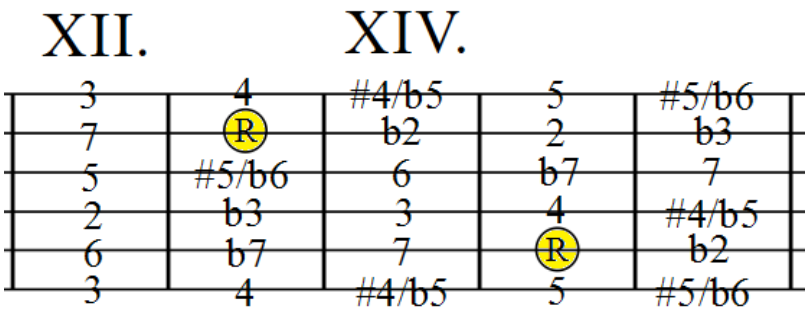
G-position interval mind map (key of C)



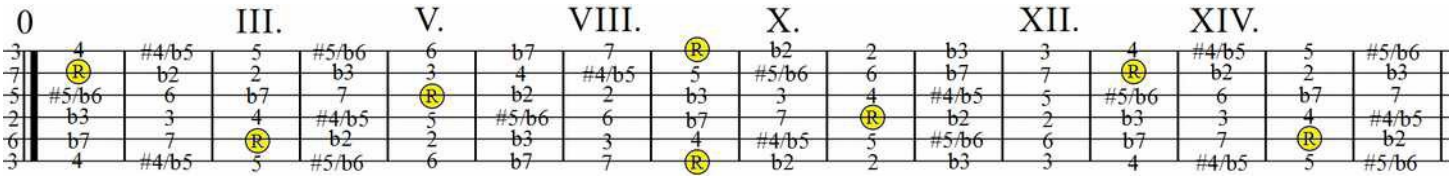
E-position interval mind map (key of C)



D-position interval mind map (key of C)



C-position interval mind map (key of C)



Full interval mindmap (key of C)

2.8. Other Fretboard Exercises

Until now, we've mostly visualised the fretboard in a very horizontal fashion. Here are a couple of ideas I want to share with you, to break away from that and further expand your freedom on the neck.

2.8.1. Single String

Playing scales, triads or any other musical device on a single string is a great way to familiarise yourself with its intervallic structure. As on a piano, you can more easily measure the distances of one note to another visually if they're laid out on one string.

Try this:

1. Pick a key, tonality and pattern of your choice. For this example let's use the G Dorian scale.
2. Play the scale on the low E-string, starting from the lowest diatonic note available (in this case the open E).
3. Go up till you've hit the root note an octave up, then descend back. Alternatively you can go up till you run out of frets, but it's not absolutely necessary since the guitar repeats beyond the 12th fret anyway.
4. Do the same thing on each string.

A thorough knowledge of CAGED is going to help you, because you can visualise the notes you're playing against different root notes as you ascend and descend.

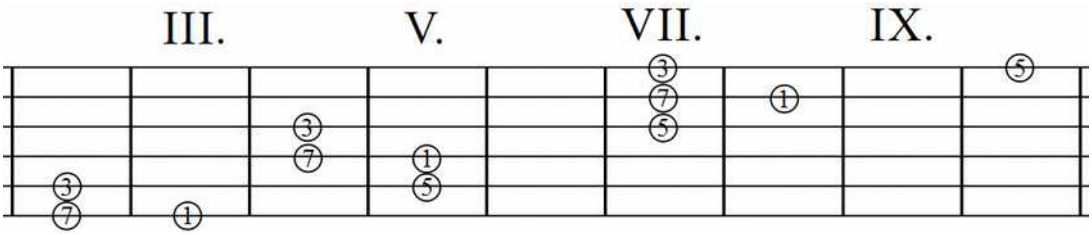
I've also found it very helpful to practise melodies on a single string that I'm already familiar with (folk or children's songs, standards, licks, etc).

Video/tab File: Single_String_Exercise

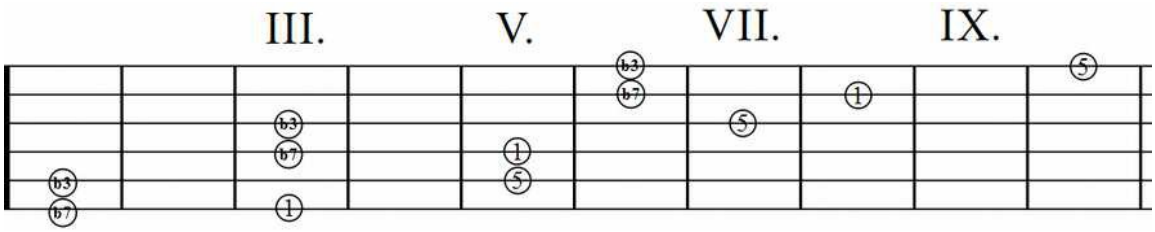
2.8.2. Diagonal movement

Due to the tuning and the layout of the guitar, it so happens, that any set of notes on the low E- and A-string repeats two frets up on the D- and G-string (an octave up) and five frets up on the B- and high E-string (2 octaves up). It's a great method to move across the neck diagonally, finding similar ideas in different positions and ascending or descending two octaves with very little effort.

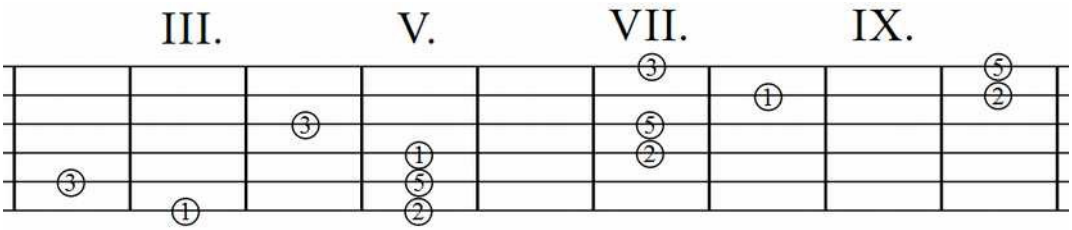
Let me give you a few examples of “cells” that can be moved up and down diagonally. I've written chords next to each of them, over which you can apply these.



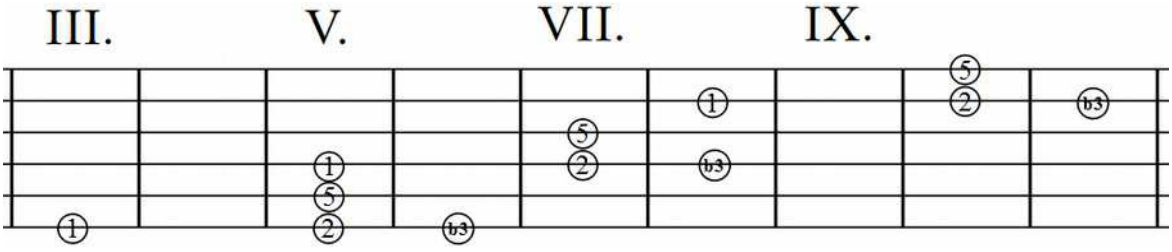
G major 7, horizontal



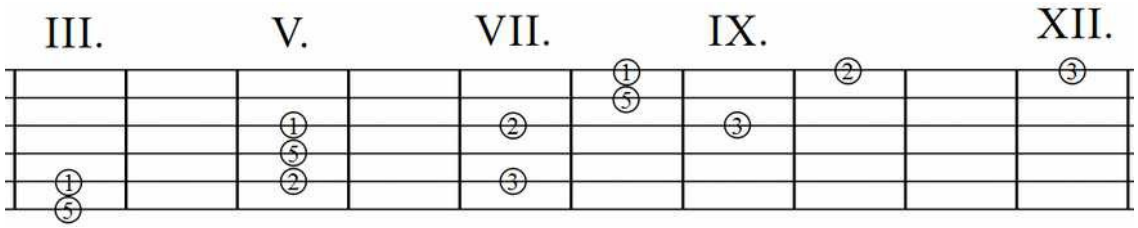
G minor 7, horizontal



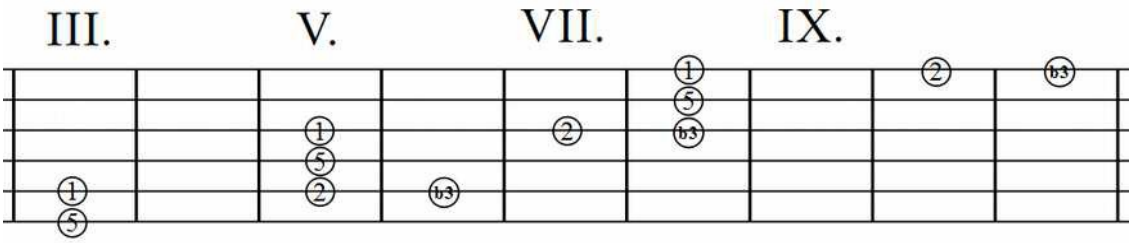
G major add9, horizontal



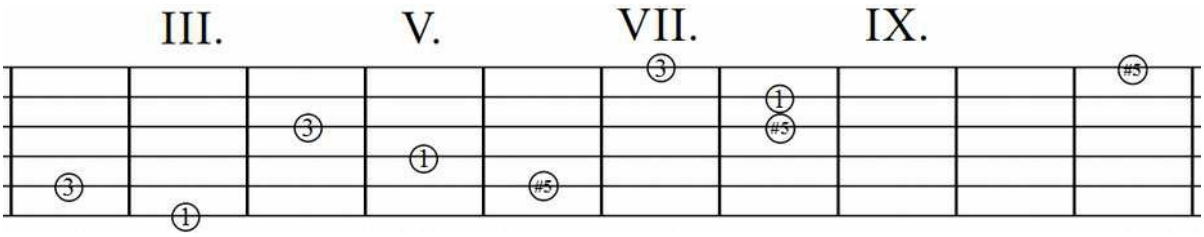
G minor add9, horizontal



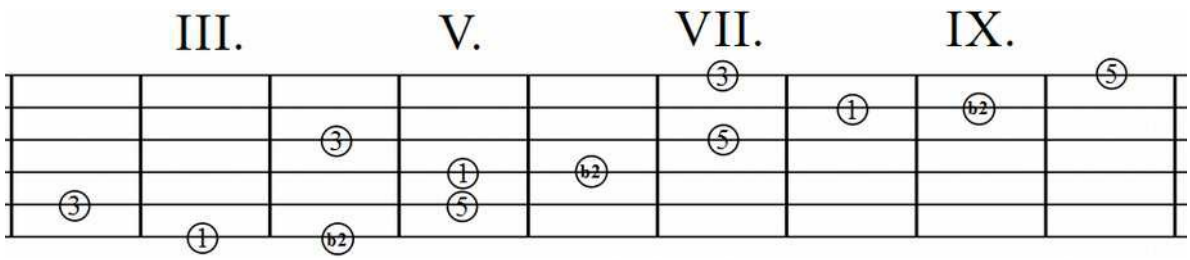
C major add9, horizontal



C minor add9, horizontal



G augmented triad, horizontal (works over G7#5, Galt, etc.)



G major triad with added b9, horizontal (works over G7b9, Galt, etc.)

Here are a few examples of how to incorporate this concept within a line:

Video/tab Files:
diagonal_lick1
diagonal_lick2
diagonal_tritone_lick
C7_Fm7_diagonal

Diagonal Lick 1
The first diagonal lick is based on a minor 7th arpeggio played over a Cm7 chord. There is an F added to two of the “cells”, turning it into an entire pentatonic played on two strings over multiple octaves.

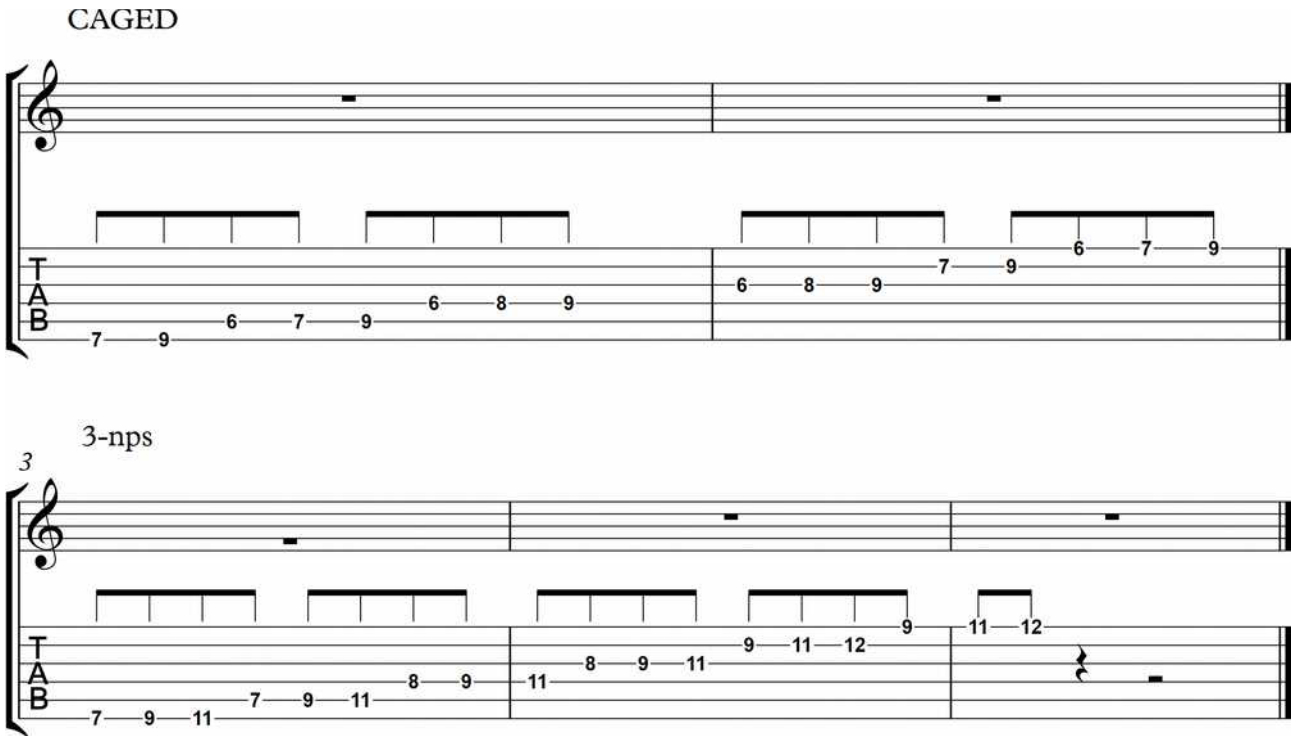
Diagonal Lick 2
... superimposes an Ebmaj7 arpeggio over a Cm7 chord, creating a Cm9 sound, due to the Ebmaj7 arpeggio containing a D. Also note how each of the cells is sequenced a little differently as opposed to just played straight up or down.

Diagonal Tritone Lick
The chord progression underneath this line is a C7 altered dominant chord resolving to the tonic Fm7. I'm moving two triads a tritone apart (C major and F# major) across 3 octaves to outline the sound of a C7b9#11 chord. This concept is derived from the half-whole diminished scale.

C7 - Fm7 Diagonal Lick
Just like in the previous example, we have a C7 chord resolving to Fm7. This time my diagonal cell consists of the notes C-Db-E-Ab, implying a C7b9#5 sound over the C7 chord.

2.8.3. The 3-note-per-string System

The 3nps system is very popular among guitar players, especially in the shred/metal-community. In my opinion, it's not very efficient as a sole method of fretboard visualisation. For one thing, you have to learn seven shapes for a 7-note scale (starting from every note on the low E-string) as opposed to five with the CAGED system. Secondly, every shape involves stretches and position shifts. As an example, play the B-major scale starting from B on the low E-string using the CAGED system first, followed by the 3nps system.



B major scale CAGED & 3nps

The CAGED version covered a 4-fret range and involved no stretches, whereas the 3nps version was full of stretches and encompassed a 6-fret range. This makes it way harder to be aware of what interval you're playing in relation to the root note, because you're moving so far away from it geometrically. It makes it harder to build an accurate and reliable interval mind map. And despite covering all that range you still need to learn 7 of these shapes. On top of that, many people find it hard to recognise triads and 7th arpeggios within those shapes, because they don't relate to a familiar chord or even a pentatonic shape. Generally, 3nps is not very efficient as a tool for visualising the neck, especially for improvisation- and melody-based playing.

With all that said, there are benefits to the 3nps system as well. The consistency of the number of notes per string makes it suitable for fast picking runs. Also, sequences will pan out very similarly from one shape to the next. Legato playing is easier because you have a higher average of notes per string. I'm all for learning as many systems and options as you can. If you already have a solid understanding of CAGED, you can further expand your fretboard knowledge by giving this system a try. Your overall proficiency with the fretboard is only going to get better from it.

2.8.4. Note Recognition

As an improviser, it is beneficial to be able to think of music in relative terms, meaning to think about the relation from one note to the next or from a note to the harmony. The upside is, you can apply a principle you learned in one key to any other key. However, it is still necessary to quickly recognise actual note names in numerous occasions, whether it's sight reading, finding the right key, communicating with other musicians or students, etc.

Most less experienced players tend to use an octave shape system to find the names of the notes they're playing on the higher strings. This means they'll only know the names of the notes on the two E-strings and the A-string. For example, if the note on the 10th fret on the B-string has to be named, they'll play that note and apply the shape of an octave, which leads them to the same note an octave lower on the 7th fret on the D-string. Then they'll do the same thing over again, by applying an octave shape yet again to find the same note (an octave lower) on the 5th fret of the low E-string. Only then will they know that they played an A.

This system does work and can serve lesser experienced players well, but it will be way too slow in many situations. You have to take three steps in the example above (going down an octave, going down another octave, recognising the note) just to name a note on the B-string. To me that is highly inefficient. I advocate training yourself to recognise any note on any string immediately without having to take any extra steps.

Here's a few tips on how to do that:

1. Practise C major scale on every string (as described in 2.8.1.) slowly, calling out the name of each note loudly. As soon as you're familiar with the location of the notes of the major scale you can find the accidentals by going up or down a half-step. As you gain experience you will also start to recognise accidentals immediately.

2. Find orientation points. One thing that helped me greatly was to remember the location of all E's on the guitar. Along with the names of the open strings, this is a good starting place and is going to save you plenty of time "counting" up and down notes.

3. Pick any random note (like E, for example) and find that same note on every string, starting on the low E string and going up string by string without skipping any. This ensures that you're not applying the octave shape system but think about the location of the notes on each string individually. When you've worked your way up to the high E-string, work your way back down again.

3. Triads

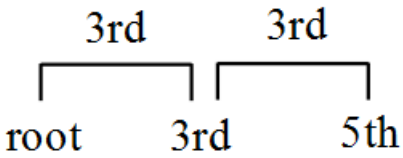
3.1. Introduction and Theory

In this chapter we're going to take a more detailed look at one of the most common and essential devices in music: the triad. We're going to examine what they are, how they work, how we find them on the guitar and apply them to soloing and chordal playing.

Again, this is no theory book, so I'm going to keep the theory portion fairly brief. I advise you to seek out resources such as the Mark Levine Jazz Theory Book or Jazzology by Hal Leonard publications. Or, even better, an accomplished theory teacher.

Most Western music is based on tertian harmony. That is, chords are constructed by stacking diatonic 3rds (and a diatonic 3rd means a 3rd interval that naturally occurs in a scale). It's often said that a true chord should consist of at least three notes. While you can stack a huge number of notes to form large, complex chords such as 13ths, the simplest chords are formed from triads, which contain three notes (as the name "triad" suggests). Triads can also be used in a linear fashion, to suggest harmony within a single note melody.

There are different ways to look at how a triad is constructed. When building a triad in its root position (root as lowest note) you can think of it as root note, 3rd (minor or major) above the root note and 5th (diminished/perfect/augmented) above the root note. Alternatively you can see it as a root note, a 3rd stacked on top of another 3rd stacked on top of the previous 3rd. Like this...



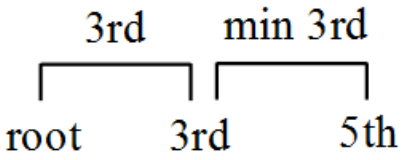
triad construction

A third + another third = a fifth! There also is a third way to form triads. You can do it by alternating notes within a diatonic scale. Here's how that works...

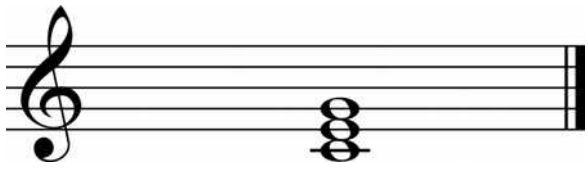
Pick a note from a diatonic scale as the root note of your triad. Then find the other two notes by skipping one note within a scale. So in a 7-note scale, you can build a triad from the 1st, 3rd and 5th note of a scale. The next triad within that scale would be build from the 2nd, 4th and 6th note, another one on the 3rd, 5th and 7th note and so on.

Depending on which combination of major and minor thirds you stack, you're going to get different triad types. Also, the combination of thirds also affects the type of 5th you create (remember, 3rd + 3rd = 5th). For now we will be dealing with close position triads. This means, as the name suggests, that the different notes are arranged in the closest, most compact order possible. The notated examples are in the key of C.

Major Triad

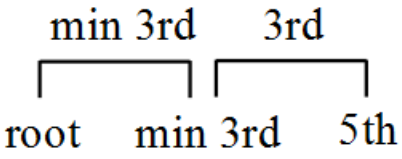


major chord construction

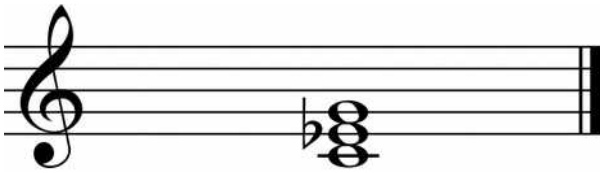


C major triad

Minor Triad

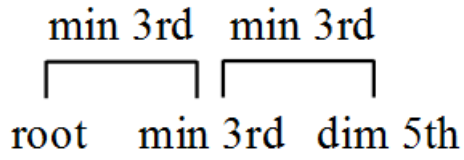


minor chord construction

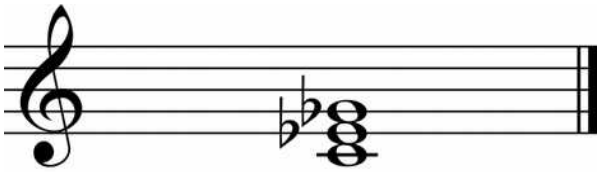


C minor triad

Diminished Triad

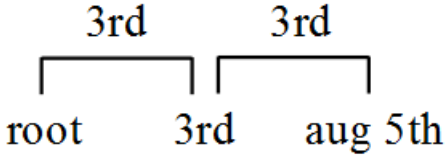


diminished chord construction

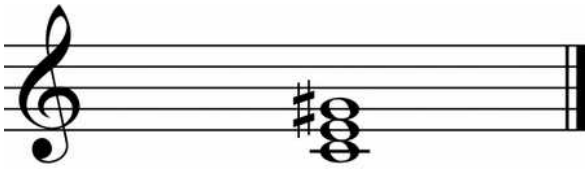


C diminished triad

Augmented Triad



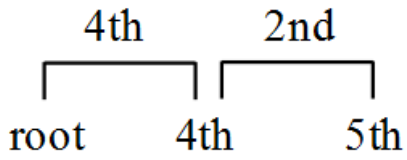
augmented chord construction



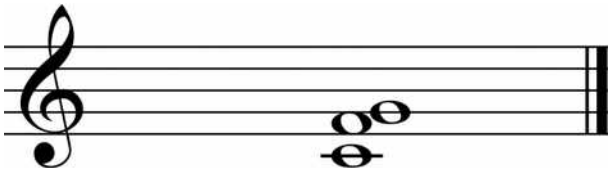
C augmented triad

Suspended 4th

In a suspended triad, the 3rd is replaced by either a 2nd or a 4th. This is a deviation from triadic harmony. In a classical sense the 4th in a sus4 triad wants to resolve to the 3rd. In modern jazz harmony both the sus2 and sus4 triads can be used in a non-resolving or modal fashion.

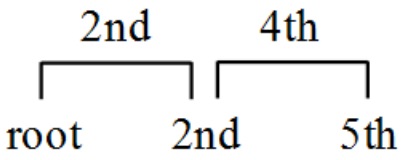


sus4 triad construction

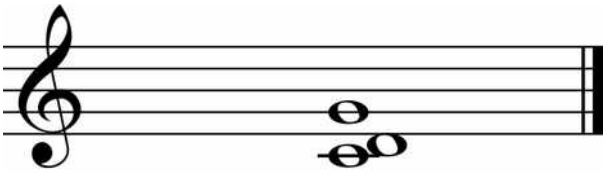


C sus4 triad

Suspended 2nd



sus2 triad construction



C sus2 triad

3.2. Inversions

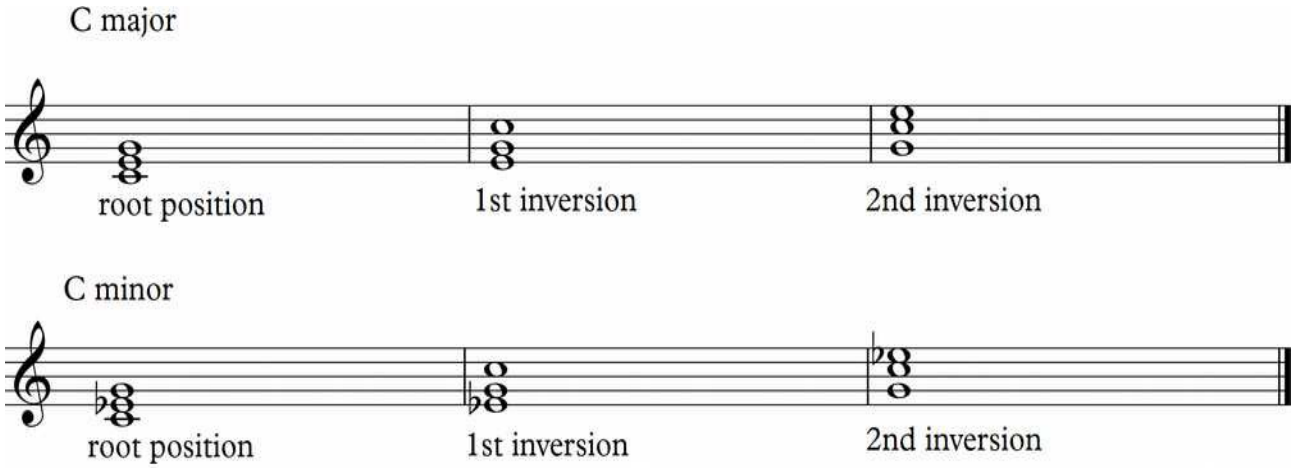
So far we've only been looking at triads in their root position, meaning that the root note is the bottom (bass) note. A triad can be inverted by putting any of the other two notes as the lowest note. This gives us a root position plus two inversions.

Root position – bottom note is the root note
1st Inversion – bottom note is the 3rd
2nd Inversion – bottom note is the 5th

In order to build the 1st inversion, use the 3rd as your bass note and find the closest possible 5th and root above it. You can also take the root position and move the root note up an octave (for a C major triad, CEG becomes EGC).

In order for you to build the 2nd inversion, use the 5th as your bass note and find the closest possible root and 5th above it. You can also take 1st inversion and transpose the 3rd up an octave (again in C major, EGC becomes GCE).

Things become more obvious when laid out on a keyboard or staff:



C major and minor triad inversions

Inversions can be applied to any of the aforementioned triad types.

All of these examples, unless specified, are in "closed position". This means that the notes are organised in the narrowest, closest possible order. Therefore, when ascending the root is always followed by a 3rd, the 3rd is always followed by a 5th and the 5th is always followed by a root. Later on we will have a look at options of covering a wider range with triads.

3.3. Triads in Single Notes

If you worked through the fretboard visualisation portion of this masterclass, you'll probably have played a lot of triads by now, as part of the CAGED workout in conjunction with pentatonics, 7th arpeggios, etc.

Now I want to take an even closer look at visualising triads specifically. The difference from the CAGED workout is that we're now going to look at triads individually, covering just one octave instead of building an arpeggio across all strings in a position.

The idea is to play triads as single notes starting on every string except the high E string, starting on either the 1st, 2nd/3rd or 3rd/4th finger. This will give us every possible (reasonable) fingering for these triads. Stretches will be allowed in this exercise. To get the maximum benefit from this, try to visualise each triad within its corresponding CAGED position.

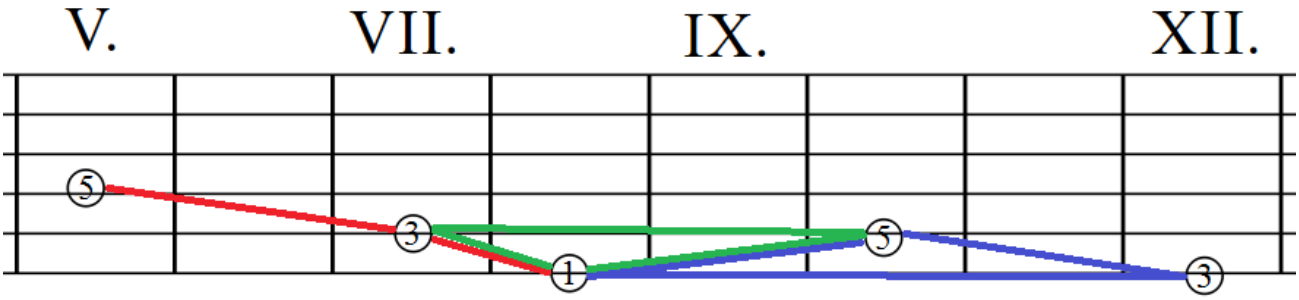
The colours in these diagrams indicate the different fingerings.

Blue: 1st finger on the root note – going visually towards the guitar bridge
Green: 2nd or 3rd finger on the root note – going visually towards the ground Red: 3rd or 4th finger on the root note – going visually towards the headstock

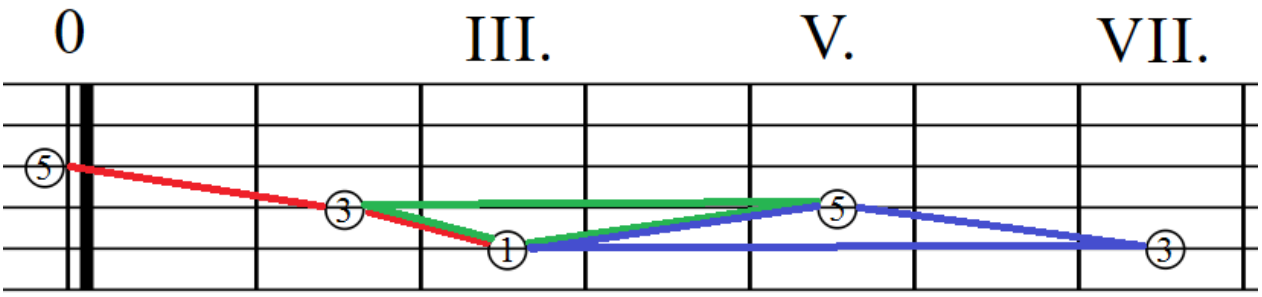
These lines also help you memorise those shapes geometrically. Notice how each diagram has all three colours, showing the three interlinked triad shapes, except the final diagrams covering the B/E strings.

3.3.1. Major Triads

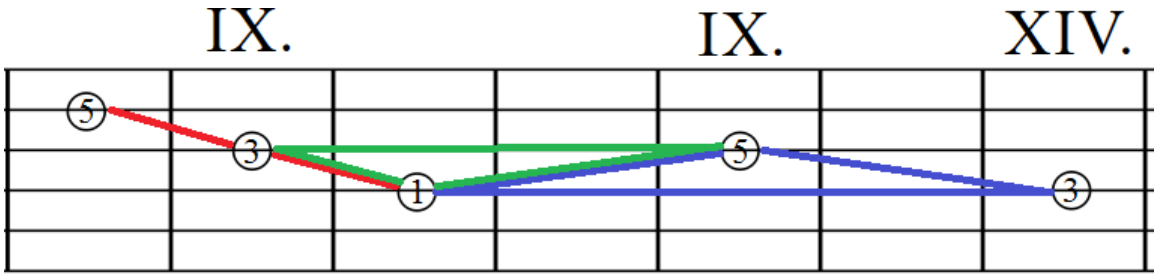
C major triad (root position)



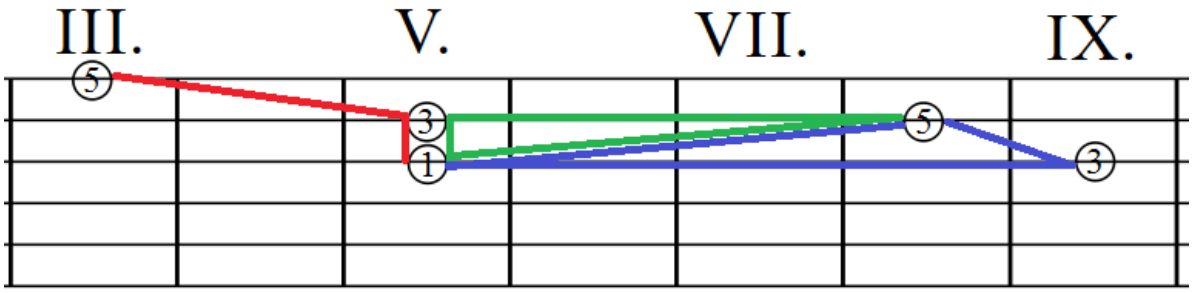
C major triad (root position) fingerings, low E-string



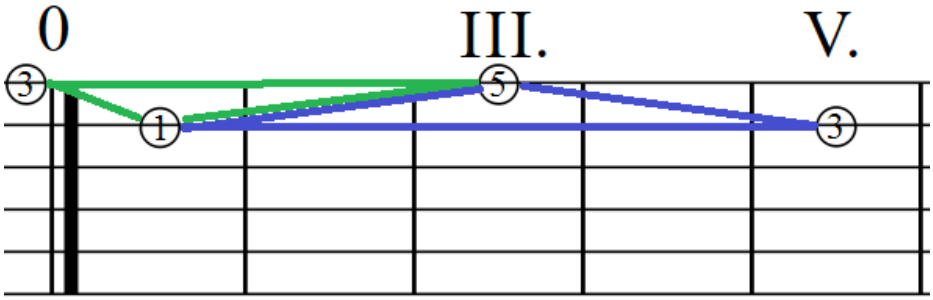
C major triad (root position) fingerings, A-string



C major triad (root position) fingerings, D-string



C major triad (root position) fingerings, G-string



C major triad (root position) fingerings, B-string

I recommend playing through every shape on every string, going from high to low, to make sure you're not skipping any triad fingering.

There is absolutely nothing wrong with doing the same thing for the two inversions. Tackling a problem from as many different angles as possible can be nothing but helpful. However I decided not to do similar diagrams for these because...

A) ... it would make this masterclass even bigger than it already is.
B) ... with the material covered so far, you should be able to find these fingerings yourself and most importantly...
C) ... if you have practised the root positions of this chapter and the larger fingerings of the previous chapter carefully, visualising the inversions should be relatively easy. Just combine the multiple fingerings you know in your mind and start playing the triad from the 3rd or the 5th, depending on what inversion you're going for.

When visualising inversions, my focal point tends to be the root note, viewing how the other notes relate to it. Sometimes I find it easier to seek out the fifth or triad. Experiment to find what works best for yourself.

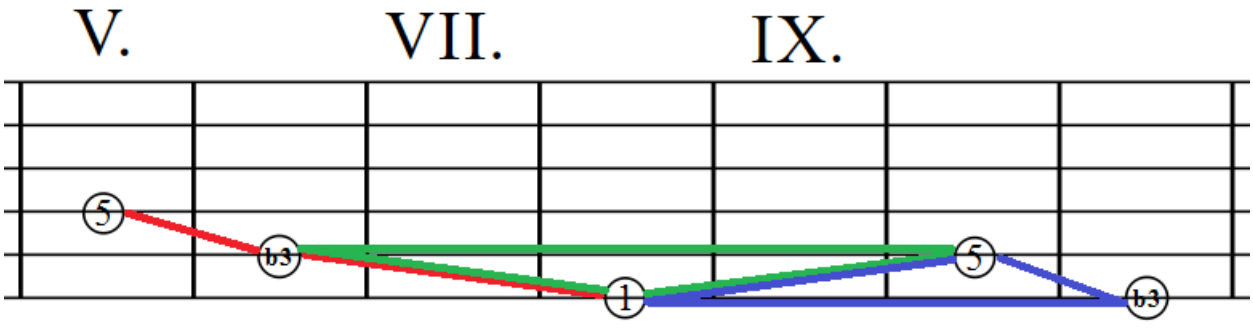
Video/tab File: Cmaj_triad_single.mp4

3.3.2. Minor Triads

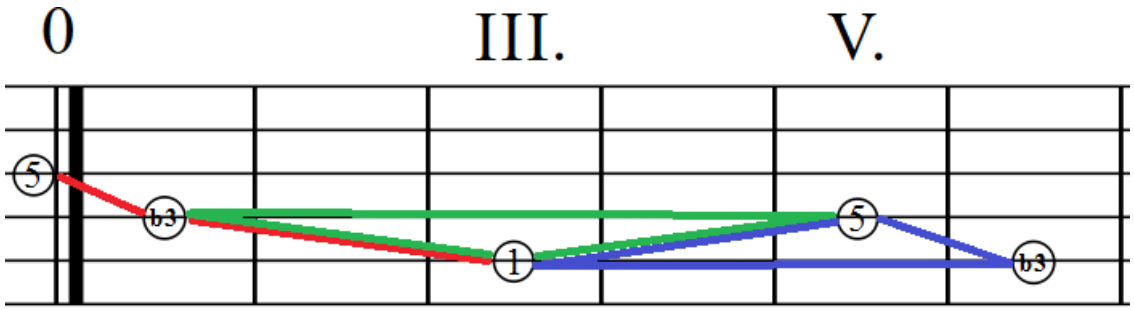
You should practise the minor triad shapes the same way as I described above.

C minor triad (root position)

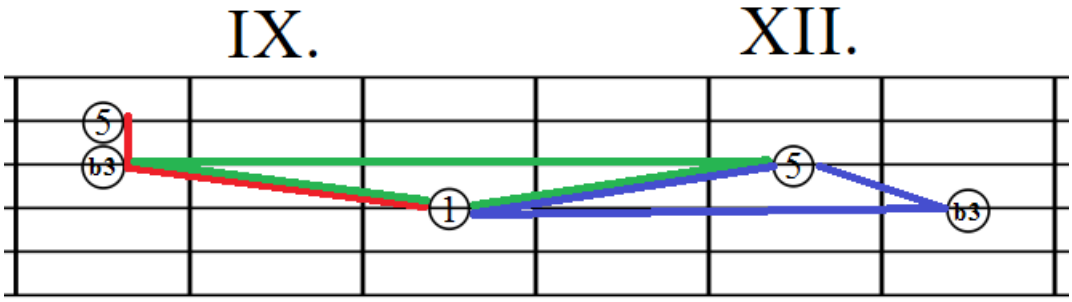
Video/tab File: Cm_triad_single



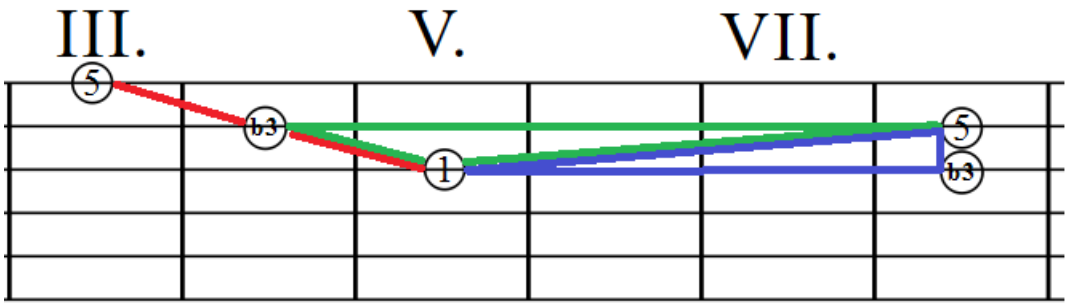
C minor triad (root position) fingerings, low E-string



C minor triad (root position) fingerings, A-string



C minor triad (root position) fingerings, D-string



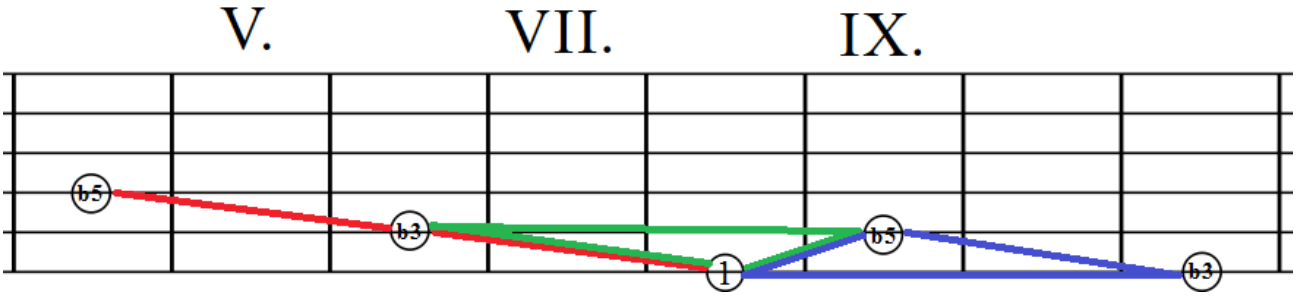
C minor triad (root position) fingerings, G-string



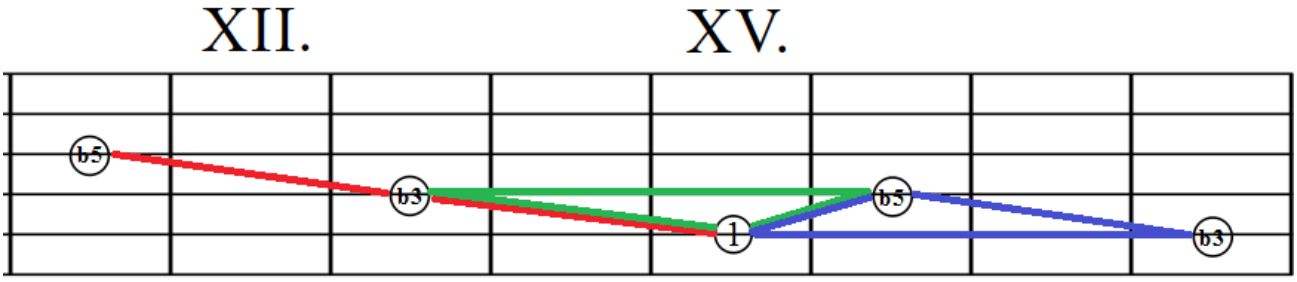
C minor triad (root position) fingerings, B-string

3.3.3. Diminished Triads

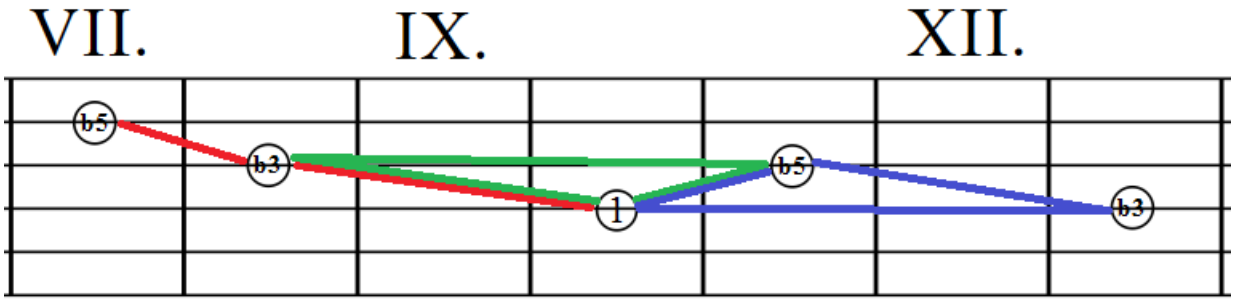
C diminished triad (root position)



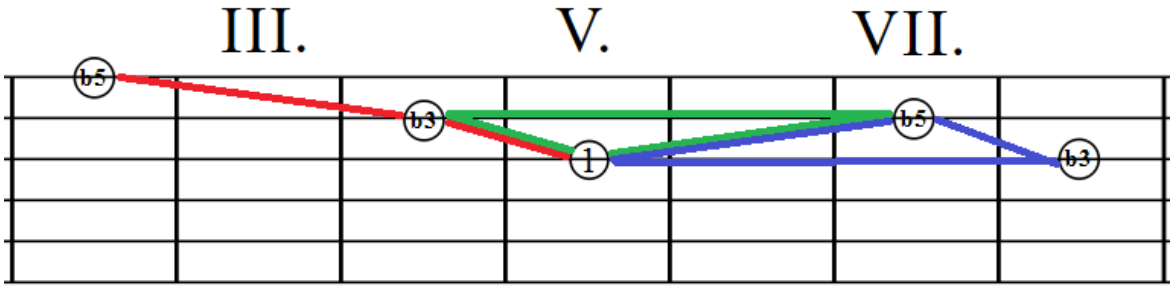
C diminished triad (root position) fingerings, low E-string



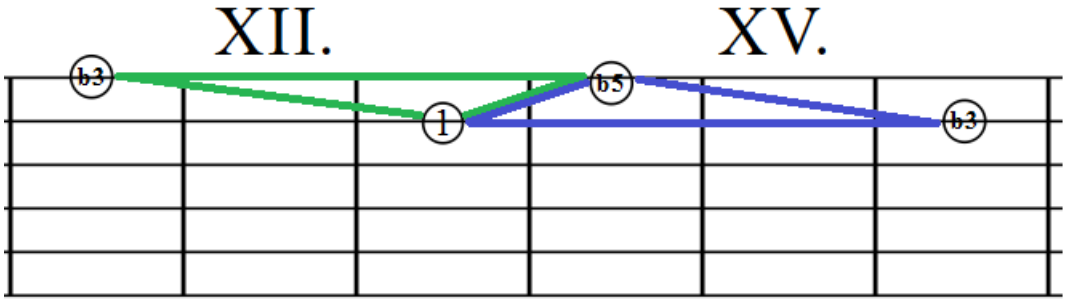
C diminished triad (root position) fingerings, A-string



C diminished triad (root position) fingerings, D-string



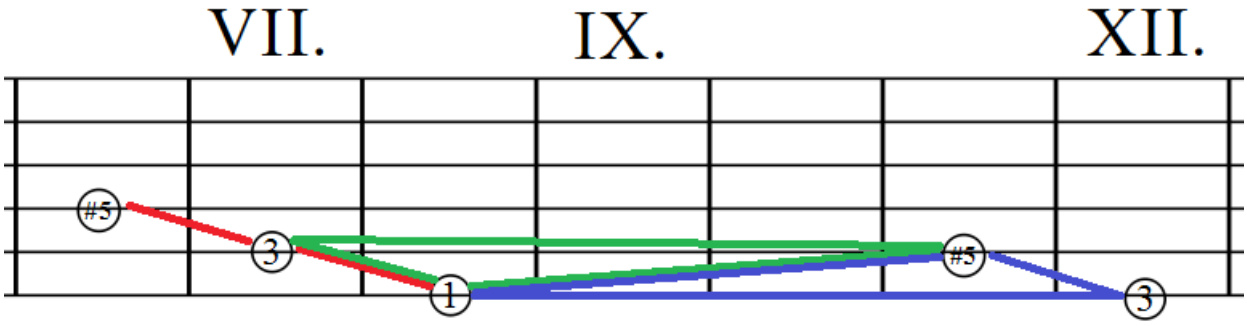
C diminished triad (root position) fingerings, G-string



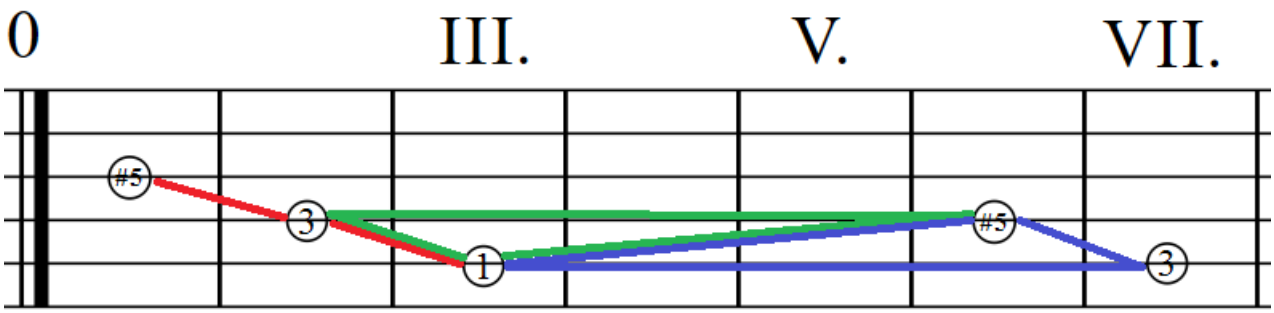
C diminished triad (root position) fingerings, B-string

3.3.4. Augmented Triads

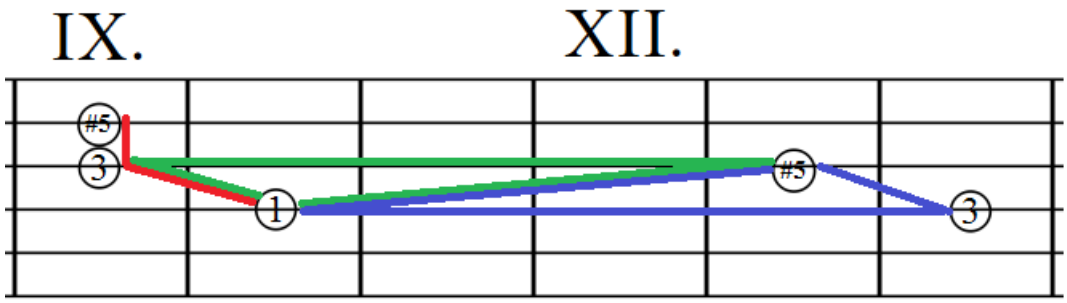
C augmented triad (root position)



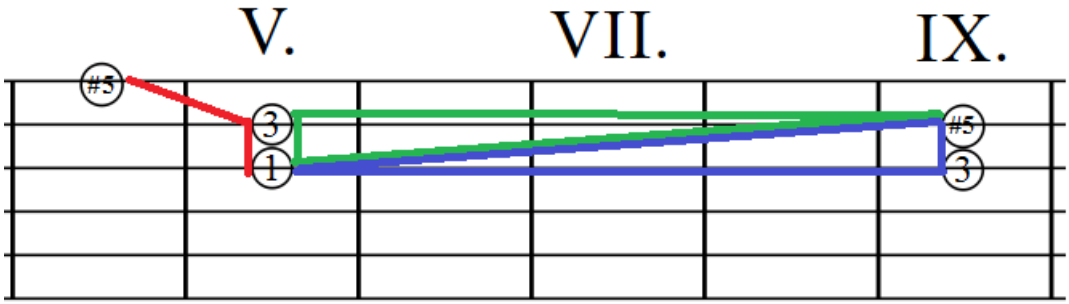
C augmented triad (root position) fingerings, low E-string



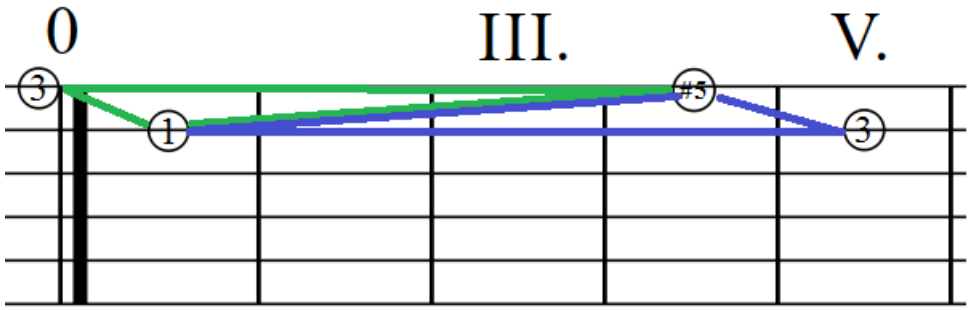
C augmented triad (root position) fingerings, A-string



C augmented triad (root position) fingerings, D-string



C augmented triad (root position) fingerings, G-string



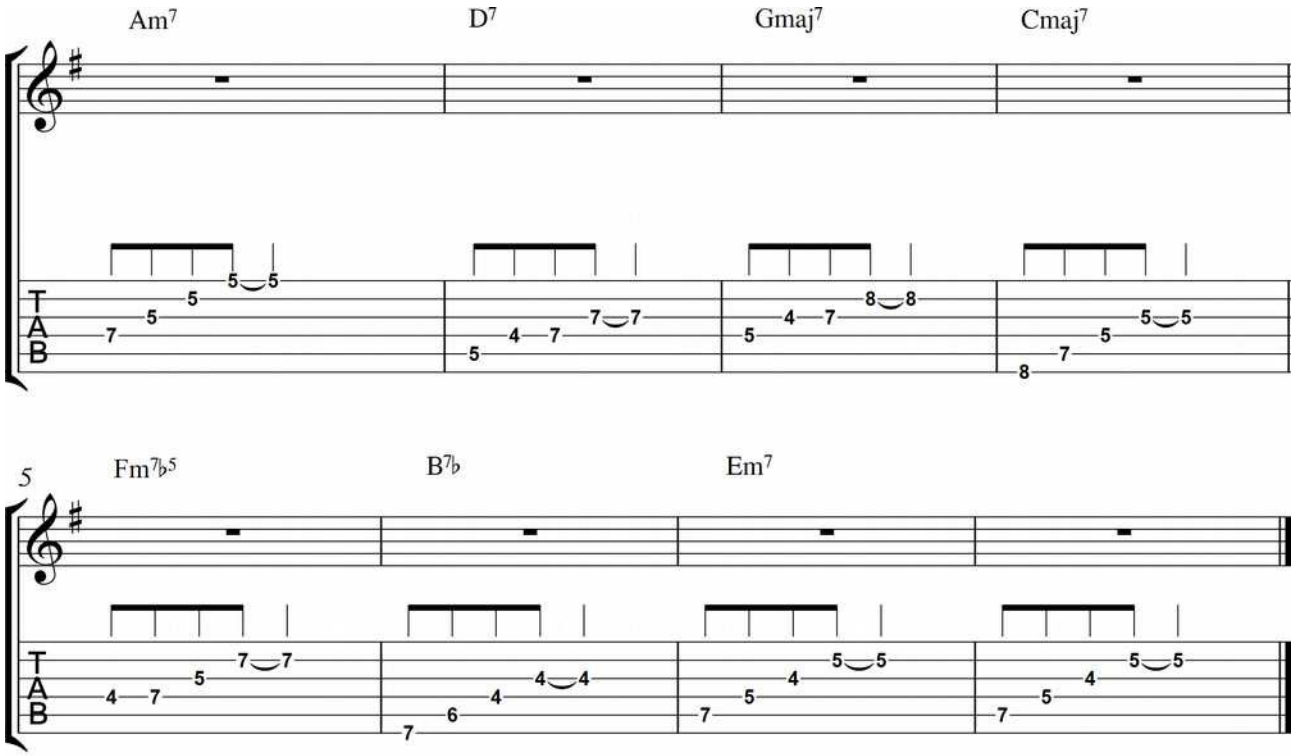
C augmented triad (root position) fingerings, B-string

Once you're comfortable with the major/minor/diminished/augmented triads, have a look at the sus2 and sus4 triad shapes, too!

3.3.5. Triad Exercise Over Jazz Cadence

In this segment I want to make a few more suggestions for challenging your triad knowledge even further. Obviously you don't want to be solely practising everything in the key of C (although that is a great method at first to compare different triad qualities with each other). Try any of these exercises:

1. Pick a random key and triad type, play through all fingerings
2. Choose a triad type, run through the cycle of fifths. Either stay in one position and shift through the keys or play through all positions in one key before moving on to the next.
3. Choose a chord progression or jazz standard and play the triads over the chords. If you want to further challenge yourself, try to stay in one position regardless of the chord changes. Here's how that could look like applied to a classic jazz cadence:



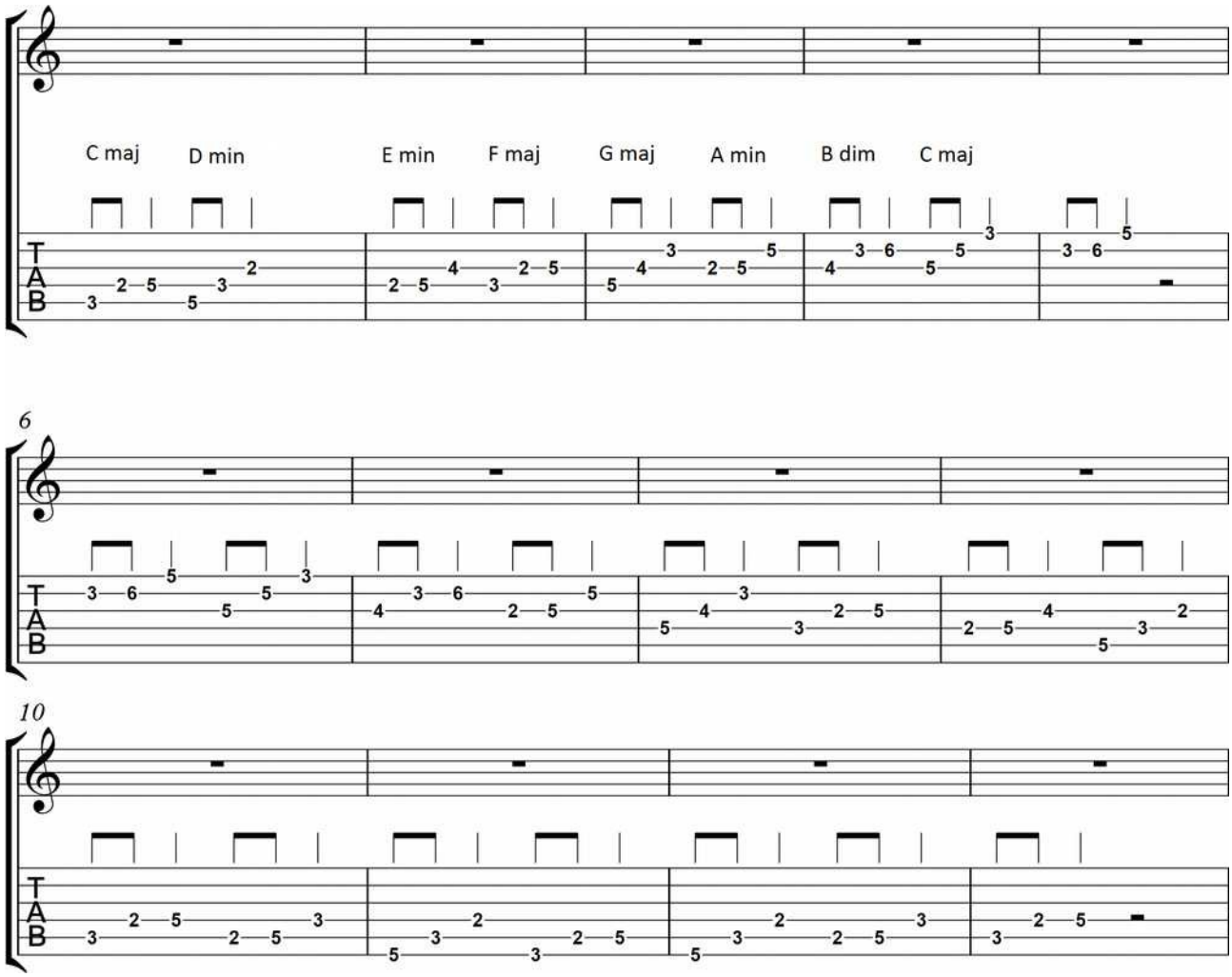
Triad exercise over jazz progression

Video/tab File: jazz_triad_ex

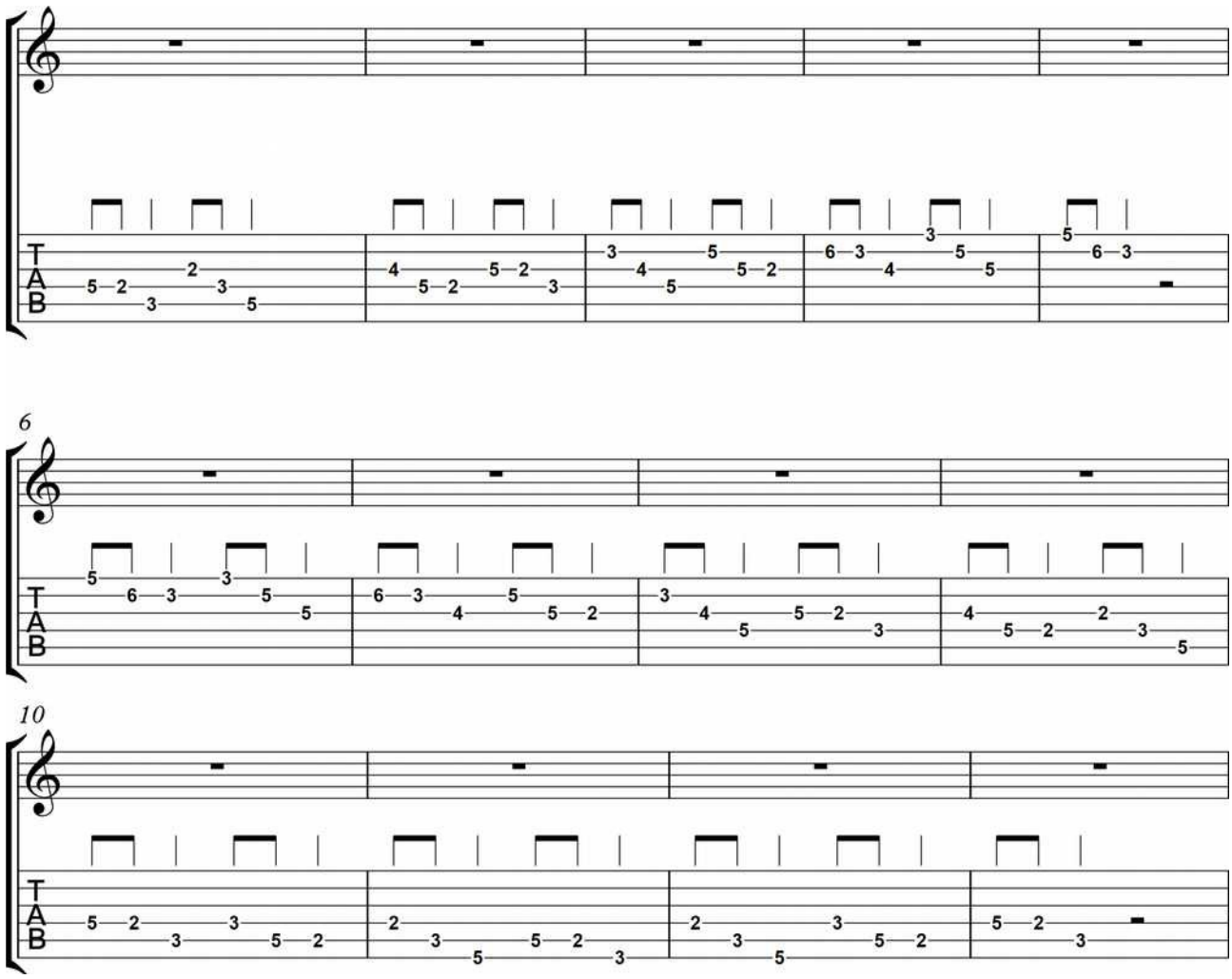
4. Pick a key, tonality and position, then play through all diatonic triads in that position. This means you build a triad from every scale degree by stacking two diatonic thirds onto it. In functional/intervallic numbers, that means the triads will be built as following:

Scale Degree	Intervallic Formula	In the key of C major
I	1-3-5	C-E-G (C maj)
II	2-4-6	D-F-A (D min)
III	3-5-7	E-G-B (E min)
IV	4-6-8	F-A-C (F maj)
V	5-7-9	G-B-C (G maj)
VI	6-8-10	A-C-E (A min)
VII	7-9-11	B-D-F (B dim)

Try both ascending and descending triads.



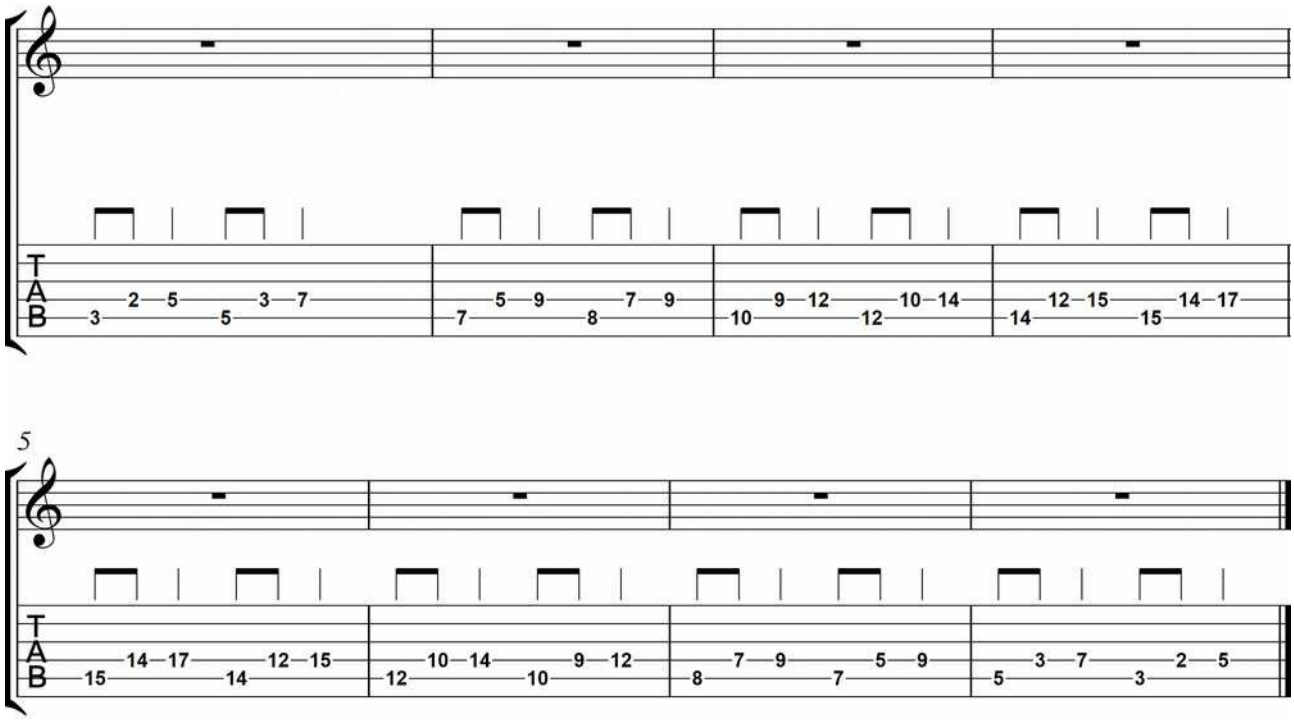
C major ascending diatonic triads exercise, A-position



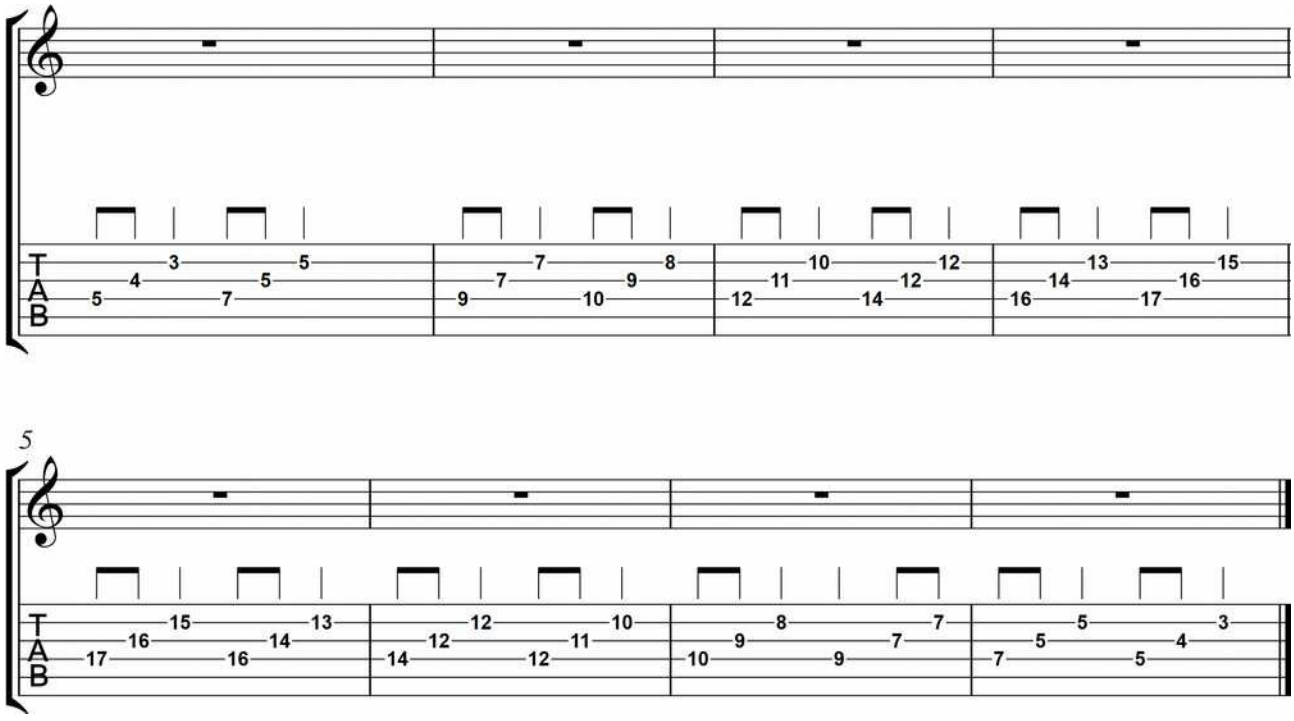
C major descending diatonic triads exercise, A-position

Video/tab Files: Cmaj_triads_up & Cmaj_triads_down

5. Do the same exercise, but now choose one triad fingering shape and move it horizontally up and down the neck. Here are two examples using different fingering types.



C major diatonic triads exercise (horizontal)



G major diatonic triads exercise (horizontal)

This exercise alone can be expanded almost indefinitely by varying the key, tonality, string set, triad inversion, flipping the notes of the triad around, etc.

Video/tab Files: Cmaj_triads_horizontal & Gmaj_triads_horizontal

3.3.6. Linear Examples using Triads

The following licks are an example on how to incorporate triads in your lead lines. While theory and fretboard visualisation is very important, you shouldn't neglect the importance of vocabulary. Once you're done with these examples, start transcribing lines from players that you like. Keep an eye out for how they make use of triads in their playing and expand on their ideas.

Video/tab Files:
Cm_triads_lick
Cm_triads_lick_2
Cmaj_triads_lick
outside_triads_lick1
outside_triads_lick2

Cm Diatonic Triads Lick
The first triad lick descends through a few diatonic triads of the C dorian scale superimposed over a Cm7 chord. Take some time to identify what these triads are and how they sound against the Cm7 harmony.

Cm Diatonic Triads Lick 2
This example employs a similar concept to the previous one, where several diatonic triads are played against a static harmony. I'm skipping over every other triad, which adds a note to the upper structure, that the line implies (9th, 11th, 13th) with every triad that goes by.

C major Triad Lick C & D Pair
Lick number 4 makes use of triad pairs to outline the underlying modal harmony, in this case C Lydian (C D E F# G A B). I'm moving up through the inversions, alternating C major (root-3-5) and D major (2-#4-6) triads. These two triads contain all the notes that define the sound of the Lydian mode.

Outside Triad Lick 1: Eb & F# Over Cm
In this triad line I've put a diatonic triad (Eb major) and a non-diatonic triad (F# major) against a Cm7 chord. That way the line starts off very inside the harmony, then creates tension and, similar to example 3, resolves back into the harmony to create interest.

Outside Triad lick 2: F# Over Cm
In this line I play an F# major triad against a Cm7 chord to create a lot of tension. When looking at these substitutions, it can be very helpful to investigate what functions their individual notes have against the underlying harmony. In case of F# over Cm7 it is the #4, b7 and b9 (F#, A#, C#). When a line creates such high levels of tension, I try to resolve it back inside the harmony smoothly, to give it the right context. In this case I do it by shifting into a little blues phrase to finish off the lick.

3.4. Triads as Chords

Triads are not only an important device in single note playing, they are also an essential part of thorough chordal knowledge and skill. They can be used in pretty much all styles of music and are a foundation for building bigger chords. Also a solid grip on triads is going to make it much easier for you to harmonise lines.

When practising triads as chords I recommend practising them on all sets of strings in all inversions. I find it more important to practise inversions specifically here than in single note playing. You need to have all three notes grouped in your mind and played simultaneously. When you need to grab a whole triad immediately, it's inefficient to find the 3rd or 5th and then start counting through the CAGED position.

For visualising triads as chords I apply the same thought process as I described for single notes:

When visualising inversions, my focal point tends to be the root note and how the other notes relate to it. Sometimes I find it easier to seek out the fifth or triad. Experiment to find what works best for yourself.

The following diagrams show the different triad types in all inversions in the key of C on all sets of strings. When playing through them, start on the lowest inversion on each string. Play up and down through all inversions before moving on to the next set of strings.

Again, the colours in these diagrams indicate the different inversions...

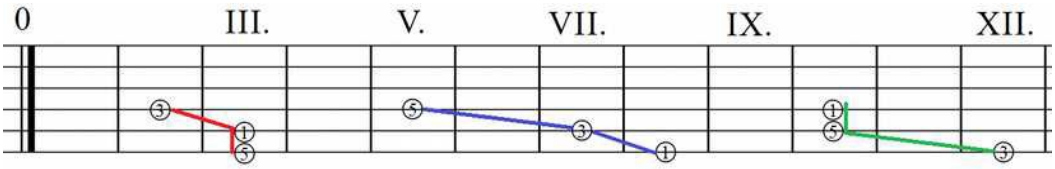
Blue: root position

Green: 1st inversion

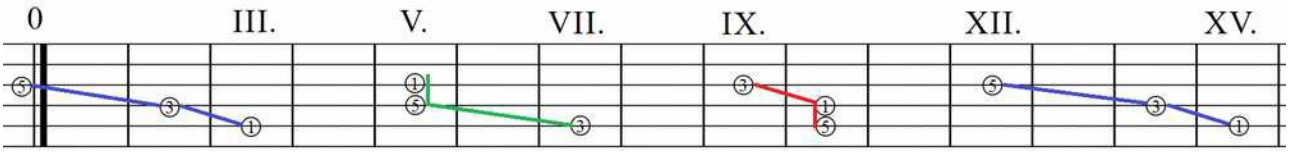
Red: 2nd inversion

3.4.1. Major Triads

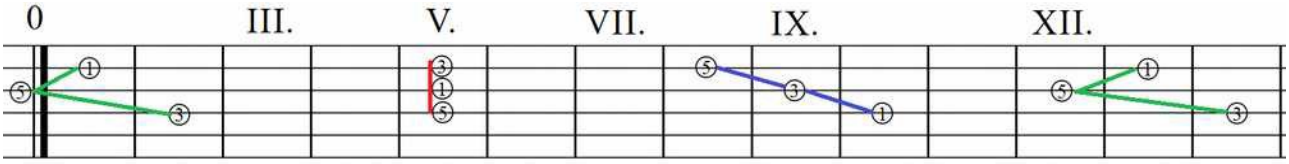
C Major



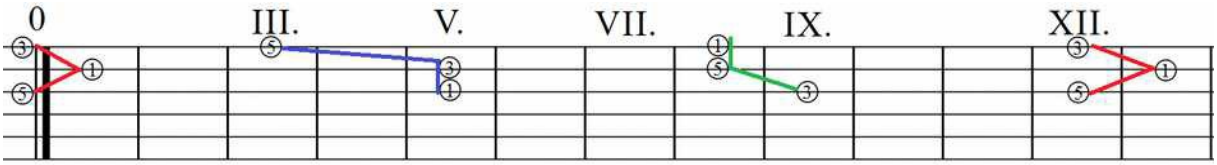
C major triad inversions, E-A-D stringset



C major triad inversions, A-D-G stringset



C major triad inversions, D-G-B stringset

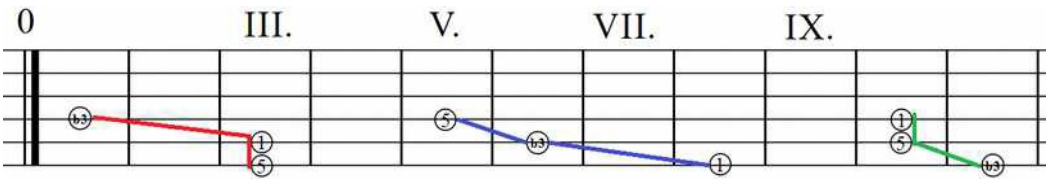


C major triad inversions, G-B-E stringset

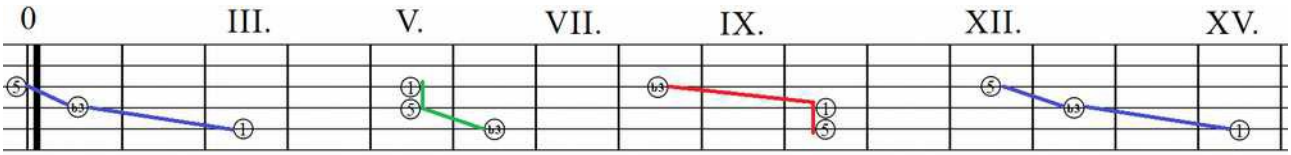
Videotab File: Cmaj_triad_chords

3.4.2. Minor Triads

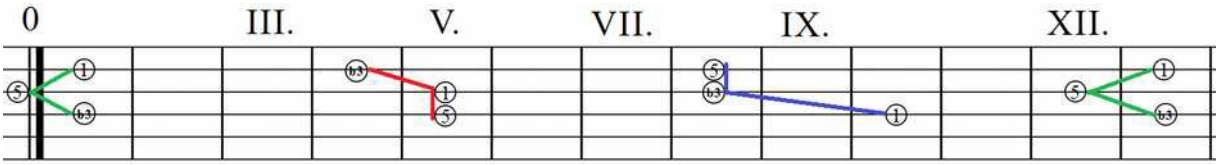
C Minor



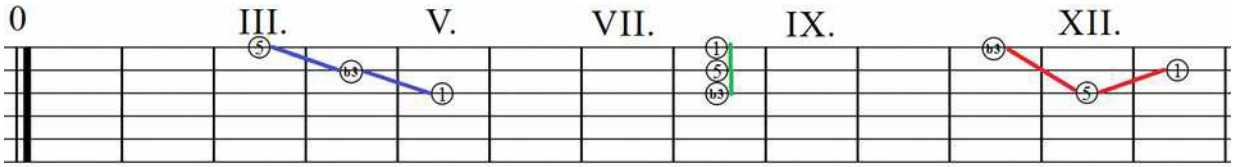
C minor triad inversions, E-A-D stringset



C minor triad inversions, A-D-G stringset



C minor triad inversions, D-G-B stringset

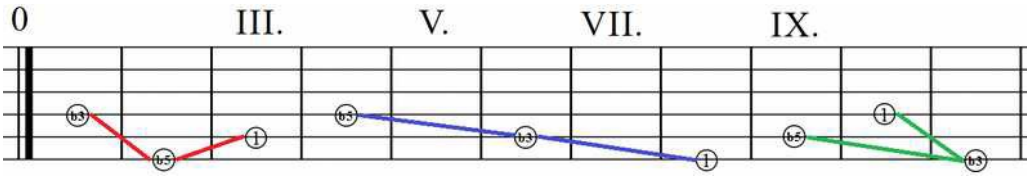


C minor triad inversions, G-B-E stringset

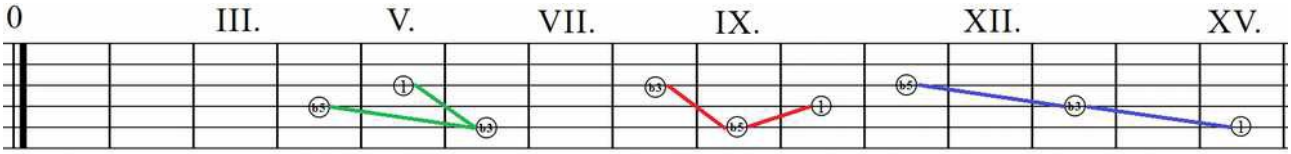
Video/tab File: Cm_triad_chords.mp4

3.4.3. Diminished Triads

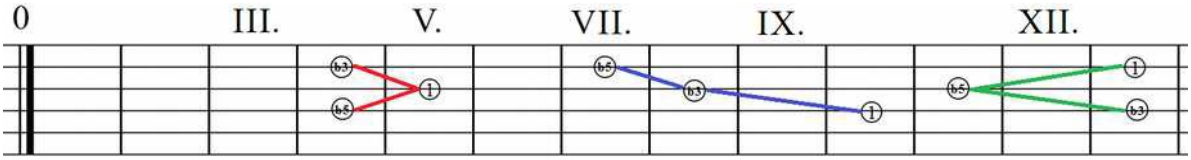
C Diminished



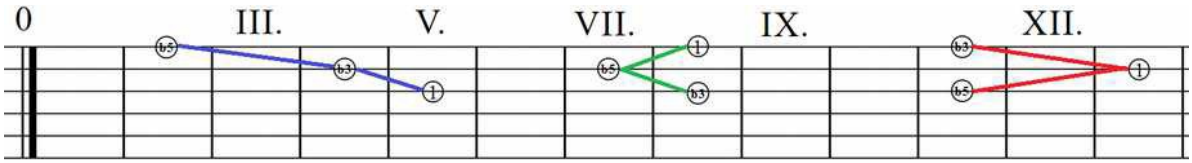
C diminished triad inversions, E-A-D stringset



C diminished triad inversions, A-D-G stringset



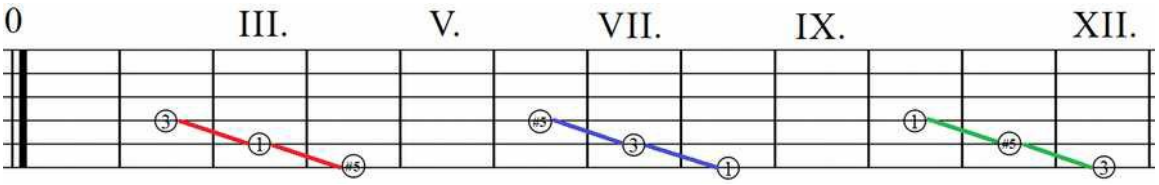
C diminished triad inversions, D-G-B stringset



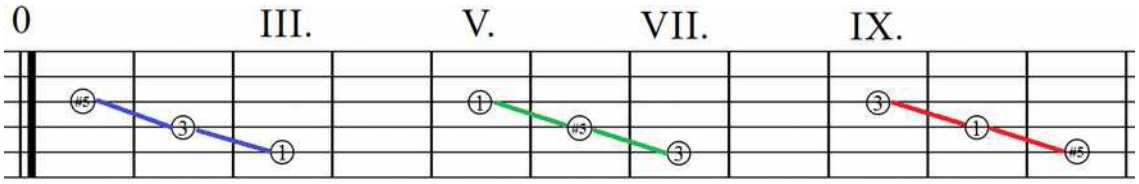
C diminished triad inversions, G-B-E stringset

3.4.4. Augmented Triads

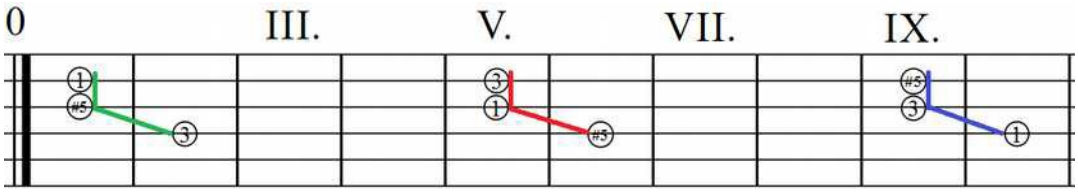
C Augmented



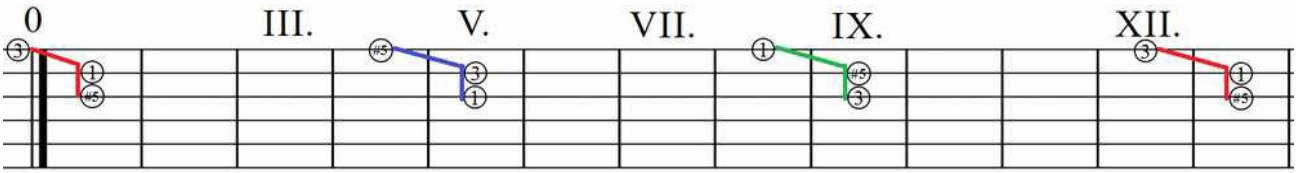
C augmented triad inversions, E-A-D stringset



C augmented triad inversions, A-D-G stringset



C augmented triad inversions, D-G-B stringset



C augmented triad inversions, G-B-E stringset

The augmented triad is symmetrical, meaning the distance from one note to the next is always identical: a major 3rd. This doesn't apply to the diminished triad, because of the distance from the b5 to the octave. Therefore every inversion has the same shape on the neck.

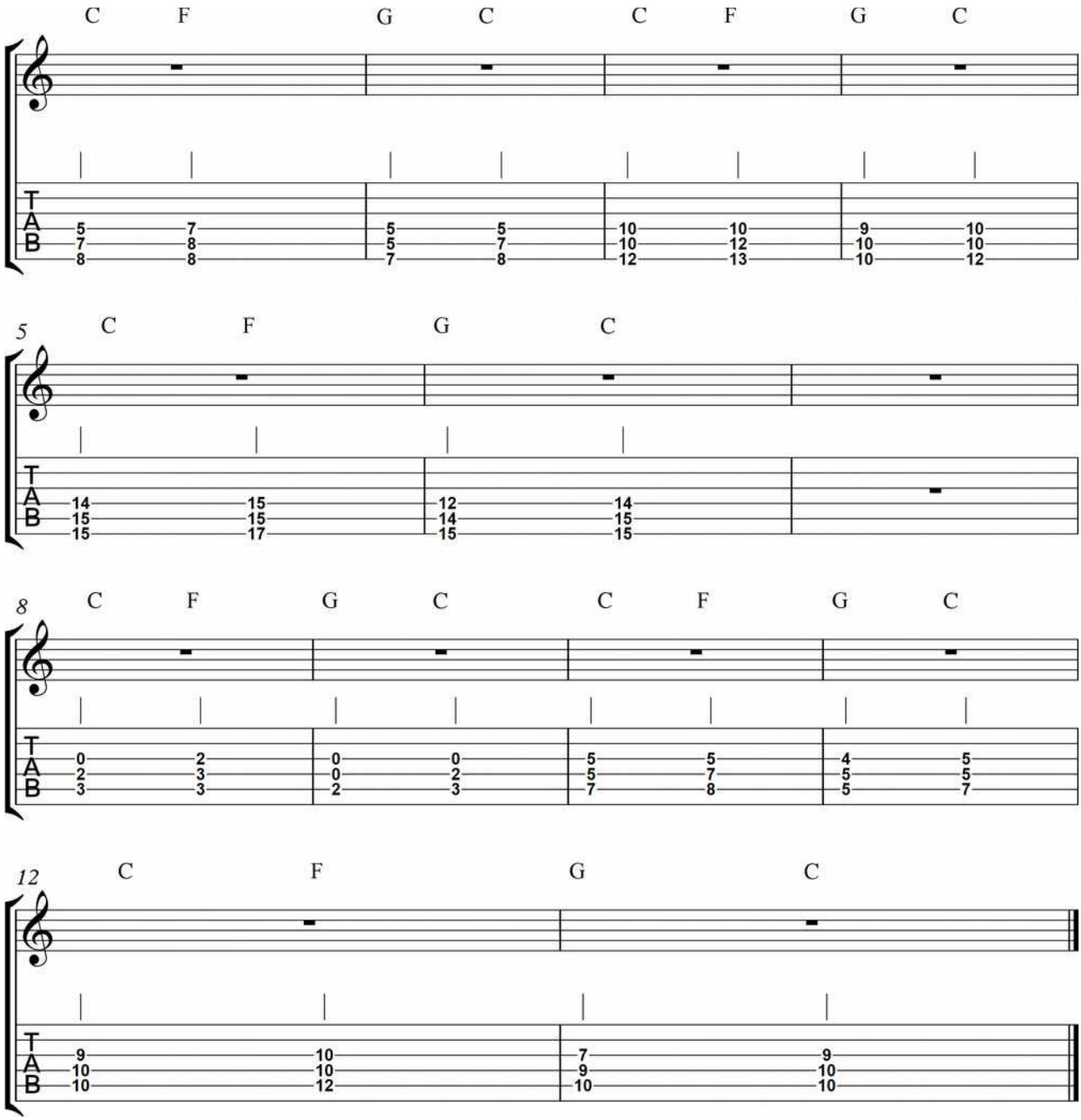
The augmented triad keeps repeating every major 3rd (or every 4 frets). This means the C augmented triad in root position is exactly the same chord as the E augmented triad in second inversion and the G# augmented triad in first inversion. C augmented contains the same notes as E and G# augmented (C E G#), but a different note is seen as the root note. Therefore the functionality of the rest of the notes is flipped too, depending on what root note you compare them to.

3.4.5. Triad Chord Practice Advice
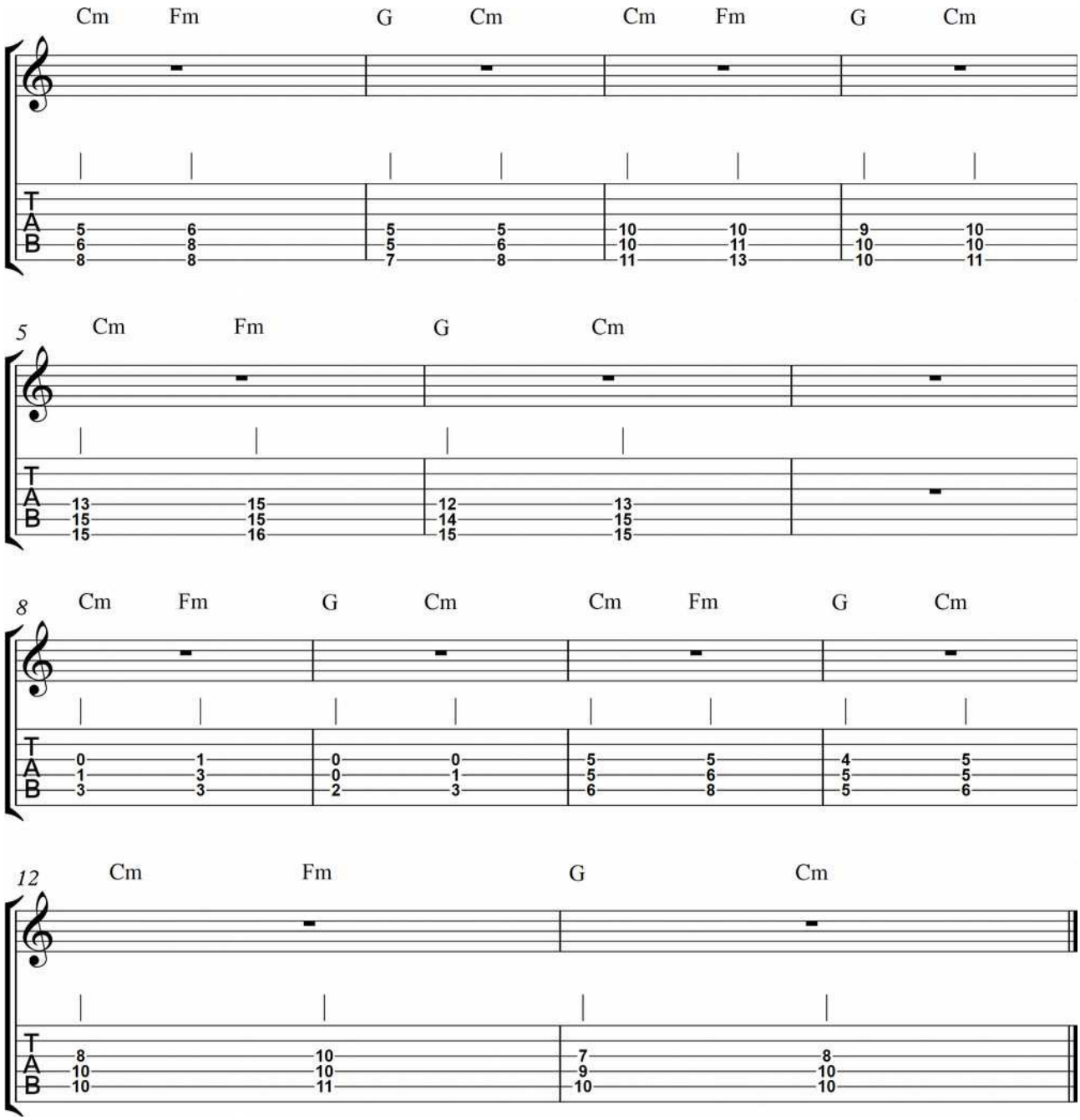
Let's look at some ways to further deepen your knowledge of triad based chords on the guitar.

1. Pick a random key and triad type, play through all fingerings on any set of strings.
2. Choose a triad type, run through the cycle of fifths. Either stay in one position and shift through the keys or play through all positions in one key before moving on to the next.
3. Choose a chord progression and play the corresponding triads. Go through several positions and string sets. Use voice-leading as much as possible - this means that each note should either stay or move to the closest note in the next chord.

Let's try this exercise over a I-IV-V-I progression in major and minor.



C major cadence triad exercise



C minor cadence triad exercise

Extend these exercises to the missing sets of strings by yourself!

Video/tab Files: Cmaj_cadence & Cmin_cadence

3.4.6. Using Triads in 7th Chord Harmony

Triads are used in all kinds of music ranging from pop to classical, country, folk and so on. Other styles of music such as jazz rarely use triad based harmony, but 7th chord harmony. We'll be covering 7th chords extensively in the next volume of my Improvisation Masterclass, but I'll give you a quick overview of what they are.

Usually a 7th chord is seen as triad with another third stacked on top. As with triads there are different qualities of 7th chords such as major 7th or minor 7th, depending on their intervalic structure. We now have three 3rd intervals, so there are more permutations.

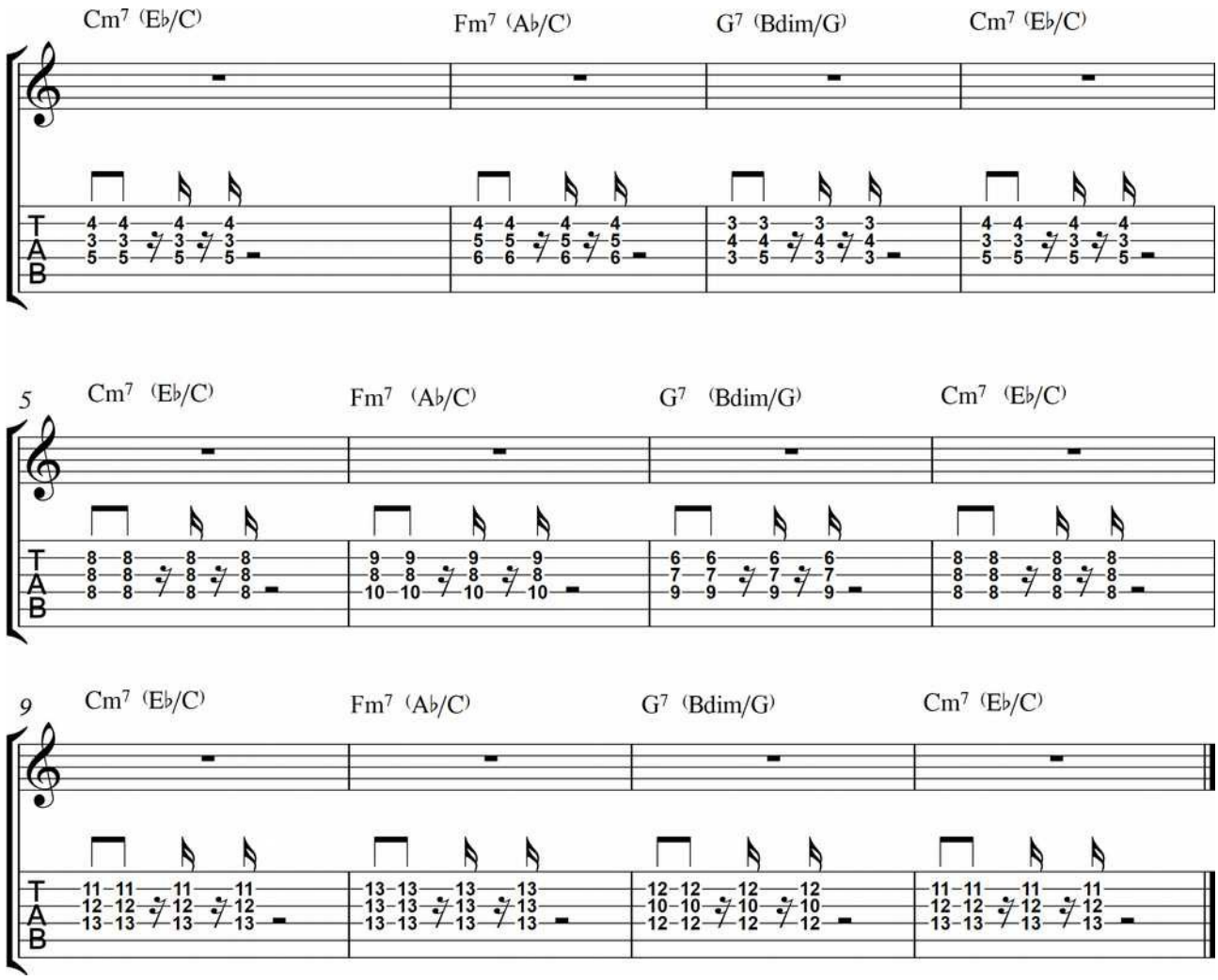
We've learned to visualise and play all these triads so it would be a real waste to not be able to use them chordally in a 7th chord context. So let's rethink the construction of 7th chord and see how we can use triads as substitutions to imply 7th chord harmony over a bass note provided by another instrument or player.

Let's assume the bassist is playing a static low C (you could record a low C drone with a keyboard and loop it when practising this). By playing different types of triads ABOVE the bass note (either a major 3rd above or a minor 3rd above) you can imply different types of 7th chords. You're basically stacking a triad onto a bass note. Here are your options:

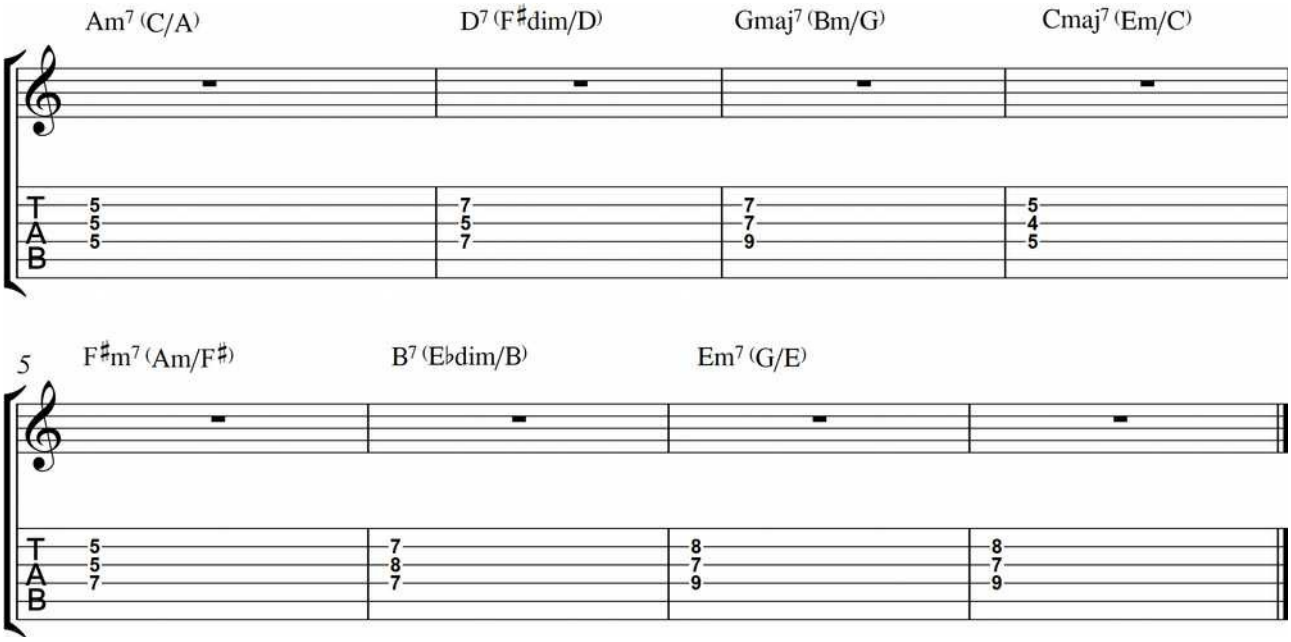
Implied Harmony	Bass Note	Stacked Triad	Formula (independent of key)
C Major 7	C	Em (E-G-B)	Bass Note + minor triad a major 3rd above
C Dominant 7	C	Edim (E-G-Bb)	Bass Note + diminished triad a major 3rd above
C Minor 7	C	Eb (Eb-G-Bb)	Bass Note + major triad a minor 3rd above
C Minor 7 b5	C	Ebm (Eb-Gb-Bb)	Bass Note + minor triad a minor 3rd above
C Diminished 7	C	Ebdim (Eb-Gb-A)	Bass note + diminished triad a minor 3rd above

We can now use this knowledge to "comp" (= to accompany) different jazz standards or play funk rhythm using triads, etc. The beauty of this is that you can easily jump back and forth between different inversions, which is something that is not as easily done with full 7th chords on the guitar.

Also a lot of times, omitting the bass note from your guitar chords can make a rhythm section sound a lot cleaner and less muddy; it avoids having too many instruments covering the same spectrum. Basically you're maximising the effect of your triad practice by using your existing knowledge in this new context. Let's have a look at two examples.



7th chord Funk exercise using triads



7th chord Jazz cadence exercise using triads

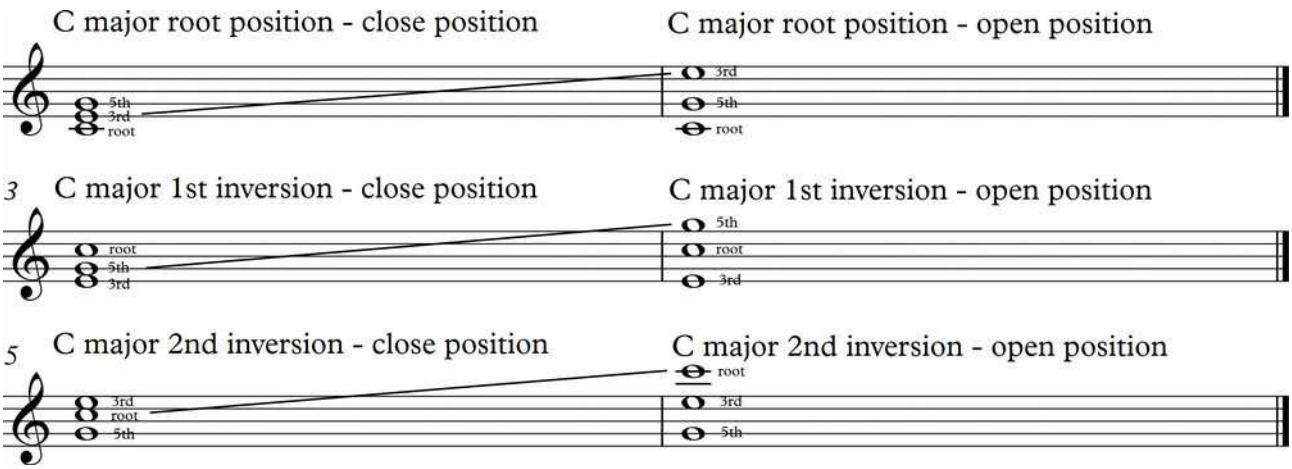
Video/tab Files: funk_inversions & jazz_cadence_ext

3.5. Open Position Triads

3.5.1. Theory

All previous examples of triads have been in close position (see 3.2 for an explanation). In this voicing type all triad notes are arranged as close together as possible. The distance between the lowest and highest note is relatively small - the notes are spaced within an octave. This covers a rather narrow range of pitch.

If you want to have a wider spread of the same intervallic functions, you can put them into open position, where the notes span more than an octave. In order to do this, take a triad in any inversion and raise the middle note by an octave. The lowest note still determines the name of the inversion. Note: the term "open position" in this instance has nothing to do with open strings on the guitar neck. It refers to the spacing of notes within a triad.



Close/Open position comparison

Even though the open position triads contain the same notes, they way they're spread out gives these triads quite a different feel. You might find they sound a bit more distinct, interesting or even dramatic. I'm sure you've heard players like Eric Johnson (Cliffs of Dover), Steve Morse (Well Dressed Guitar) or Pat Metheny using these triads in their playing and compositions with great effect.

This concept can be applied to any triad type mentioned so far (maj, aug, sus etc). In the next few pages I'll concentrate on just the major and minor variants. As you'll probably expect from me by now, I urge you to discover the other triad types on your own, following the process described here.

3.5.2. Visualisation

Open position triads can be used in chordal and single note contexts, just as we did with close position triads in the previous pages. The wide intervallic nature means we never have two notes on the same string. Therefore we don't have to create two sets of shapes - we use the same shapes for chords and single notes. To make visualisation easier, I came up with a system based on different combinations of strings. This way you won't miss out on any usable shape. Practise all of them and then memorise and use the ones you find the most convenient to play. I skipped a few inversions with certain string combinations, because I found them unusable due to their "stretchiness".

Here's the list of string sets to try these triads with:

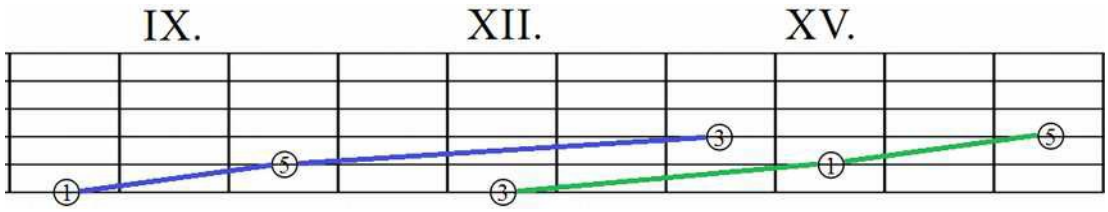
E-A-D; E-A-G; E-D-G; E-D-B
A-D-G; A-D-B; A-G-B; A-G-E
D-G-B; D-G-E; D-B-E;
G-B-E

Try every combination and decide for yourself which fingerings work for you visually, sonically and physically.
The colors in these diagrams indicate the different inversions.

Blue: root position
Green: 1st inversion
Red: 2nd inversion

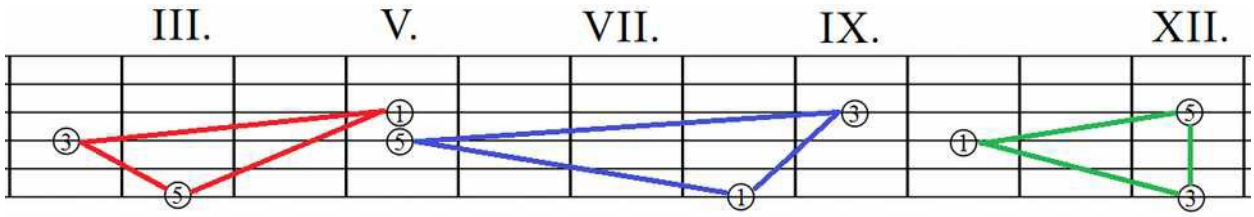
3.5.3 C Major Triads (Open Position)

E-A-D Strings



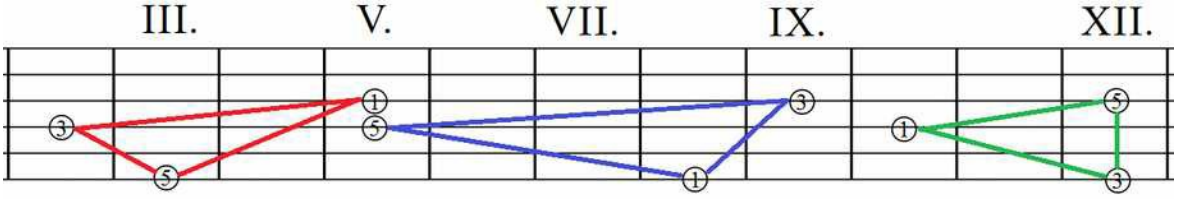
C major inversions (open position), E-A-D strings

E-A-G Strings



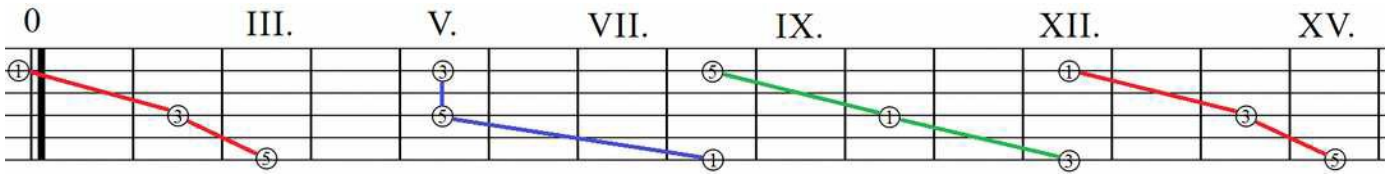
C major inversions (open position), E-A-G strings

E-D-G Strings



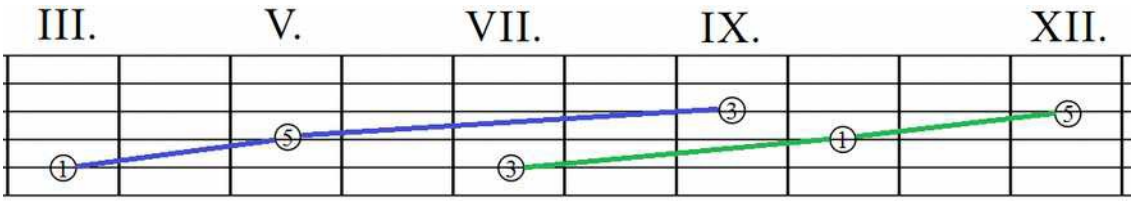
C major inversions (open position), E-D-G strings

E-D-B Strings



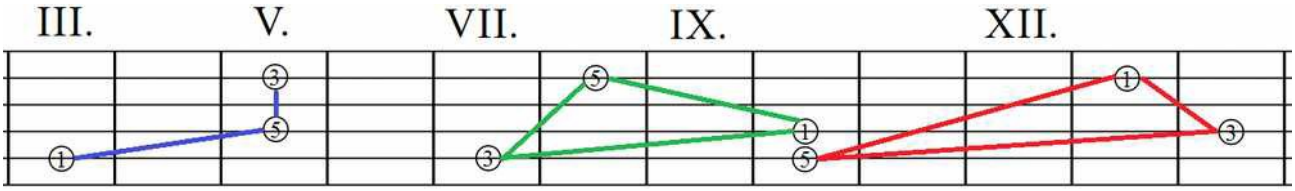
C major inversions (open position), E-D-B strings

A-D-G Strings



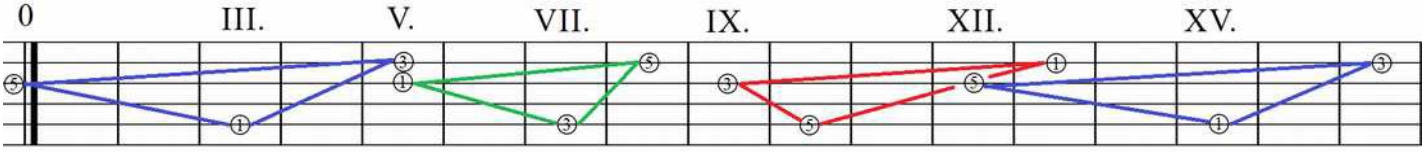
C major inversions (open position), A-D-G strings

A-D-B Strings



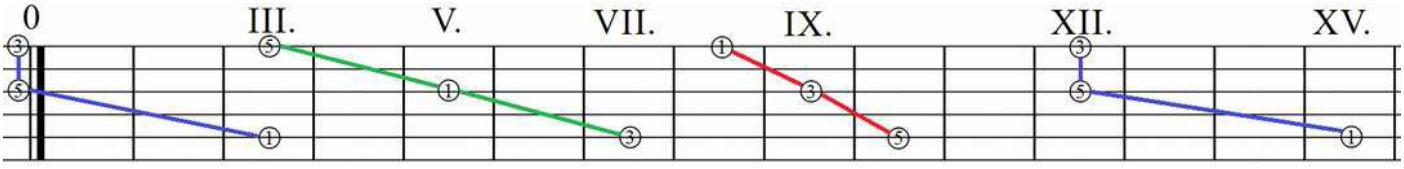
C major inversions (open position), A-D-B strings

A-G-B Strings



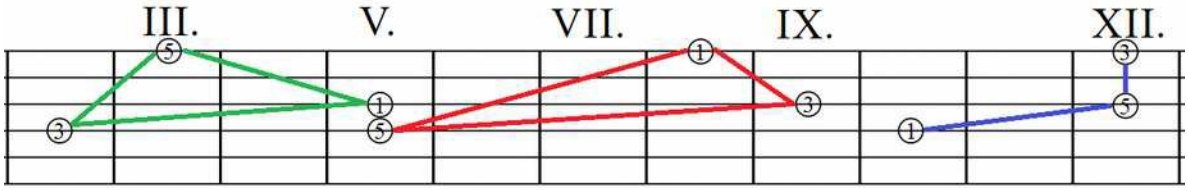
C major inversions (open position), A-G-B strings

A-G-E Strings



C major inversions (open position), A-G-E strings

D-G-E Strings



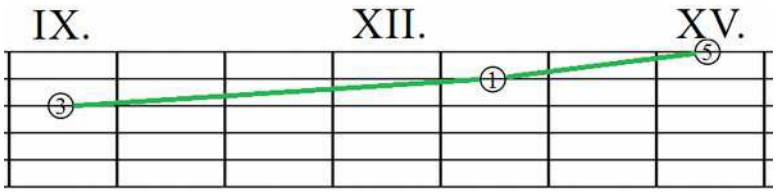
C major inversions (open position), D-G-E strings

D-B-E Strings



C major inversions (open position), D-B-E strings

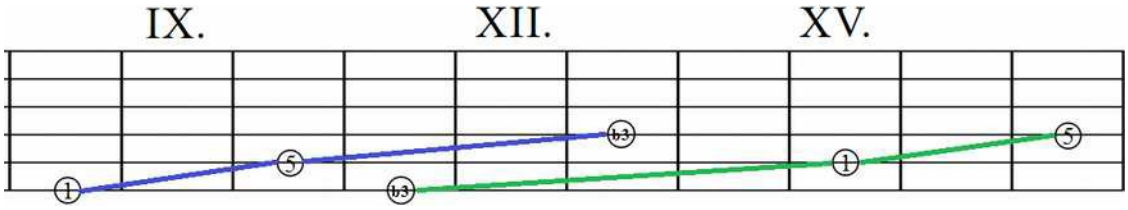
G-B-E Strings



C major inversions (open position), G-B-E strings

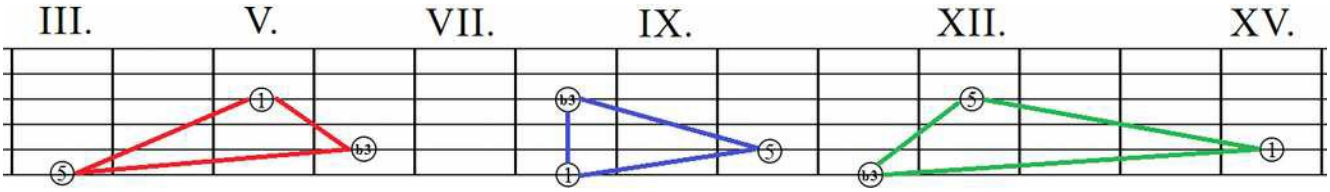
3.5.4 C Minor Triads (Open Position)

E-A-D Strings



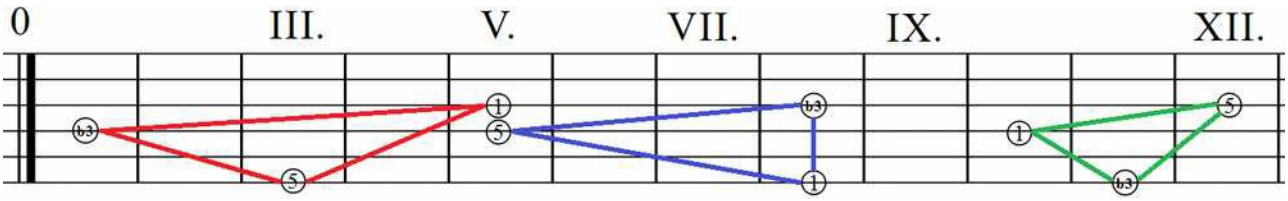
C minor inversions (open position), E-A-D strings

E-A-G Strings



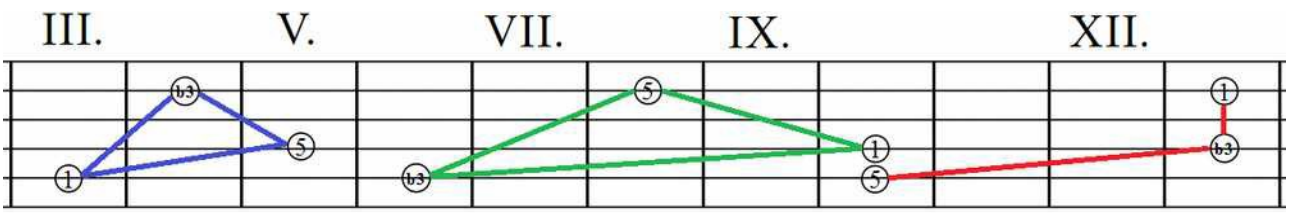
C minor inversions (open position), E-A-G strings

E-D-G Strings



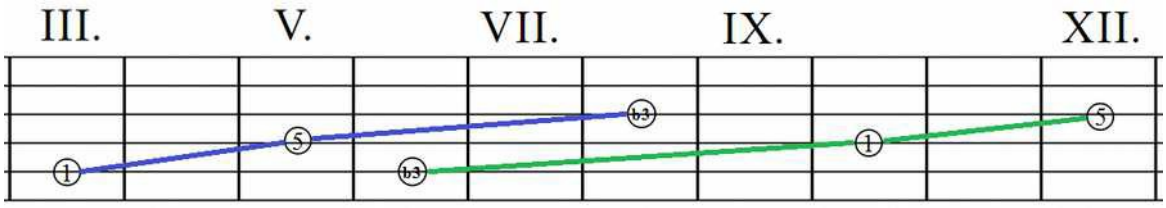
C minor inversions (open position), E-D-G strings

E-D-B Strings



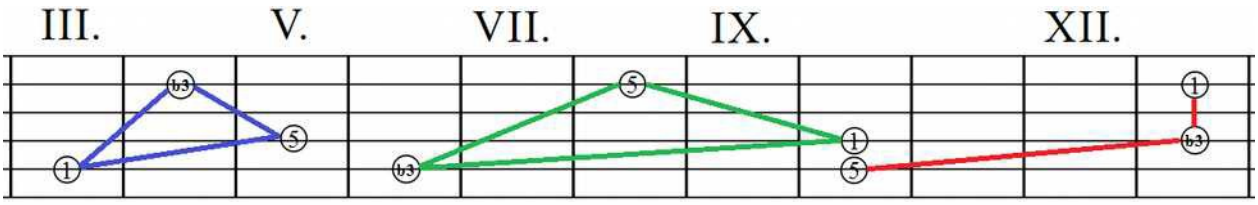
C minor inversions (open position), E-D-B strings

A-D-G Strings



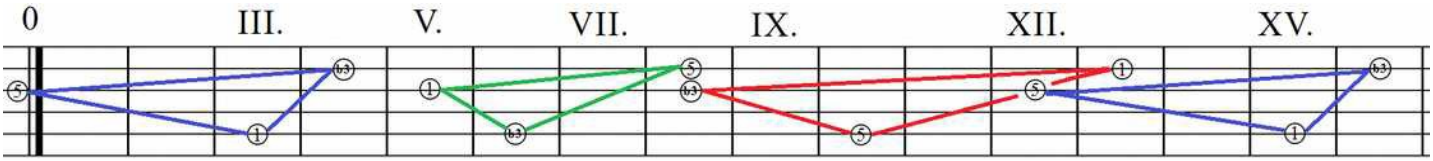
C minor inversions (open position), A-D-G strings

A-D-B Strings



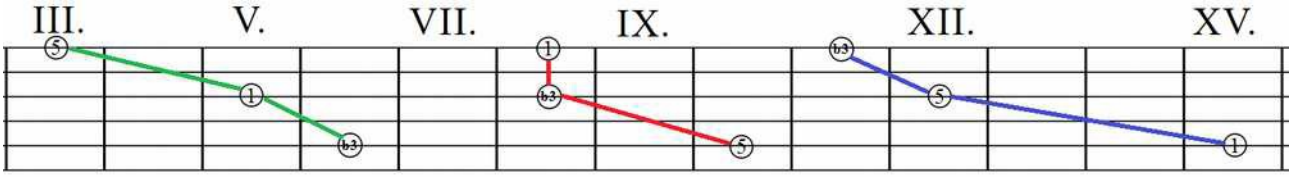
C minor inversions (open position), A-D-B strings

A-G-B Strings



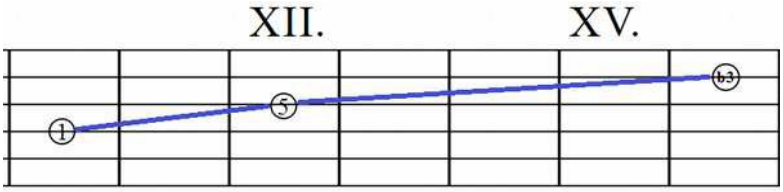
C minor inversions (open position), A-G-B strings

A-G-E Strings



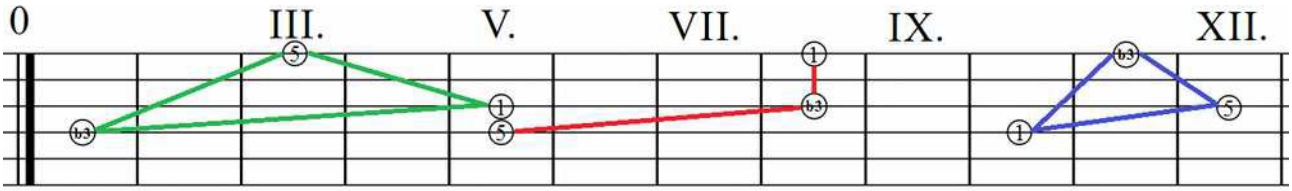
C minor inversions (open position), A-G-E strings

D-G-B Strings



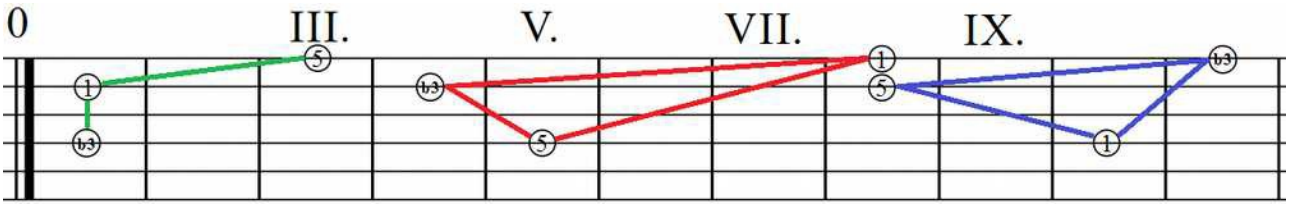
C minor inversions (open position), D-G-B strings

D-G-E Strings



C minor inversions (open position), D-G-E strings

D-B-E Strings
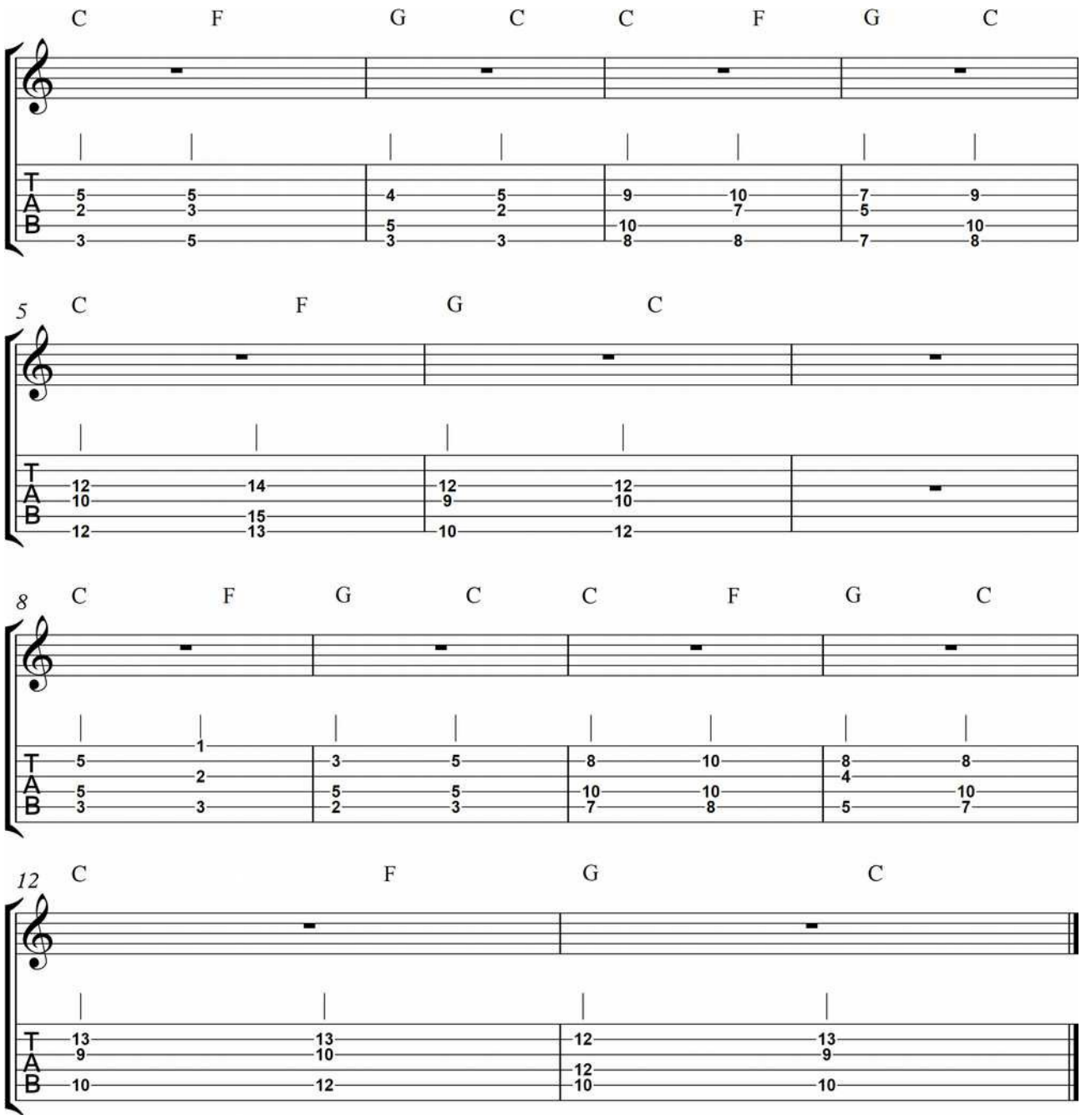


C minor inversions (open position), D-B-E strings

3.5.5 Open Position Triad Practice Advice
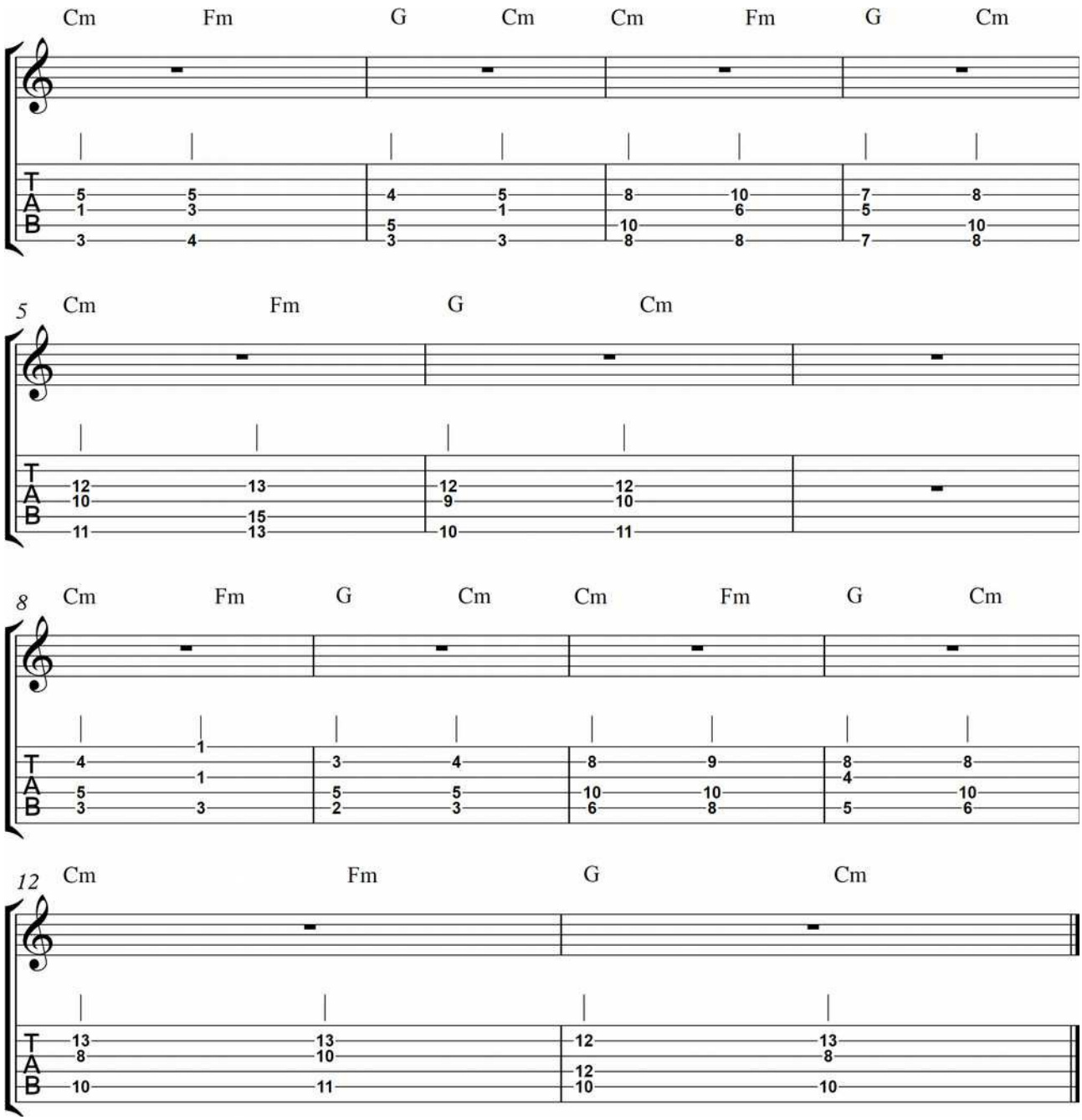
The practice methods for open position will be similar to the close position suggestions I've made. There will be no real distinction between single note and chordal exercises as the fingerings are the same. So try any exercises notated as single notes in a chordal fashion and vice versa.

1. Pick a random key and triad type, play through all fingerings on any set of strings.
2. Choose a triad type, run through the cycle of fifths. Either stay in one position and shift through the keys or play through all positions in one key before moving on to the next.
3. Choose a chord progression and play the corresponding triads. Go through several inversions and string sets. Voice lead as closely as possible when going from one triad to the next, meaning the notes move as little as possible.

Let's try this exercise over a I-IV-V-I progression in major and minor.



C major cadence triad exercise (open position)
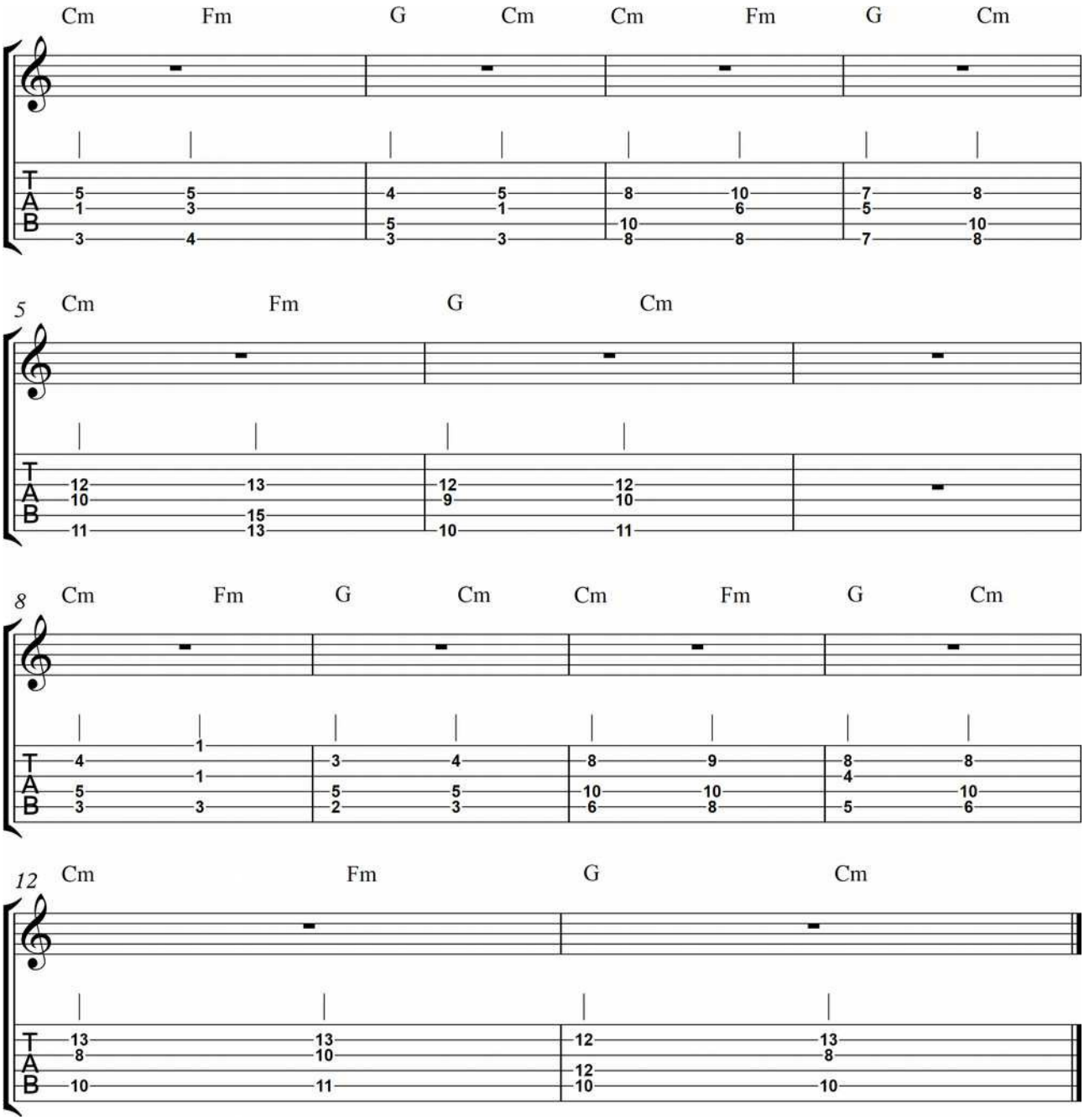


C minor cadence triad exercise (open position)

Extend these exercises to the missing sets of strings by yourself!

Video/tab Files: Cmaj_cadence_open & Cmin_cadence_open

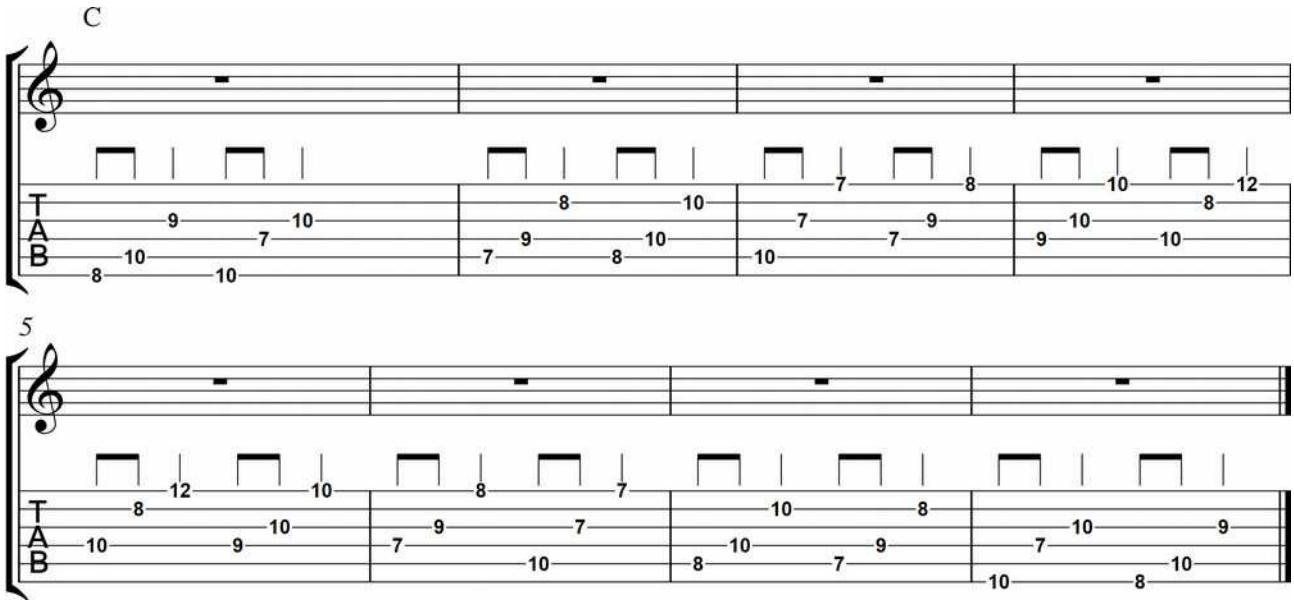
... and here's the same concept applied as single notes to a popular jazz cadence, voice led very closely. Try the same thing starting from the next higher or lower inversion and on different string sets. After you exhausted those possibilities, transpose and then move on to the next progression. Keep in mind that this process, depending on your current state of development, can take weeks, months or years. It's nothing to be "brushed over". All of this needs time until it sits with you.



Jazz cadence cadence triad exercise (open position)

Video/tab File: jazz_cadence_chords

4. Pick a key, tonality and position, then play through all diatonic triads in that position. You can apply the exact same exercises as described in 3.3.5. to the open position triads. Don't forget to do this with all types of inversions. Due to the more intricate nature of this type of voicing, I find it necessary to practise every inversion separately. Unlike with single note, close position triads, I don't look at inversions as a result of having learned a fingering across multiple octaves and then playing up the fingering, starting from the 3rd or 5th.



C major ascending diatonic triads exercise (open position)

Video/tab Files: open_triads

3.5.6. Linear Examples using Open Position Triads

To finish our intense study of triads I want to provide you with a few examples of how open position triads can be used in a musical context.

Video/tab Files:
open_triads_lick1
open_ triads_lick2
jazz_cadence_notes

Open Position Triad Lick 1
Here we have the same concept as triad lick #4 applied to a C major 7th chord, but using open position triad inversions as opposed to close voiced triads.

Open Position Triad Lick 2
The next example superimposes diatonic triads over a static chord similar to triad licks #1 and #2 but utilising open position, sequenced in a very interesting fashion.

Open Position Jazz Cadence (single notes)
The last line is an example that shows how you can use several open position triad inversions over the jazz cadence we've studied throughout this masterclass. It gives the phrase a very inside, almost classical sound.

4. Bonus Solo Transcription

To finish off this masterclass, I want to provide you with two solo examples, that you can study, analyse and play on your own. Despite the importance of conceptual and intellectual studying, one should never neglect the importance of learning and assimilating vocabulary and performing music (learning by doing). When going through these solos, try to recognise the concepts we've covered so far. For example keep an eye on which triads I use against the chords or what CAGED positions I tend to use when playing through these chord changes.

Video/tab Files: solo1 & solo2

Now I leave you to practise and hope to see you in the next instalment of the Improvisation Masterclass. Enjoy learning and see you soon!